

Theory of the Diffraction of Electrons by Crystals

H. Bethe

Annalen der Physik 392, 55–129 (1928)

English translation

Translated by Yueming Guo using Muse Spark 1.3 (Meta)

dr.yueming.guo@gmail.com

To our best knowledge, this paper has never been translated into English before. Yet the German original has been cited by many electron microscopists and diffraction physicists, most of whom today cannot read German. The translator does not read German. Made possible by modern AI, this translation aims to put Bethe's founding work in the hands of every practitioner in the field.

How to cite: Y. Guo, English translation of H. Bethe,
"Theorie der Beugung von Elektronen an Kristallen," Ann. Phys. 392, 55–129 (1928).

Contents of this edition

Translation

Translator's note and Bethe's original contents list	1
The paper, §§1–12 and Summary	2–74
(journal pp. 55–129 marked in the margins throughout)	

Appendices and reference material

Appendix: Printing errors in the 1928 original	75–76
Original figures 1–8 (German facsimile)	77–84
Original tables 1–4 (Tabellen 1–4, German facsimile)	85–88
Appendix: Historical terminology and modern equivalents	89

The cover and this contents page are unnumbered. Within the translation, “p. 55”, “p. 56”, etc. in the margins refer to the original journal pagination.

English translation

Translator's note

This is an English translation of Hans Bethe's 1928 paper "Theorie der Beugung von Elektronen an Kristallen" -- his doctoral dissertation, founding the dynamical theory of electron diffraction. The translation follows the original page by page; the German journal page numbers are marked in the margin as "p. 55", "p. 56", etc.

- **Equations** keep Bethe's original numbering, shown as (1), (2), ... (94) at the right margin,
- Symbols follow Bethe's original notation; source-print anomalies are preserved as printed and documented in the audit appendix, while transcription errors have been corrected.
- German decimal commas have been converted to English decimal points (0,063 -> 0.063). "lg" denotes the base-10 logarithm.
- Footnotes keep their original numbering within each page.

Translated by Yueming Guo using Muse Spark 1.3 (Meta).

Muse Spark 1.3: translation and about ten rounds of audit;
GPT-5.6 (OpenAI): about ten rounds of audit and cross-check.

Contents

- § 1. The Refractive Index for de Broglie Waves — p. 55
- § 2. The Field of the de Broglie Waves Inside the Crystal — p. 63
- § 3. Formulation of the Boundary Conditions — p. 69
- § 4. Reduction of the dynamical equations to two — p. 73
- § 5. The dispersion equation — p. 79
- § 6. Calculation of the intensities — p. 82
- § 7. Sketch of a third approximation — p. 91
- § 8. Relation to the continuum theories of X-ray reflection. Extinction laws — p. 94
- § 9. Symmetric Reflection. Coordinate Transformation — p. 100
- § 10. Calculation and Discussion of the Work Function — p. 105
- § 11. Calculation of the Fourier coefficients of the potential — p. 115
- § 12. Numerical calculations of the quantities of electron diffraction theory — p. 123
- Summary — p. 129

Theory of the Diffraction of Electrons by Crystals

H. Bethe

Annalen der Physik, Fourth Series, Vol. 87 (Vol. 392), pp. 55–129 (1928).

— p. 55 —

§ 1. The Refractive Index for de Broglie Waves

In recent times the wave nature of the electron has been directly demonstrated many times through beautiful experimental investigations. In all of these experiments we are dealing with the *crystal diffraction* of electrons either by *single crystals* [Davisson and Germer 1)] or by *microcrystalline* metals [Thomson 2), Rupp 3)]. Here we shall be concerned chiefly with the theoretical interpretation of the first-mentioned works, but the results are also in part applicable to diffraction by irregularly ordered microcrystals.

In the experiments of Davisson and Germer an electron beam of homogeneous velocity v falls perpendicularly upon the (111) face (octahedral face) of a nickel single crystal, and the electrons reflected back toward the entrance side are observed. It turns out that reflection occurs *selectively*, just as with X-rays: For a given de Broglie wavelength $\Lambda = \frac{h}{mv}$ (in other words, given velocity v) of the incident beam, one generally observes only continuous scattering in *all* directions; only at certain *discrete* incident wavelengths are the electrons preferentially diffracted into one *particular* direction 4) or into several symmetry-related directions.

Footnotes:

- 1) C. J. Davisson and Germer, Nature 119. p. 558. 1927; Phys. Review 30. p. 705. 1927; Proc. Nat. Ac. Amer. 14. p. 317. 1928. (D. and G. I, II, III.)
- 2) G. P. Thomson, Proc. Roy. Soc., A., 117. p. 600. 1928.
- 3) E. Rupp, Ann. d. Phys. 85. p. 981. 1928.
- 4) We specify the direction of the reflected beam by the angle Θ between it and the crystal normal, and by an azimuthal angle φ . Following Davisson and Germer, we denote the latter by the Miller indices of the simplest crystal face whose normal lies in the plane of reflection determined by the incident and reflected beams.
For longer electron waves, selective reflection comes into consideration chiefly in the (100) azimuth ($\varphi = 0, 120, 240^\circ$), in the (111) azimuth ($\varphi = 60, 180, 300^\circ$) and in the (110) azimuth ($\varphi = 30, 90, 150, 210, 270, 330^\circ$).

— p. 56 —

From this phenomenon of selective reflection one may conclude that the electron waves are diffracted by the crystal lattice of nickel.

Qualitatively, then, there is complete analogy between the reflection of electrons and that of X-rays; quantitatively, however, the analogy is only partial: The relation between the *direction* of the diffracted beams and the de Broglie wavelength in vacuum, $\Lambda = \frac{h}{mv}$, remains unchanged

$$n\Lambda = d \sin \Theta, \quad (1)$$

where n is the order of the reflection, d the distance between two adjacent rows of atoms running in the crystal surface perpendicular to the plane of reflection. 1) Relation (1) follows immediately from the lattice structure of the *crystal surface*, which indeed forms a simple surface grating, by means of the ordinary wave-kinematical reasoning employed in deriving the diffraction patterns of a line or surface grating; it is entirely independent of the diffraction phenomena in the *interior* of the crystal. The latter come into question only for the explanation of *selective* reflection, i.e. the determination of the magnitude of the wavelengths that can be selectively reflected. With regard to this magnitude there is, as is well known, a *discrepancy* between the wavelengths Λ computed from the measured velocity v of the incident electrons by the de Broglie equation, and the selectively reflectable wavelengths λ to be expected from Laue's theory of space-lattice diffraction; each observed wavelength Λ lies between two wavelengths to be expected from lattice theory, the longest observed wavelength being shorter than the longest to be expected in each individual azimuth. Fig. 1 represents these relations schematically,

Footnotes:

- 1) Ni is known to have a cubic face-centered lattice (side of the elementary cube 3.52 Å), and the atoms are therefore arranged in the form of equilateral triangles in an octahedral [(111)] face. For reflection in the (100) and (111) azimuths the sides of the triangles, for that in the (110) azimuth the altitudes, act as "grating lines."

— p. 57 —

For the observed discrepancy there are from the outset two possible explanations: a change in the crystal dimensions or a change in the wavelength of the incident electrons. Davisson and Germer at first invoked the first explanation on purely phenomenological grounds 1) and computed from their experiments that the spacings of the lattice planes parallel to the surface near the crystal surface would have to be contracted to 70–95 percent of their normal value in the interior of the crystal in order to account for the deviation in the selectively reflectable wavelengths. Of course no reality can be ascribed to such a contraction of the crystal, varying so extraordinarily strongly (and, moreover, quite irregularly) with the electron velocity, and the contraction factor was probably intended by Davisson and Germer from the outset only as a phenomenological description of their experiments.

[Fig. 1: schematic plot with abscissa in Angströms (0, 0.5, 1.0, 1.5, 2.0); crosses (×) mark the observed wavelengths Λ (after D. and G. II), circles (⊙) the wavelengths λ expected from diffraction theory, in the (111) azimuth; solid connecting lines mark the "new", dashed lines the old assignment.]

× Observed wavelengths Λ (after D. and G. II) } in the (111) azimuth ⊙ Wavelengths λ expected from diffraction theory } The connecting lines — mark the "new", the lines - - - - the old assignment Fig. 1

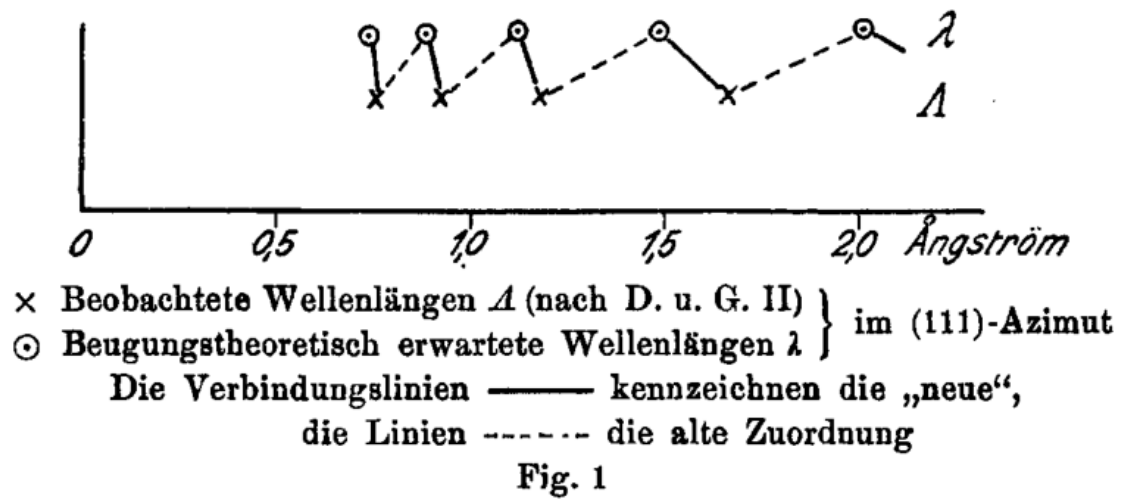


Fig. 1. × Observed wavelengths Λ (after Davisson and Germer II);
 o wavelengths λ expected from diffraction theory — in the (111) azimuth.
 The solid connecting lines denote the “new” assignment,
 the dashed lines the old assignment.

The other possible assumption is that the wavelength of the de Broglie wave, which outside the crystal is Λ , has a different value, λ , inside it, i.e. that the crystal must be assigned a refractive index

$$\mu = \frac{\Lambda}{\lambda} \quad (2)$$

for de Broglie waves. Such a refractive index was first introduced on purely phenomenological grounds by Eckart 2). The author 3) has independently of this

Footnotes:

- 1) A paper by Patterson (Nature 120. p. 46. 1927) deals theoretically with the possibility of a contraction factor.
- 2) C. Eckart, Proc. Nat. Acad. Washington 12. p. 460. 1927.
- 3) H. Bethe, Naturwissenschaften 15. p. 787. 1927 (cited as "Bethe I").

— p. 58 —

shown that a refractive index is directly demanded by wave mechanics, and that the assumption of such an index is indeed in exact agreement with the experimental results: Relation (1) between vacuum wavelength Λ and diffraction angle Θ is preserved, because it already follows from the surface structure of the crystal alone; the magnitude of the selectively reflectable wavelengths, on the other hand, which depends on the three-dimensional action of the atoms in the interior of the *crystal*, is of course influenced by a refractive index attributed to the interior of the crystal. 1) For now the wavelength λ in the crystal must naturally have the magnitude following from lattice-diffraction theory and no longer the vacuum wavelength Λ . Since the vacuum wavelengths Λ follow from the measured electron velocity, and the crystal wavelengths λ can be computed from the known lattice of nickel, μ is determined as soon as it is decided which λ corresponds to a given Λ . Here again two possibilities arise: either one may assign to each wavelength that is selectively reflected according to the observations the next *larger* in the scale of the wavelengths λ that can be selectively reflected according to lattice theory: this assignment, indicated by dots in Fig. 1, was first proposed by Davisson and Germer 2) and yielded refractive indices $\mu < 1$. Against this, however, there are very grave objections, connected in large part with the wave-mechanical theory of the refractive index to be discussed presently 3), but also the circumstance that for $\mu < 1$ no electrons could emerge grazing from the crystal into the vacuum, whereas such "grazing beams" were observed by Davisson and Germer. 4) Consequently the experiments of Davisson and Germer admit only the possibility of relating the observed wavelengths Λ to the next *smaller* of the expected wavelengths λ .

Footnotes:

- 1) Cf. H. Bethe I, loc. cit.
- 2) The "new assignment" was presented by Davisson and Germer II, Phys. Rev. 30. p. 705 as possible but improbable. According to a letter of early June 1928, D. and G. have now likewise adopted the new assignment.
- 3) H. Bethe, Naturwissenschaften 16. p. 333. 1927. Cited as "Bethe II". [Tr.: misprint for "1928".]

4) Cf. H. Bethe II, Naturwissenschaften 16.

— p. 59 —

This “new” assignment, indicated in Fig. 1 by solid lines, yields refractive indices $\mu > 1$.

Theoretically the existence and magnitude of the refractive index follow at once from the Schrödinger differential equation, which the electron must of course satisfy. For our purposes we write it in the form 1)

$$\Delta\psi + \frac{8\pi^2me}{h^2}(E + V)\psi = 0. \quad (3)$$

Here E denotes the accelerating potential, V the electrostatic *potential*, hence Ee the total energy and $-Ve$ the potential energy of the electron. If in vacuum we set $V = 0$, and in the crystal, to a first approximation, V constant equal to V_0 , we obtain as solution of the differential equation plane waves with the wavelengths

$$\begin{cases} \Lambda = \sqrt{\frac{h}{2meE}} & \text{in vacuum and} \\ \lambda = \sqrt{\frac{h}{2me(E + V_0)}} & \text{in the crystal.} \end{cases} \quad (4)$$

In other words, one obtains a refractive index

$$\mu = \frac{\Lambda}{\lambda} = \sqrt{1 + \frac{V_0}{E}}. \quad (5)$$

Of V_0 one can say from the outset that it must be *positive*. For $-eV_0$ represents the potential energy of an electron in the crystal, referred to vacuum as zero, and hence also the potential energy of the conduction electrons already present in the crystal. If now $-eV_0$ were positive, the conduction electrons would have to leave the crystal spontaneously, whereas according to the Richardson effect positive work must be expended to this end. V_0 is therefore necessarily positive, and hence $\mu > 1$, as corresponds to our “new” assignment: The electrons are accelerated upon entering the crystal, the group velocity of the de Broglie wave *increases* from $\sqrt{\frac{2e}{m}E}$ to $\sqrt{\frac{2e}{m}(E + V_0)}$, the phase velocity, and hence — because of the constant total energy — the wavelength too, decreases in the same ratio.

Footnotes:

- 1) In the two communications in the Naturwissenschaften I used the Schrödinger notation, which differs from that used here by the sign of V and the factor e in E and V .

— p. 60 —

Table 1 contains 1) the selectively reflectable wavelengths observed by Davisson and Germer 2) as well as the direction 3) in which they are reflected, then the Miller indices h_1 , h_2 , h_3 of the reflecting lattice plane (according to the

[Table 1, see p. 8]

Footnotes:

- 1) The table is already contained in part in Bethe II.
- 2) Cf. Davisson and Germer I and II, loc. cit.
- 3) Cf. p. 55, footnote 4.

— p. 61 —

Table 1. Observed electron reflections (after Davisson and Germer),
corresponding refractive indices μ and crystal potentials V_0

Observational results					Theoretical conclusions				
E (V)	Azimuth	Order	Θ	Λ (Å)	Miller (hkl)	λ (Å)	μ	V_0 old	V_0 new
54	[111]	1	50°	1.670	(331)	1.494	1.118	-18	+13
65	[100]	1	44°	1.520	(422)	1.353	1.123	-21	+16.5
106	[111]	1	28°	1.190	(442)	1.128	1.048	-39	+11.5
126	[100]	1	28°	1.093	(533)	1.040	1.047	-44.5	+11
143	[110]	1	56°	1.024	(531)	1.044	(0.979)	-43	-7
160	[100]	2	60°	0.968	(622)	0.923	1.045	-26	+14
[170]	[110]	[1]	[46°]	[0.940]	[642]	[0.871]	[1.075]	[-34]	[+26]*
174	[111]	1	22°	0.928	(553)	0.893	1.038	-56.5	+13
(179)	(110)	(1)	(44.5°)	(0.915)	(642)	(0.871)	(1.050)	(-43)	(+17)*
181	[111]	2	55°	0.910	(551)	0.877	1.036	-37	+13
[188]	[110]	[1]	[43°]	[0.894]	[642]	[0.871]	[1.024]	[-52]	[+8]*
190	[100]	1	20°	0.889	(644)	0.836	1.060	-53	+23
230	[100]	2	46°	0.807	(733)	0.788	1.023	-56	+10
248	[111]	2	44°	0.778	(662)	0.748	1.036	-54	+19
248	[110]	1	34°	0.778	(753)	0.734	1.057	-52	+28
258	[111]	1	<20°	0.763	(664)	0.739	1.032	-71	+15
292	[100]	1	<20°	0.717	(755)	0.696	1.028	-79	+15
310	[100]	3	70°	0.696	(822)	0.676	1.028	-21	+16
312	[100]	2	37°	0.693	(844)	0.676	1.023	-72	+14
343	[111]	2	34°	0.662	(773)	0.645	1.022	-76	+15
					(660)	0.676	0.971	-58	[-21]
347	[111]	3	62°	0.658	(771)	0.615	1.068	-21	[+48]
					Mean	0.646	1.019	-40	+14
370	[100]	3	57°	0.637	(933)	0.615	1.035	-44	+25

*) Two beams were observed at 170 and 188 volts. For the calculation of the mean of V_0 , 179 volts was used as the basis.

Mean +14.8

new assignment), referred to cubic axes, and the corresponding crystal wavelengths λ , computed according to the lattice-diffraction formula

$$\lambda = \frac{2a}{\sqrt{3}} \frac{h_1 + h_2 + h_3}{h_1^2 + h_2^2 + h_3^2} \quad (6)$$

in which a denotes the side of the elementary cube (for Ni 3.52 Å). The third column from the right contains the refractive indices μ computed according to (2), the last the crystal potentials V_0 following from them according to (5). The mean value of V_0 is 14.8 volts. 1) In the second column from the right, V_0 according to the “old assignment” is also given for comparison.

To be sure, the “new assignment” at first gives rise to a small difficulty 2): The longest selectively reflectable wavelengths λ to be expected from lattice theory would have no partners among the observed electron wavelengths Λ (cf. Fig. 1). The electron waves in question, however, already have *in the crystal* a wavelength λ that approaches the lattice constant d of the “surface lattice”. Because of the generally valid relation

$$n\lambda = d \sin \vartheta \quad (1a)$$

the angle ϑ between the reflected beam and the normal to the crystal surface already becomes very large *in the crystal*, i.e. the beam runs nearly parallel to the surface. Because of the refractive index $\mu > 1$ the beam can then no longer emerge from the crystal into the vacuum, but total reflection at the crystal surface sets in. 3) This also manifests itself in that the vacuum wavelengths Λ belonging to the longest expected wavelengths λ in the crystal, according to the equation

$$\Lambda = \lambda\mu$$

become larger than the lattice constant d , so that the surface-grating relation (1) can no longer be satisfied with any angle Θ . Table 2 lists for

Footnotes:

- 1) On the relation of this quantity to the Richardson effect cf. H. Bethe II, loc. cit., Rosenfeld and Witmer, Ztschr. f. Phys. 49. p. 736. 1928; this work § 10, p. 113ff.
- 2) This difficulty moved Davisson and Germer to give up the new assignment. (Davisson and Germer II, loc. cit. p. 732). Cf. Bethe II, loc. cit. p. 334.
- 3) Cf. the paper Bethe II, and especially p. 80 of this work.

Table 2. Electron reflections not observed

Order n	Azimuth	Reflection indices	λ in crystal	E+V0 (eV)	$\sin \theta$	μ	$\sin \Theta$	Θ^*
1	[111]	220	2.03	36	0.945	1.31	1.24	—
1	[100]	311	1.85	44	0.86	1.23	1.04	ca. 90°
1	[110]	420	1.22	100	0.985	1.08	1.06	—
1	[331]	620	0.81	226	0.992	1.032	1.03	—
1	[210]	640	0.78	244	0.96	1.03	0.99	ca. 80°
2	[111]	440	1.02	144	0.95	1.052	1.00	ca. 90°
2	[100]	511	1.06	134	0.985	1.06	1.05	—
3	[111]	$\overline{551}$	0.72	289	1.00	1.022	1.025	—
3	[100]	711	0.72	289	1.00	1.025	1.025	—

*) — means the beam is not observable due to total reflection.

each azimuth the longest selectively reflectable waves to be expected from lattice theory: In the (111) and (110) azimuths the longest waves of first order must suffer total reflection, and thus cannot be observed at all, whereas the corresponding electron beam in the (100) azimuth must emerge grazing into the vacuum. Precisely here, however, Davisson and Germer observed an especially strong

The agreement with theory is not quite so good in the higher orders.

Footnotes:

- 1) “Grazing beams” (grazingly emerging electron beams) were observed by Davisson and Germer in each of the three principal azimuths. Their occurrence is explained by Davisson and Germer in general by the fact that the path of the electrons from one atomic layer to the next is very large for grazing beam propagation, and that therefore a large part of the electrons scattered by the deeper layers of the crystal is absorbed before reaching the surface. Consequently the main part of the intensity of the observed beam is supplied by the electrons diffracted at the surface layer, and for the production of a grazingly emerging beam it is therefore not absolutely necessary that the waves scattered by all layers of the crystal be in phase with one another, i.e. that the Laue equations of space-lattice diffraction be satisfied. In this case, however, the intensity of the observed grazingly emerging beam will be especially large, and this is precisely the case in the (100) azimuth.

— p. 63 —

Thus the longest second-order beam reflected by the (440) plane in the (111) azimuth should emerge grazing, but is not observed. This is probably due to the fact that in the second order the intensity falls off much more rapidly with decreasing Θ than in the first, and already at $\Theta = 70^\circ$, where reliable observations first become possible, the scattered waves of the uppermost atomic layer have exactly the opposite phase to those of the second. The absence of the beams to be expected at 232 and 248 volts in the (210) and (331) azimuths is also understandable because of their presumably small intensity. Another question is the irregularity of the third-order reflections in the (111) azimuth, which should agree with those in the (100) azimuth. The explanation for this follows only from the dynamical theory (§ 9). On the whole one may well say that the phenomenon of the Laue spots missing because of *total reflection* and the *intensity* of the “grazing beams” form a further support of the “new assignment”, but in no case speak for the old.

§ 2. The Field of the de Broglie Waves Inside the Crystal

The elementary theory developed in the preceding section is based on Laue's purely wave-kinematical theory of the refractive index.

Like Laue's theory, the theory developed here can represent only a first *approximation*. This is already evident from the fact that we have summarily assumed the potential of the electrons in the crystal to be *constant*, whereas consistently we must represent it in a spatial *Fourier series*.¹⁾ If this circumstance is taken into account, one is led to a *dynamical* theory corresponding to Ewald's dynamical theory of X-ray reflections²⁾, and largely following its method.

The course of the solution is roughly as follows: One represents the wave function as a superposition of plane waves, mathematically speaking: One expands ψ in a similar

Footnotes:

- 1) At the center of the atoms V even becomes infinite as $1/r$, and consequently the wavelength becomes infinitely small as $r^{1/2}$. The ansatz we use here contains this fact implicitly.
- 2) P. P. Ewald, Ann. d. Phys. 54. p. 514. 1917 (cited as Ewald III). [Tr.: misprint for "p. 519".]

— p. 64 —

Fourier series to that of the potential, but with the wave number of the incident beam in the crystal, k_0 , still undetermined. Then between the amplitudes of the individual plane waves (the coefficients of the Fourier expansion of ψ) just as many equations must be satisfied as there are plane waves. Into these equations enter, besides the Fourier coefficients of the potential, which we must assume known, the known vacuum wave number K and the unknown wave number in the crystal (k_0) of the incident wave. In addition, doubly infinitely many boundary conditions must be satisfied. The solution of the indicated system of equations proceeds in such a way that the two amplitudes of the incident and the reflected wave are assumed *large* compared with the amplitudes of the remaining plane waves of the ψ -field, and all remaining amplitudes are expressed through these two. One then obtains two homogeneous equations between the two amplitudes assumed "large". The vanishing of their determinant yields, as usual, the dispersion equation, from which two different values result for the as yet unknown wave number k_0 . By a combination of two solutions with the two possible wave numbers k_0 , the boundary conditions can be satisfied, as in Ewald's theory. From the solution thus obtained one then determines the more accurate magnitude of the selectively reflectable wavelengths, the intensity of the reflected beam, and the resolving and reflecting power of the crystal.

As the origin of our coordinate system we choose the nucleus of any atom of the crystal surface. As coordinate axes a_1 , a_2 and a_3 we choose, say, the usual crystallographic axes, which moreover is not absolutely necessary. If we are dealing with a crystal of a system with orthogonal axes, we may of course assume a Cartesian coordinate system X , Y , Z . We shall also do this occasionally later, but where possible we choose the vectorial notation, which can be applied to any obliquely-angled axis system.

Footnotes:

- 1) The wave number (per unit length) is defined as 2π divided by wavelength.

— p. 65 —

With the chosen axes the Fourier expansion of the potential reads simply

$$\frac{8\pi^2 me}{h^2} V(\mathbf{r}) = U(\mathbf{r}) = \sum_g v_g e^{2\pi i(\mathbf{g}\mathbf{r})}. \quad (7)$$

Here, in the known manner, $\mathbf{r}$ is the position vector and $\mathbf{g}$ a radius vector in Ewald's reciprocal lattice 1)

$$\mathbf{g} = g_1 \mathbf{b}_1 + g_2 \mathbf{b}_2 + g_3 \mathbf{b}_3, \quad (8)$$

where g_1, g_2, g_3 are integers. The reciprocal axes are known to be defined by

$$\mathbf{b}_1 = \frac{[\mathbf{a}_2 \mathbf{a}_3]}{(\mathbf{a}_1 [\mathbf{a}_2 \mathbf{a}_3])} \quad (9)$$

and cyclic permutations, and satisfy the relations

$$(\mathbf{a}_i \mathbf{b}_k) = \delta_{ik} = \begin{cases} 0, & i \neq k \\ 1, & i = k \end{cases} \quad (9a)$$

$\mathbf{b}_3$ is, according to our assumptions, perpendicular to the crystal surface.

The index g in the Fourier expansion (7) at v_g of course represents in each case an index triple g_1, g_2, g_3 , and (7) is really to be written as a triple sum over g_1, g_2, g_3 . That the expansion (7) really possesses the required periodicity, i.e. returns to the same value upon a displacement of the field point $\mathbf{r}$ by a multiple of the identity distances a_1, a_2, a_3 , say by

$$\mathbf{r}' = r_1 \mathbf{a}_1 + r_2 \mathbf{a}_2 + r_3 \mathbf{a}_3$$

follows at once from (8) and (9a).

If the origin of coordinates is a center of symmetry of the crystal, then

$$V(\mathbf{r}) = V(-\mathbf{r})$$

and consequently

$$v_g = v_{-g}. \quad (7a)$$

This reduces the expansion (7) to a sum of cosines

$$U(\mathbf{r}) = \sum_g 2 v_g \cos 2\pi(\mathbf{g}\mathbf{r}) \quad (7b)$$

Footnotes:

- 1) Ewald III, loc. cit., p. 536, and especially Ewald, Ztschr. f. Kristallographie 56. p. 129. 1921 (cited as “Ewald IV”).

— p. 66 —

with real coefficients v_g . For crystals with orthogonal axes (rhombic, tetragonal, cubic, and, with suitable choice of axes, also hexagonal system) and identity distances a_1 , a_2 and a_3 , the reciprocal axes become

$$\mathbf{b}_i = \frac{\mathbf{a}_i}{a_i^2}, \quad b_i = \frac{1}{a_i} \quad (9b)$$

and the potential becomes simply:

$$U(x, y, z) = \sum_g v_g e^{2\pi i \left(\frac{g_1 x}{a_1} + \frac{g_2 y}{a_2} + \frac{g_3 z}{a_3} \right)} \quad (7c)$$

We now wish to consider the field of the de Broglie waves in the crystal. To do so we first assume the crystal to be infinitely extended, i.e. we regard the ansatz (7) for the potential as valid in all space. Later we shall then set up the conditions that are added by the existence of a surface of the crystal.

The most straightforward approach would be to solve the Schrödinger differential equation

$$\Delta\psi + \frac{8\pi^2 me}{h^2} (E + V)\psi = 0 \quad (3)$$

by the ansatz of a plane wave

$$\psi_0 = c_0 e^{i(\mathbf{f}_0 \mathbf{r})} \quad (10)$$

Here, following Ewald 1), we call $\mathbf{f}_0$ the propagation vector of the plane wave (10). It gives the direction of propagation of the plane wave; its absolute value k_0 is called the wave number and stands in the simple relation to the wavelength λ_0 of the plane wave

$$k_0 = \frac{2\pi}{\lambda_0} \quad (10a)$$

Substituting the ansatz (10) into the Schrödinger equation (3) and taking into account the Fourier expansion (7) of the potential, we w

$$\left(\frac{8\pi^2 me}{h^2} E - k_0^2 \right) c_0 e^{i(\mathbf{f}_0 \mathbf{r})} + \sum_g c_0 v_g e^{i(\mathbf{f}_0 \mathbf{r}) + 2\pi i(\mathbf{g} \mathbf{r})} = 0.$$

This equation cannot be satisfied simultaneously at all field points $\mathbf{r}$. Instead, it turns out

Footnotes:

1) Ewald III, loc. cit., p. 525.

— p. 67 —

that each plane wave $c_0 e^{i(\mathbf{f}_0 \cdot \mathbf{r})}$ in the periodic potential field $V(\mathbf{r})$ of the crystal is conceivable at all only in conjunction with other plane waves

$$c_h e^{i(\mathbf{f}_0 + 2\pi \mathbf{h} \cdot \mathbf{r})}$$

This is simply due to the fact that V occurs multiplied by ψ .

To take account of this circumstance, we set

$$\psi = \sum_g \psi_g e^{i(\mathbf{f}_0 + 2\pi \mathbf{g} \cdot \mathbf{r})} \quad (11)$$

We thereby regard ψ as the superposition of a triply infinite family of plane waves with the propagation vectors

$$\mathbf{f}_g = \mathbf{f}_0 + 2\pi \mathbf{g} \quad (11a)$$

and the amplitudes ψ_g , the index g of course again standing for the index triple g_1, g_2, g_3 . The waves $\mathbf{f}_g$ are identical with all those waves that can arise from a “primary wave” $\psi_0 e^{i(\mathbf{f}_0 \cdot \mathbf{r})}$ according to the elementary interference theory. The propagation vector of the “primary beam” inside the crystal, $\mathbf{f}_0$, remains undetermined until the dispersion theory is carried through in § 5; but once $\mathbf{f}_0$ is given, one need only, following Ewald, seek the point of propagation, i.e. the point $-\mathbf{f}_0/2\pi$ in the reciprocal lattice, and draw from there the radius vectors to all lattice points $\mathbf{g}$ of the reciprocal lattice, and one has at one stroke all propagation vectors $\mathbf{f}_g/2\pi = \mathbf{f}_0/2\pi + \mathbf{g}$ of all ψ -waves in the crystal. In each case the direction of the propagation vector $\mathbf{f}_g$ gives the direction of propagation, its absolute value k_g the wave number (equal to 2π divided by wavelength) of each plane wave.

We can now substitute the ansätze (7) and (10) into the Schrödinger differential equation (3), in which we also set for abbreviation:

$$K = \sqrt{\frac{8\pi^2 m e}{h^2} E} = \frac{2\pi}{h} \sqrt{2m e E} \quad (12)$$

(K denotes the wave number for $V = 0$.)

Then (3) goes over into

$$\Delta\psi + (K^2 + U)\psi = 0. \quad (3a)$$

If we now enter here with (7) and (10), we get

$$\sum_h \psi_h (K^2 - k_h^2) e^{i(\mathbf{f}_h \cdot \mathbf{r})} + \sum_p \psi_p e^{i(\mathbf{f}_0 + 2\pi \mathbf{p} \cdot \mathbf{r})} \cdot \sum_g v_g e^{2\pi i(\mathbf{g} \cdot \mathbf{r})} = 0.$$

Since this equation holds for every field point $\mathbf{r}$, the coefficient of each exponential must vanish separately. This yields, with a corresponding change of the summation indices:

$$\psi_h(K^2 - k_h^2) + \sum_g v_g \psi_{h-g} = 0$$

or, if we extract from the sum the term $v_0 \psi_h$:

$$\psi_h(K^2 + v_0 - k_h^2) + \sum_g' v_g \psi_{h-g} = 0, \quad (13)$$

where the prime denotes that v_{000} is to be excluded.

- (13) holds for every index triple h_1, h_2, h_3 and thus represents a triply infinite scheme of equations valid inside the crystal, in each equation the amplitude of a ψ -oscillation ψ_h occurring multiplied by a resonance factor, all other ψ -amplitudes multiplied each by a Fourier coefficient of the potential. The system of equations represents the *exact* solution of the Schrödinger equation inside crystals, only inelastic collisions being disregarded. This distinguishes our method fundamentally from Born's 1), which seeks to grasp the processes in the collision of an electron with an atom by successive approximation. That Born's method is not adapted to the treatment of crystal diffraction is shown by the fact that Zwicky 2) could explain from the first approximation according to Born's method only the occurrence of crystal diffraction at all, but *not* the refractive index for the de Broglie wave. On the other hand, the method developed here is *not* applicable to diffraction by a *single atom* 3), but is essentially conditioned by the regular *arrangement* of the atoms in the crystal, as a consequence of which *diffraction by a single atom* whose structure is so far not sufficiently known, *recedes* behind diffraction by the *arrangement* of the atoms.

Our system of equations (13) is the necessary condition for the existence of “dynamical equilibrium” in the crystal, i.e. for the wave field ψ being continually regenerated by diffraction at the crystal atoms. The only remaining freedom is

Footnotes:

- 1) M. Born, Ztschr. f. Phys. 38. p. 803. 1926.
- 2) F. Zwicky, Proc. Nat. Acad. Amer. 12. p. 518. 1927. [Tr.: sic; correct is Sci. 13, pp. 518-525.]
- 3) Closest to our method comes that developed by Faxén and Holtsmark for diffraction by a single atom, which likewise refers to the actual ψ -field of the diffracted electron in the diffracting atom. Ztschr. f. Phys. 45. p. 307. 1927.

the choice of the propagation vector $\mathbf{f}_0$ and the amplitude ψ_0 of the primary wave; but only so long as the crystal is really infinitely extended and nowhere comes into contact with another medium. The surface phenomena treated in the following section determine the constants of the “primary beam”.

§ 3. Formulation of the Boundary Conditions

We now no longer assume the crystal to be infinitely extended, but bounded by the surface

$$z = 0 \quad (14)$$

and by z we understand a coordinate in the direction of the crystal normal. The crystal is in the half-space $z > 0$, the half-space $z < 0$ is vacuum. Then

$$\begin{cases} V = 0 & \text{for } z < 0 \\ \frac{8\pi^2 me}{h^2} V = U = \sum_g v_g e^{2\pi i(\mathbf{g}\mathbf{r})} & \text{for } z > 0 \end{cases} \quad (15)$$

For each of the two regions we can give the solution of the Schrödinger equation

$$\Delta\psi + (K^2 + U)\psi = 0 \quad (3a)$$

In the crystal the solution of (3a) is, as we have seen, a superposition of triply infinitely many plane waves

$$\psi = \sum_g \psi_g (e^{i(\mathbf{f}_g \mathbf{r})}) \quad (z > 0) \quad (11)$$

whose amplitudes ψ_g are coupled to one another through (13). In the vacuum we obtain as solution plane waves $c e^{i(\mathbf{K}\mathbf{r})}$ with the wave number $|\mathbf{K}| = K$, which can be superposed arbitrarily;

$$\begin{cases} \Psi = \sum_g \Psi_m e^{i(\mathbf{K}_m \mathbf{r})} \\ |\mathbf{K}_m| = K, \quad \Psi_m \text{ arbitrary} \end{cases} \quad (z < 0) \quad (16)$$

[Note: the summation index in (16) is printed as g in the original.]

The question is in what relation the wave fields in the crystal and in the vacuum stand to one another. — p. 70 —

According to Faxén and Holtsmark¹), on both sides of a potential jump both the wave function ψ and its normal derivative $\partial\psi/\partial n$ have the same value:

$$\psi = \Psi, \quad \frac{\partial\psi}{\partial n} = \frac{\partial\Psi}{\partial n}. \quad (17)$$

If, then, $\mathbf{r}'$ denotes a vector in the crystal surface, the following must hold:

$$\begin{cases} \sum_m \Psi_m e^{i(\mathbf{K}_m \mathbf{r}')} = \sum_g \psi_g e^{i(\mathbf{f}_g \mathbf{r}')} \\ \sum_m \Psi_m (\mathbf{K}_m \mathbf{n}) e^{i(\mathbf{K}_m \mathbf{r}')} = \sum_g \psi_g (\mathbf{f}_g \mathbf{n}) e^{i(\mathbf{f}_g \mathbf{r}')} \end{cases} \quad (18)$$

$\mathbf{n}$ is a unit vector in the direction of the crystal normal.

Equations (19)* hold for every point $\mathbf{r}'$ of the crystal surface. Hence the coefficients of the individual periodic functions of $\mathbf{r}'$ must vanish separately. Now two exponentials, e.g. $e^{i(\mathfrak{K}_m \mathbf{r}')}$ and $e^{i(\mathfrak{f}_g \mathbf{r}')}$, evidently have the same periodicity in the surface of the crystal only if the projections of the propagation vectors $\mathfrak{K}_m$ and $\mathfrak{f}_g$ onto the crystal surface coincide. We shall therefore resolve the propagation vectors into a component normal to the surface and a tangential component:

$$\begin{cases} \mathfrak{K}_m = A_m \mathfrak{T}_m + \Gamma_m \mathbf{n} \\ \mathfrak{f}_g = \alpha_g \mathfrak{t}_g + \gamma_g \mathbf{n} \end{cases} \quad (19)$$

Here $\mathfrak{T}_m$ and $\mathfrak{t}_g$ shall be unit vectors in the direction of the projections of the propagation vectors $\mathfrak{K}_m$ and $\mathfrak{f}_g$ onto the crystal surface. The surface conditions (18) must then be satisfied separately for all those waves whose propagation vectors have the same projection onto the surface. Among the vacuum waves there are now, since $|\mathfrak{K}_m| = K$ is fixed, only two at all with given tangential component $A_m \mathfrak{T}_m$. Their propagation vectors $\mathfrak{K}_m$ and $\mathfrak{K}'_m$ differ in the sign of their normal component

$$\Gamma_m = -\Gamma'_m. \quad (20)$$

Footnotes: 1) Faxén and Holtsmark, Ztschr. f. Phys. 45, p. 307, 1927. The same boundary conditions can be obtained by Ewald's method (Ann. d. Phys. 49, pp. 1–38 and pp. 117–143, 1916). We cite these papers as “Ewald I and II”.

— p. 71 —

For finite amplitudes, however,

$$(\Psi_m + \Psi'_m) e^{i A_m (\mathfrak{T}_m \mathbf{r}')} = 0$$

$$(\Psi_m \Gamma_m - \Psi'_m \Gamma'_m) e^{i A_m (\mathfrak{T}_m \mathbf{r}')} = 0$$

can never be satisfied simultaneously. Consequently, simultaneously with the incident ($\mathfrak{K}_m$) and reflected ($\mathfrak{K}'_m$) vacuum waves, refracted waves $\mathfrak{f}_g$ must *always* also occur in the crystal, which coincide with the vacuum waves with respect to the projection of their propagation vectors onto the crystal surface, i.e. for which

$$A_m \mathfrak{T}_m = \alpha_g \mathfrak{t}_g \quad (21)$$

holds.

The number of these crystal waves is not, like that of the vacuum waves, limited to two¹⁾, because the wave number $\mathfrak{f}_g$ of the crystal waves is arbitrary. We thus obtain

$$\begin{cases} \Psi_m + \Psi'_m = \sum_{A_m \mathfrak{T}_m = \alpha_g \mathfrak{t}_g} \psi_g \\ \Psi_m - \Psi'_m = \sum_{A_m \mathfrak{T}_m = \alpha_g \mathfrak{t}_g} \frac{\gamma_g}{\Gamma_m} \psi_g \end{cases} \quad (22)$$

* Tr.: The source prints “(19)”; Eq. (19) is introduced only below, so the reference is evidently to Eq. (18).

On the right, the sum is to be taken over all crystal waves for which (21) is satisfied. For the vacuum waves we obtain

$$\begin{cases} \Psi_m = \sum_{A_m \mathfrak{T}_m = \alpha_g \mathfrak{t}_g} \frac{\Gamma_m + \gamma_g}{2\Gamma_m} \psi_g \\ \Psi'_m = \sum_{A_m \mathfrak{T}_m = \alpha_g \mathfrak{t}_g} \frac{\Gamma_m - \gamma_g}{2\Gamma_m} \psi_g \end{cases} \quad (23)$$

These, together with (21), are the desired boundary conditions, i.e. the relations between the directions and amplitudes of the vacuum and crystal waves.

From the agreement of the vacuum and crystal waves with respect to the tangential components of the propagation vectors, the law of refraction follows for each individual pair of crystal wave and vacuum wave. For if we denote the angle between the crystal normal and the direction of a crystal wave by ϑ_h and the corresponding angle for the vacuum wave by Θ_h , then according to the foregoing, evidently

$$A_h = K \sin \Theta_h = k_h \sin \vartheta_h = \alpha_h$$

Footnotes: 1) If the surface of the crystal is a lattice plane, infinitely many crystal waves with given tangential component occur.

— p. 72 —

and from this follows the law of refraction

$$\frac{\sin \Theta_h}{\sin \vartheta_h} = \frac{k_h}{K} = \mu_h \quad (24)$$

of course with a refractive index varying from case to case.

If, furthermore,

$$\Psi_0 e^{i(\mathfrak{K}_0 \mathfrak{r})} = \Psi_0 e^{i(A_0 \mathfrak{T}_0 + \Gamma_0 \mathfrak{n}, \mathfrak{r})} \quad (25)$$

denotes a primary beam incident from the vacuum, then the primary beam in the crystal must be representable in the form

$$\psi_0 e^{i(\mathfrak{f}_0 \mathfrak{r})} = \psi_0 e^{i(A_0 \mathfrak{T}_0 + \gamma_0 \mathfrak{n}, \mathfrak{r})} \quad (25a)$$

By lattice diffraction, only the crystal waves

$$\psi_g e^{i(\mathfrak{f}_g \mathfrak{r})} = \psi_g e^{i(A_0 \mathfrak{T}_0 + \gamma_0 \mathfrak{n} + 2\pi \mathfrak{g}, \mathfrak{r})} \quad (26)$$

can now arise from this primary beam, and consequently only such vacuum waves

$$\Psi_m e^{i(\mathfrak{K}_m \mathfrak{r})} = \Psi_m e^{i(A_m \mathfrak{T}_m + \Gamma_m \mathfrak{n}, \mathfrak{r})} \quad (26a)$$

whose propagation vectors are given by

$$\begin{cases} A_m \mathfrak{T}_m = A_0 \mathfrak{T}_0 + 2\pi \mathbf{g}_{\text{tang}} \\ \Gamma_m^2 = K^2 - A_m^2 \end{cases} \quad (27)$$

Every vector $\mathbf{g}$ of the reciprocal lattice is, as is well known, the normal to a family of lattice planes

$$(\mathbf{g}\mathbf{r}) = Gn$$

where G is the greatest common divisor of the three indices and n an integer running number. The surface of the crystal $(\mathbf{n}\mathbf{r}) = 0$ cuts the first lattice plane of the family ($n = 1$) in the line of intersection

$$(\mathbf{g}_{\text{tang}}\mathbf{r}) = (\mathbf{g} - (\mathbf{g}\mathbf{n})\mathbf{n}, \mathbf{r}) = G. \quad (28)$$

The position and spacing of the grating lines formed by the atoms of the surface thus determine all vectors $\mathbf{g}_{\text{tang}}$, and thereby all diffracted vacuum waves (26) arising from the primary beam (24). This again yields the theorem already stated in § 1: For fixed given direction of incidence and vacuum wavelength, the direction of any beams that may occur, reflected back into the vacuum, is given solely by the structure of the crystal surface and is independent of all processes inside the crystal, in particular of a refractive index, even though it be direction-dependent; only the de-

— p. 73 —

termination of the wavelength at which selective reflection can occur at all (for given direction of incidence), and of the intensity of the reflected beam, is influenced by the processes in the crystal (cf. p. 58).

§ 4. Reduction of the dynamical equations to two

The system of equations

$$\psi_h(K^2 + v_0 - k_h^2) = - \sum \psi_g v_{h-g} \quad (13)$$

of the dynamical equations cannot, of course, be solved *exactly*. But we may assume that the sum on the right-hand side assumes particularly *large* values for at least no triple of values h_1, h_2, h_3 , but in general particularly *small* values for none either.¹⁾ The magnitude of the amplitude ψ_h will thus be determined essentially by the magnitude of the *resonance factor*

$$K^2 + v_{000} - k_{h_1 h_2 h_3}^2$$

This factor can, in general, i.e. for arbitrary K , be small for only *a single* ψ -oscillation. But since the incident (primary) wave has a large amplitude under all circumstances, the resonance factor

$$K^2 + v_{000} - k_{000}^2$$

must be small under all circumstances. Then, however, there exist no further (reflected) waves with small resonance factor, i.e. large amplitude, and outwardly only the continuous scattering

originating from the thermal motion of the crystal atoms will be observable. But for quite definite wave numbers K of the incident wave, *two resonance factors become small simultaneously.*²⁾ There then occurs the “selective reflection” in a definite direction, treated wave-kinematically in § 1. The condition set up here of the simultaneous vanishing of two resonance factors is identical with the condition set up in § 1 for selective reflection, namely that the Laue equations modified by a refractive index are satisfied.

Footnotes: 1) On the becoming small and vanishing of the sum cf. § 8, extinction laws. 2) Possibly more, but we exclude this case from consideration because of the mathematical difficulties connected with it. On the case of symmetric reflection cf. § 9.

— p. 74 —

In order first to study the geometrical relations for two strong beams, we again use the illustration of the propagation vectors in the Ewald reciprocal lattice. As already remarked in § 2, the propagation vectors

$$\mathbf{f}_h = \mathbf{f}_0 + 2\pi\mathbf{h} \quad (11)$$

can be represented as the radius vectors that can be drawn from the propagation point A with coordinates $-\alpha/2\pi, -\beta/2\pi, -\gamma/2\pi$ to all lattice points

$$\mathbf{h} = h_1\mathbf{b}_1 + h_2\mathbf{b}_2 + h_3\mathbf{b}_3$$

of the reciprocal lattice. Here the propagation point A is in general *no* lattice point. The strict Laue equations with the modification by a refractive index would now require that the two resonance factors of the primary and secondary beams be exactly zero:

$$k_0^2 = k_h^2 = K^2 + v_0 = \kappa^2. \quad (29)$$

Geometrically this means that the lattice points $(0, 0, 0)$ and (h_1, h_2, h_3) of the reciprocal lattice must both lie on the propagation sphere, i.e. on the sphere of radius $\kappa/2\pi = \frac{1}{2\pi}\sqrt{K^2 + v_0}$ about the propagation point A . But we shall still obtain two strong waves even when the two points of the reciprocal lattice lie in the neighbourhood of the propagation sphere, and the intensity ratio of the two beams will depend precisely on the distance of the two points of the reciprocal lattice from the propagation sphere. These distances

$$\begin{cases} \varepsilon_1 = \kappa - k_0 \\ \varepsilon_2 = \kappa - k_h \end{cases} \quad (30)$$

we call *resonance errors*. They stand in a geometrical relation to one another, because according to the preceding paragraph the propagation vector of the wave incident from the vacuum

$$\mathbf{K} = A\mathbf{T} + \Gamma\mathbf{n} \quad (31)$$

which we have to presuppose as known, agrees with the propagation vector of the primary wave in the crystal

$$\mathbf{f}_0 = \alpha \mathbf{t} + \gamma \mathbf{n} \quad (31a)$$

— p. 75 —

with respect to the two components parallel to the surface:

$$A\mathfrak{T} = \alpha \mathbf{t}$$

The propagation point is thus already fixed to a quite definite normal to the crystal surface (at the reciprocal lattice) drawn through the point $-\alpha \mathbf{t}/2\pi$. On this parallel there exists a point L , which we designate with Ewald as the Laue point, for which the *origin* of the reciprocal lattice comes to lie *exactly* on the

[Figure 2: Ewald construction in the reciprocal lattice — an arc of the propagation sphere through the origin O and the “reflection point” H ; the Laue point L below on the dashed vertical $\mathbf{b}_3$ (crystal normal) line; reciprocal lattice points shown as dots; vectors $\mathfrak{C}/2\pi$ from L to O and $(\kappa - \zeta)/2\pi$ drawn from the Laue point.]

L Laue point; O origin of the reciprocal lattice; H “reflection point” (h_1, h_2, h_3) . $\mathbf{b}_3$ direction of the crystal normal (geometric locus for Laue and propagation point). $\kappa/2\pi$, $(\kappa - \zeta)/2\pi$: propagation vectors drawn from the Laue point.

Fig. 2

propagation sphere. In this case the lattice point $\mathbf{h}$ will possess a quite definite distance from the propagation sphere, which we designate as the *excitation error*. If we denote the radius vector from the Laue point to the origin of the reciprocal lattice by $\mathfrak{C}/2\pi$, then:

$$\mathfrak{C} = A\mathfrak{T} + \gamma_0 \mathbf{n}; \quad \gamma_0 = \sqrt{\kappa^2 - A^2}, \quad |\mathfrak{C}| = \kappa \quad (32)$$

$$|\mathfrak{C} + 2\pi \mathbf{h}| = \kappa - \zeta. \quad (32a)$$

ζ represents the resonance error ε_2 for the case that ε_1 is zero. ζ stands in a simple relation to κ , if we, in agree-

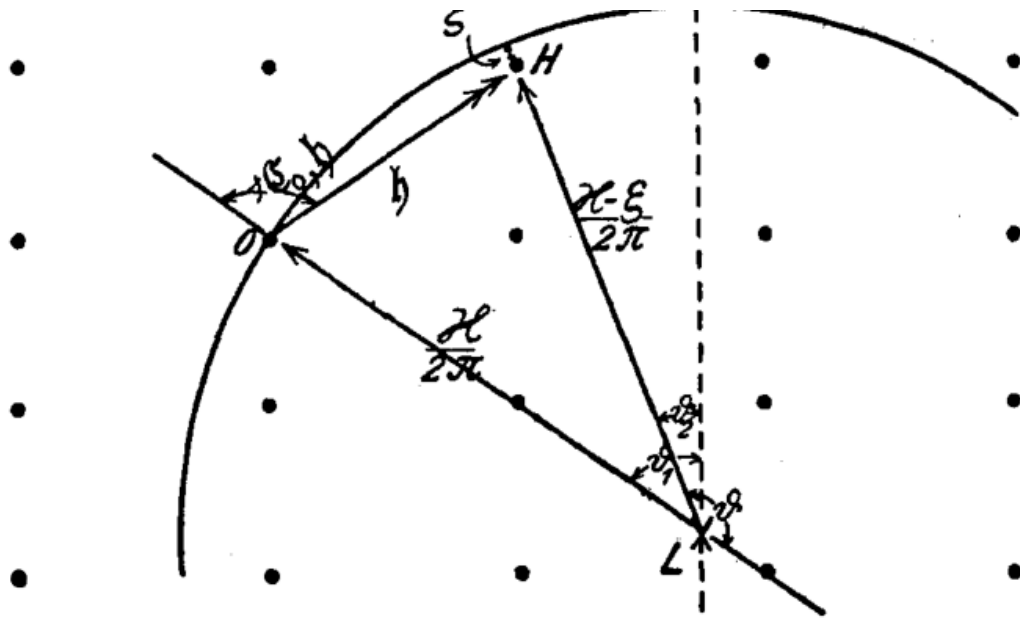
— p. 76 —

ment with the experiments of Davisson and Germer, assume that the *direction* of the incident beam in the crystal remains *constant* once and for all. For then the excitation error must *vanish* for a quite definite wave number κ_0 , i.e. there must be a wave number κ_0 which satisfies the Laue equations *exactly* and for which consequently both lattice points $(0, 0, 0)$ and (h_1, h_2, h_3) lie simultaneously on the propagation sphere. But now from (32) and (32a), neglecting ζ^2 , there follows

$$|\mathfrak{C} + 2\pi \mathbf{h}|^2 = \kappa^2 + 4\pi(\mathfrak{C}\mathbf{h}) + 4\pi^2 h^2 = \kappa^2 - 2\zeta\kappa$$

and for $\kappa = \kappa_0$

$$|\mathfrak{C}_0 + 2\pi \mathbf{h}|^2 = \kappa_0^2 + 4\pi(\mathfrak{C}_0 \mathbf{h}) + 4\pi^2 h^2 = \kappa_0^2$$



L Lauepunkt; O Nullpunkt des reziproken Gitters; H „Reflexionspunkt“ (h_1, h_2, h_3). b_3 Richtung der Kristallnormale (geom. Ort für Laue- und Ausbreitungspunkt). $\frac{\kappa}{2\pi}, \frac{\kappa - \zeta}{2\pi}$: vom Lauepunkt aus gezogene Ausbreitungsvektoren

Fig. 2. L , Laue point; O , origin of the reciprocal lattice; H , “reflection point” (h_1, h_2, h_3). b_3 , direction of the crystal normal (geometric locus of Laue and propagation points). $\kappa/2\pi, (\kappa - \zeta)/2\pi$: propagation vectors drawn from the Laue point.

and from this

$$\zeta\kappa = 2\pi(\mathfrak{C}_0 - \mathfrak{C}, \mathfrak{h}) = 2\pi(\kappa_0 - \kappa)h \cos(\mathfrak{C}_0, \mathfrak{h}).$$

From Fig. 2 one readily reads off

$$2\pi h = -2\kappa \cos(\mathfrak{C}_0, \mathfrak{h})$$

$$\sphericalangle(\mathfrak{C}_0, \mathfrak{h}) = \frac{\pi}{2} + \frac{\vartheta_2 - \vartheta_1}{2} = \pi - \frac{\vartheta}{2}$$

$$2\pi h \cos(\mathfrak{C}_0, \mathfrak{h}) = -2\kappa \cos^2(\mathfrak{C}_0, \mathfrak{h}) = -\kappa(1 + \cos \vartheta)$$

so that we finally obtain for the excitation error

$$\zeta = (\kappa - \kappa_0)(1 + \cos \vartheta) \quad (33)$$

where ϑ is the angle between the primary and secondary beams in the crystal.

By means of the excitation error ζ we can now, however, easily reduce the two resonance errors ε_1 and ε_2 to the distance $d/2\pi$ between the propagation point A and the Laue point L . The propagation point shall thus possess in the reciprocal lattice the coordinates

$$-\frac{\mathfrak{f}_0}{2\pi} = -\frac{\alpha \mathfrak{t}}{2\pi} - \frac{(\gamma_0 - d)\mathbf{n}}{2\pi}$$

Then one reads off directly for small “adaptations” d from Fig. 3 the relations:

$$\begin{cases} \varepsilon_1 = d \cos \vartheta_1 \\ \varepsilon_2 = d \cos \vartheta_2 + \zeta \end{cases} \quad (34)$$

and in what follows it will be a matter of computing the “adaptation” d for given excitation error.

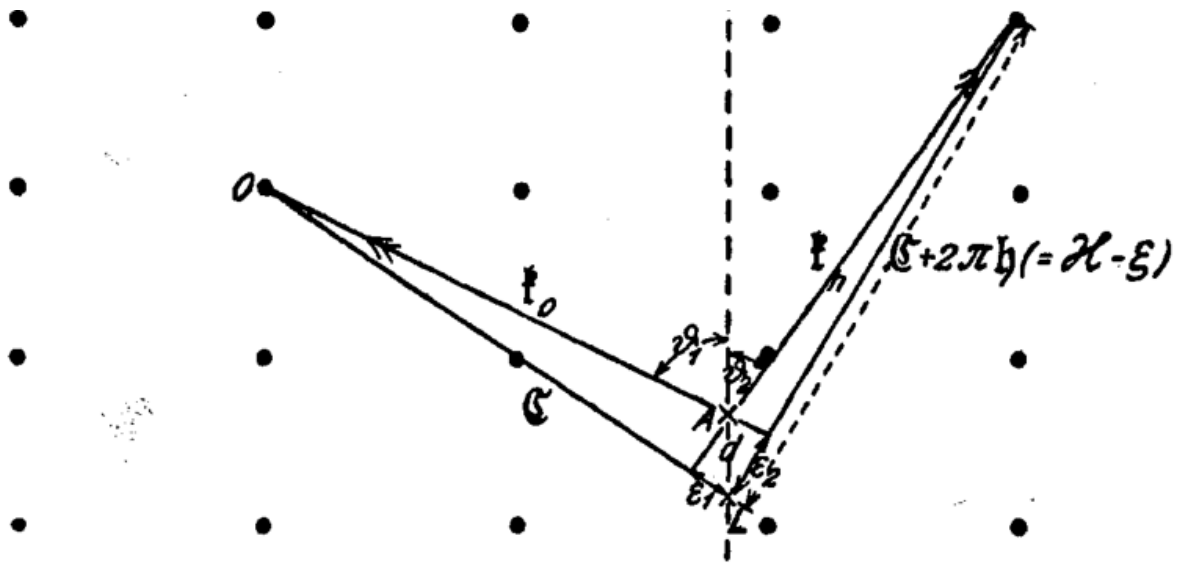
— p. 77 —

We now proceed to the treatment of the system of equations (13). For this we assume, as already indicated in § 2, p. 63, the amplitudes of the two waves with the propagation vectors

$$\begin{cases} \mathfrak{f}_0 = \alpha \mathfrak{t} + (\gamma_0 - d)\mathbf{n}, & k_0 = \kappa - \varepsilon_1 \\ \mathfrak{f}_h = \mathfrak{f}_0 + 2\pi \mathfrak{h}, & k_h = \kappa - \varepsilon_2 \end{cases} \quad (34a)$$

to be large compared with all others, and for emphasis call them c_1 and c_2 . We then compute all the remaining amplitudes from the equations

$$\psi_g(\kappa^2 - k_g^2) = - \sum_p v_p \psi_{g-p}. \quad (13)$$



Anpassung d und Resonanzfehler ϵ_1 und ϵ_2

Fig. 3

Fig. 3. Adaptation d and resonance errors ϵ_1 and ϵ_2 .

[Figure 3: reciprocal lattice (points as dots) with the dashed vertical $\mathfrak{b}_3$ crystal-normal line; the Laue point L , origin O , reflection point H ; vectors $\mathfrak{f}_0$ and $\mathfrak{f}_0 + 2\pi\mathfrak{h}$ ($= \mathfrak{C} - \zeta$) drawn; the adaptation d and the resonance errors $\varepsilon_1, \varepsilon_2$ marked near L .]

Adaptation d and resonance errors ε_1 and ε_2 .

Fig. 3

in such a way that in the sums on the right-hand side we take into account only those terms which contain the two large amplitudes c_1 and c_2 of the two principal waves. We thereby obtain for the “small” amplitudes the values

$$\psi_g = -\frac{c_1 v_g + c_2 v_{g-h}}{\kappa^2 - k_g^2} \quad (35)$$

which we can now insert into those two equations of the system (13) which refer to the two principal waves. These are the equations:

$$\psi_0(\kappa^2 - k_0^2) = 2c_1 \kappa \varepsilon_1 = -\sum \psi_g v_{-g}$$

$$\psi_h(\kappa^2 - k_h^2) = 2c_2 \kappa \varepsilon_2 = -\sum \psi_g v_{h-g}.$$

— p. 78 —

We obtain from this

$$\begin{cases} c_1(2\kappa\varepsilon_1 - V_{11}) - c_2 V_{12} = 0 \\ c_2(2\kappa\varepsilon_2 - V_{22}) - c_1 V_{21} = 0 \end{cases} \quad (36)$$

with the abbreviations

$$\begin{cases} V_{11} = \sum \frac{v_g v_{-g}}{\kappa^2 - k_g^2} \\ V_{12} = -v_{-h} + \sum \frac{v_{-g} v_{-h+g}}{\kappa^2 - k_g^2} \\ V_{21} = -v_h + \sum \frac{v_g v_{h-g}}{\kappa^2 - k_g^2} \\ V_{22} = \sum \frac{v_{g-h} v_{h-g}}{\kappa^2 - k_g^2} \end{cases} \quad (37)$$

If the crystal has a centre of symmetry, the expressions (37) simplify, with regard to the relation

$$v_h = v_{-h} \quad (7a)$$

to:

$$\begin{cases} V_{11} = \sum_g \frac{v_g^2}{\kappa^2 - k_g^2} \\ V_{12} = V_{21} = \sum_g \frac{v_g v_{g-h}}{\kappa^2 - k_g^2} - v_h \\ V_{22} = \sum_g \frac{v_{g-h}^2}{\kappa^2 - k_g^2} \end{cases} \quad (37a)$$

By taking into account on the right-hand side of equations (13) only the contributions of the amplitudes of the principal waves, we have now reduced the infinite system of equations (13) to the two homogeneous equations (36) between the amplitudes of the two principal waves. Of course one could carry the approximation further. But this would change only the numerical value of the quantities V_{11} , V_{12} , V_{22} , not, however, the structure of equations (36), which alone matters in principle. With the introduction of the adaptation d according to (34), these equations read:

$$\begin{cases} c_1(2\kappa d \cos \vartheta_1 - V_{11}) - c_2 V_{12} = 0 \\ c_2(2\kappa(d \cos \vartheta_2 + \zeta) - V_{22}) - c_1 V_{12} = 0 \end{cases} \quad (36a)$$

— p. 79 —

§ 5. The dispersion equation

Determination of the propagation vectors

In the two equations (36a) which we obtained in the preceding paragraph, besides the two amplitudes c_1 and c_2 of the principal waves, the quantity d decisive for the determination of the propagation vectors of the dynamical waves is still unknown. It is determined in the usual way from the fact that the determinant of the two equations (36a) must vanish:

$$\left(d - \frac{V_{11}}{2\kappa \cos \vartheta_1}\right) \left(d + \frac{\zeta}{\cos \vartheta_2} - \frac{V_{22}}{2\kappa \cos \vartheta_2}\right) = \frac{V_{12}^2}{4\kappa^2 \cos \vartheta_1 \cos \vartheta_2} \quad (38)$$

(38) is, in the approximation we are aiming at, the *dispersion equation for electron waves in a crystal*. Its solution is

$$\begin{cases} d = \frac{V_{11}}{4\kappa \cos \vartheta_1} + \frac{V_{22}}{4\kappa \cos \vartheta_2} - \frac{\zeta}{2 \cos \vartheta_2} \\ \pm \sqrt{\left(\frac{V_{11}}{4\kappa \cos \vartheta_1} - \frac{V_{22}}{4\kappa \cos \vartheta_2} + \frac{\zeta}{2 \cos \vartheta_2}\right)^2 + \frac{V_{12}^2}{4\kappa^2 \cos \vartheta_1 \cos \vartheta_2}} \end{cases} \quad (39)$$

By the form (39) of the solution, in which a square root occurs, we are led, as Ewald in the dynamical theory of X-ray reflections¹⁾, to distinguish two cases:

1. *Laue case*: Primary and secondary beams are both directed *away* from the entrance face of the primary beam. Then $\cos \vartheta_1$ and $\cos \vartheta_2$ are both *positive*, so the two solutions for d and hence all propagation vectors are always *real*. This case is realized in the experiments of Thomson²⁾

and Rupp³). To be sure, a somewhat different case is present there insofar as microcrystalline metals are concerned. But one can say definitely that a complex propagation vector, which in the observation of the electrons passing through the crystal would be connected with strong attenuation of the beam to be observed⁴), cannot occur.

Footnotes: 1) P. P. Ewald III, loc. cit., pp. 571ff. and especially pp. 585ff. 2) G. P. Thomson, loc. cit. 3) E. Rupp, loc. cit. 4) For complex propagation vectors the amplitude (and intensity) decreases exponentially on advancing into the interior of the crystal, as will be shown in the next paragraph.

— p. 80 —

2. *Bragg case*: The secondary beam emerges again on the entrance side of the primary beam, so $\cos \vartheta_2$ is *negative*. Then, in the case that

$$\left| \frac{V_{11}}{4\kappa \cos \vartheta_1} - \frac{V_{22}}{4\kappa \cos \vartheta_2} + \frac{\zeta}{2 \cos \vartheta_2} \right| < \frac{|V_{12}|}{2\kappa \sqrt{-\cos \vartheta_1 \cos \vartheta_2}} \quad (40)$$

there result *complex* values for d , and hence also complex resonance errors and propagation vectors. As we shall see in the next paragraph, in this case the entire intensity of the incident beam goes over into the reflected one. We shall call this case, which occurs for a certain range of values of ζ , the case of selective reflection, in precision of this designation already used in § 1. Our region of “selective reflection” represents, so to speak, the core of a Laue spot.

The analogue of selective reflection naturally occurs also for X-rays. Ewald designates the phenomenon as “total reflection”, but this designation could lead with us to confusions with the total reflection discussed at the end of § 1, which occurs when the electrons emerge from the crystal and is connected with the refractive index $\mu > 1$. The latter phenomenon is the exact analogue of optical total reflection and rests wave-theoretically on the fact that the normal component Γ_h of the emerging vacuum wave (§ 3) becomes purely imaginary, while the normal component of the associated crystal wave is real. For the propagation vector in vacuum is indeed

$$|\mathfrak{K}_h|^2 = A_h^2 + \Gamma_h^2 = K^2,$$

[while] for the propagation vector in the crystal the component parallel to the crystal surface remains the same:

$$|\mathfrak{f}_h|^2 = A_h^2 + \gamma_h^2 = k_h^2 \approx \kappa^2 = K^2 + v_0.$$

If therefore

$$|\gamma_h| < v_0, \quad (41)$$

then $K^2 < A_h^2$, hence Γ_h purely imaginary. This can also be expressed corpuscularly: If the component of the velocity of an electron normal to the surface, ξ , is too small to overcome the potential difference V between metal and vacuum, i.e. if

$$\frac{m}{2}\xi^2 < eV_0 \quad (41a)$$

— p. 81 —

then the electron cannot emerge into the vacuum.¹⁾ But now the propagation vector $\mathbf{f}$ of the de Broglie wave is in magnitude and direction equal to $2\pi/h$ times the momentum vector $m\mathbf{v}$ of the electron, and accordingly electrons cannot emerge if

$$\frac{1}{2m} \left(\frac{h}{2\pi} \right)^2 \gamma_h^2 < eV_0 \quad (41b)$$

Since now according to (7)

$$\frac{8\pi^2 me}{h^2} V = U, \quad \frac{8\pi^2 me}{h^2} V_0 = v_0,$$

the inequality (41a) is identical with the condition

$$|\gamma_h| < v_0. \quad (41)$$

It is interesting that for electron waves selective reflection lets the entire intensity incident from the vacuum return into the vacuum, whereas total reflection, because of the refractive index $\mu > 1$, on the other hand retains all the electrons in the crystal. For X-rays, because of $\mu < 1$, both phenomena act in the same sense, namely that the whole intensity of the X-rays returns into the vacuum.

In order to orient ourselves more closely about the region of selective reflection, we take, for a second approximation¹⁾, the quantities V_{11} , V_{12} , V_{22} constant in the whole wavelength region to be considered. We then see that the region of selective reflection extends symmetrically to both sides about a mean excitation error

$$\zeta_0 = \frac{V_{22}}{2\kappa} + \frac{V_{11}}{2\kappa} \cdot \frac{-\cos \vartheta_2}{\cos \vartheta_1} \quad (42)$$

up to the excitation errors

$$\zeta_1 = \zeta_0 + w, \quad \zeta_{-1} = \zeta_0 - w,$$

where the half width of the region of selective reflection, expressed in units of the excitation error, is equal to

$$w = \frac{|V_{12}|}{\kappa} \sqrt{-\frac{\cos \vartheta_2}{\cos \vartheta_1}} \quad (43)$$

(“selection width”). If we go over according to (33) to the incident wave numbers and take, with Davisson and Germer, the direction of incidence normal to the surface ($\vartheta_1 = 0$, $\vartheta_2 = \pi - \vartheta$),

Footnotes: 1) As a first approximation we shall consider the simple theory with refractive index. An indication of a third approximation we give in § 7.

— p. 82 —

we obtain the result that the region of selective reflection extends about a mean wave number

$$\kappa_m = \kappa_0 + \frac{V_{11} \cos \vartheta + V_{22}}{2\kappa_0(1 + \cos \vartheta)} \quad (42a)$$

to either side by the wave number difference

$$\Delta\kappa = \frac{w}{1 + \cos \vartheta} = \frac{|V_{12}|}{\kappa} \cdot \frac{\sqrt{\cos \vartheta}}{1 + \cos \vartheta}. \quad (43a)$$

The resolving power of the crystal is then found to be of the order of magnitude

$$\frac{\lambda}{\Delta\lambda} = \frac{\kappa_0}{2\Delta\kappa} = \frac{\kappa_0^2}{2|V_{12}|} \frac{1 + \cos \vartheta}{\sqrt{\cos \vartheta}} \approx \frac{\kappa_0^2}{|V_{12}|}. \quad (44)$$

We shall come back to this in §§ 6 and 12. For arbitrary directions of incidence, $\sqrt{-\cos \vartheta_2 / \cos \vartheta_1}$ takes the place of $\sqrt{\cos \vartheta}$.

The “adaptation” important for the determination of the propagation vectors is written most simply if we introduce the selection error

$$W = \frac{\zeta_0 - \zeta}{w} = \frac{V_{22} - V_{11} \frac{\cos \vartheta_2}{\cos \vartheta_1} - 2\kappa\zeta}{2|V_{12}| \sqrt{-\frac{\cos \vartheta_2}{\cos \vartheta_1}}} \quad (45)$$

For then

$$d = \frac{V_{11}}{2\kappa \cos \vartheta_1} - \frac{w}{2 \cos \vartheta_2} (W \pm \sqrt{W^2 - 1}). \quad (46)$$

§ 6. Calculation of the intensities

The dispersion equation (38) obtained from the dynamical equations (13) yields two values for the “adaptation” d :

$$\begin{cases} d' = \frac{V_{11}}{2\kappa \cos \vartheta_1} - \frac{wW}{2 \cos \vartheta_2} - \frac{w}{2 \cos \vartheta_2} \sqrt{W^2 - 1} \\ d'' = \frac{V_{11}}{2\kappa \cos \vartheta_1} - \frac{wW}{2 \cos \vartheta_2} + \frac{w}{2 \cos \vartheta_2} \sqrt{W^2 - 1}. \end{cases} \quad (46a)$$

To each value of d there corresponds a pair of propagation vectors of the primary and secondary beams, which, with the introduction of

$$\begin{cases} \gamma = \gamma_1 = \gamma_0 - \frac{V_{11}}{2\kappa \cos \vartheta_1} + \frac{wW}{2 \cos \vartheta_2} & \text{with } \gamma'_0 = \sqrt{\kappa^2 - A^2} \\ \omega = \frac{w}{2 \cos \vartheta_2} \sqrt{W^2 - 1} \end{cases} \quad (47)$$

we can write according to (34a) in the form

— p. 83 —

$$\begin{cases} \mathbf{f}'_1 = \mathbf{f}_1 + \omega \mathbf{n} \\ \mathbf{f}''_1 = \mathbf{f}_1 - \omega \mathbf{n} \\ \mathbf{f}_1 = A \mathfrak{T} + \gamma \mathbf{n} = \alpha_1 \mathbf{t}_1 + \gamma_1 \mathbf{n} \\ \mathbf{f}'_2 = \mathbf{f}_2 + \omega \mathbf{n} \\ \mathbf{f}''_2 = \mathbf{f}_2 - \omega \mathbf{n} \\ \mathbf{f}_2 = \mathbf{f}_1 + 2\pi \mathbf{h} = \alpha_2 \mathbf{t}_2 + \gamma_2 \mathbf{n} \end{cases} \quad (48)$$

Here $\mathbf{t}_i$ denotes, as before, the unit vector in the direction of the projection of $\mathbf{f}_i$ onto the crystal surface, $\mathbf{n}$ that in the direction of the crystal normal.

For each pair of propagation vectors in (48) there now follows from the 1st equation (36a)

$$c_2 = \frac{2\kappa d \cos \vartheta_1 - V_{11}}{V_{12}} c_1$$

a definite ratio between the amplitudes of the primary and secondary beams, which, according to (46a) and (43), reads:

$$\begin{cases} c'_2 = c'_1 \sqrt{-\frac{\cos \vartheta_1}{\cos \vartheta_2} (W + \sqrt{W^2 - 1})} \\ c''_2 = c''_1 \sqrt{-\frac{\cos \vartheta_1}{\cos \vartheta_2} (W - \sqrt{W^2 - 1})}. \end{cases} \quad (49)$$

But the question remains open so far as to how the intensity of a given wave incident from the vacuum

$$\psi_0 = C_0 e^{i(\mathfrak{K}_0 \mathbf{r})} = C_0 e^{i(A_0 \mathfrak{T}_0 + \Gamma_0 \mathbf{n}, \mathbf{r})} \quad (50)$$

is distributed over the two dynamically possible pairs of primary and secondary waves.

The surface conditions derived in § 3 now give the answer to this question. We found there relations between the amplitudes of the vacuum waves and of the crystal waves whose propagation vectors agree with respect to the projection onto the surface. In our intensity calculations we shall now, in accordance with the degree of our approximation, take into account only those vacuum waves which correspond to the two “principal waves” in the crystal. Strictly speaking, according to (23), there correspond to both the primary and the secondary wave, at the entrance side and at the exit side, two each, namely one incident from the vacuum and one penetrating into the vacuum,

— p. 84 —

wave. But of these two only one is ever of appreciable intensity. For k_1 and k_2 become nearly equal to $\kappa = \sqrt{K^2 + v_0}$, i.e. for not too “soft” beams ($K \gg v_0$) also approximately $k_1 = k_2 = K$. But since the components parallel to the surface of corresponding crystal and vacuum waves agree:

$$\alpha_g = A_g,$$

for the principal waves the corresponding normal components are also either approximately equal or opposite to one another:

$$\gamma_1 \approx \pm \Gamma_1, \quad \gamma_2 \approx \pm \Gamma_2.$$

Then one of the two vacuum waves in equations (23) always becomes nearly zero (we shall neglect it in our approximation) and there remains only, on the entrance side an incident, on the exit side of each crystal beam an emergent, vacuum wave.

On the entrance side of the primary waves f'_1 and f''_1 , ψ_0 is incident from the vacuum. From (23) there follows

$$C_0 = c'_1 \cdot \frac{\Gamma_1 + \gamma_1 + \omega}{2\Gamma_1} + c''_1 \cdot \frac{\Gamma_1 + \gamma_1 - \omega}{2\Gamma_1}$$

If we neglect here the small quantity ω and write, for the arithmetic mean of the two nearly equal quantities Γ_1 and γ_1 , the geometric mean:

$$\frac{\Gamma_1 + \gamma_1}{2} \approx \sqrt{\Gamma_1 \gamma_1}$$

we obtain

$$C_0 = \sqrt{\frac{\gamma_1}{\Gamma_1}} (c'_1 + c''_1). \quad (51)$$

On the entrance side of the secondary wave no wave is incident from the vacuum, hence

$$0 = c'_2 \cdot \frac{\Gamma_2 - \gamma_2 - \omega}{2\Gamma_2} e^{i(f'_2 \mathbf{r}')} + c''_2 \cdot \frac{\Gamma_2 - \gamma_2 + \omega}{2\Gamma_2} e^{i(f''_2 \mathbf{r}')}.$$

where $\mathbf{r}'$ denotes a place on the entrance face of the secondary beam. If this face is parallel to the entrance face of the primary beam and the distance of the two is H , there follows, with the same neglects as for the primary wave,

$$0 = c'_2 e^{i\omega H} + c''_2 e^{-i\omega H}.$$

— p. 85 —

or, using (49),

$$c'_1 (W + \sqrt{W^2 - 1}) e^{i\omega H} + c''_1 (W - \sqrt{W^2 - 1}) e^{-i\omega H} = 0. \quad (52)$$

If we further set

$$W = \text{Cos } \eta \quad (52a)$$

then the two amplitudes of the two primary waves are determined from the two boundary conditions (51) and (52):

$$\begin{cases} c'_1 = -C \cdot \sqrt{\frac{\Gamma_1}{\gamma_1}} \cdot \frac{e^{-\eta-i\omega H}}{2 \sin(\eta+i\omega H)} \\ c''_1 = +C \cdot \sqrt{\frac{\Gamma_1}{\gamma_1}} \cdot \frac{e^{+\eta+i\omega H}}{2 \sin(\eta+i\omega H)} \end{cases} \quad (53)$$

The total primary wave becomes¹⁾ :

$$\begin{cases} \psi_1 = C_0 \cdot \sqrt{\frac{\Gamma_1}{\gamma_1}} \cdot e^{i(\mathbf{k}_1 \mathbf{r})} \cdot \frac{\sin(\eta-i\omega(H-z))}{\sin(\eta+i\omega H)} \\ = C_0 \cdot \sqrt{\frac{\Gamma_1}{\gamma_1}} \cdot e^{i(\mathbf{k}_1 \mathbf{r})} \cdot \frac{\sqrt{W^2-1} \cos \omega(H-z) + iW \sin \omega(H-z)}{\sqrt{W^2-1} \cos \omega H + iW \sin \omega H} \end{cases} \quad (54)$$

Correspondingly we obtain for the secondary wave, if we take into account that $h_2 \approx k_{1z}$, i.e.

$$\frac{\cos \vartheta_1}{\cos \vartheta_2} = \frac{\gamma_1}{\gamma_2}$$

is, the expression

$$\psi_2 = C_0 \cdot \sqrt{-\frac{\Gamma_1}{\gamma_2}} e^{i(\mathbf{k}_2 \mathbf{r})} \cdot \frac{i \sin \omega(H-z)}{\sin(\eta+i\omega H)}. \quad (54a)$$

Waves emerge into the vacuum both in the direction of the continuation of the primary beam and in the direction of the secondary beam, whose amplitudes are computed from those of the primary and secondary beams in the crystal by means of the boundary conditions (23):

$$\Psi_1 = C_0 \cdot e^{i(\mathbf{R}_1 \mathbf{r})} \cdot \frac{\sqrt{W^2-1}}{\sqrt{W^2-1} \cos \omega H + iW \sin \omega H} \cdot e^{i(\gamma_1-\Gamma_1)H}. \quad (55)$$

$$\Psi_2 = C_0 \cdot e^{i(\mathbf{R}_2 \mathbf{r})} \cdot \frac{i \sin \omega H}{\sqrt{W^2-1} \cos \omega H + iW \sin \omega H}. \quad (55a)$$

The last beam is the one observed in the experiments of Davisson and Germer as the “reflected” beam.

Footnotes: 1) z denotes a coordinate in the direction of the normal to the surface pointing into the crystal.

— p. 86 —

However, what interests us less now is the mathematical expression for the various electron waves, but above all the number of electrons transported per unit time through the cross section of each beam, the “current” S . This is computed as the product of the electron velocity with the cross section of the electron beam and the number of electrons present per unit volume. The electron velocity is directly proportional to the wave number (inversely to the wavelength); the cross section

of the electron beam is directly proportional to the cosine of the angle between it and the normal to the crystal surface (cf. Fig. 4), the

[Figure 4: diagram of two beam bundles (incident and diffracted) meeting a horizontal crystal surface, with the surface normal drawn dashed and the angles ϑ_1 , ϑ_2 between the beams and the normal indicated.] Cross section of the beam bundles and direction cosines: $\frac{q_1}{q_2} = \frac{\cos \vartheta_1}{\cos \vartheta_2}$. Fig. 4

product of both quantities is therefore proportional to the normal component of the propagation vector. In this wave description, the electron number density is proportional to the squared modulus of the de Broglie-wave amplitude. We denote the particle flux of the primary beam in vacuum

$$C_0^2 \Gamma_1 = S_0 \quad (56)$$

and now have to distinguish *two cases*: 1. Outside the region of selective reflection¹⁾ we have $W > 1$, hence η and ω real. The intensities of the primary beam

$$s_1 = \frac{W^2 - \cos^2 \omega(H - z)}{W^2 - \cos^2 \omega H} \cdot S_0 \quad (56a)$$

Footnotes: 1) The same considerations always hold for the Laue case (cf. p. 79).

— p. 87 —

and of the secondary beam

$$s_2 = \frac{\sin^2 \omega(H - z)}{W^2 - \cos^2 \omega H} \cdot S_0 \quad (56b)$$

are then subject to periodic fluctuations in the crystal, in such a way that in the end the number of electrons transported through a cross section of the crystal in the direction of the primary beam,

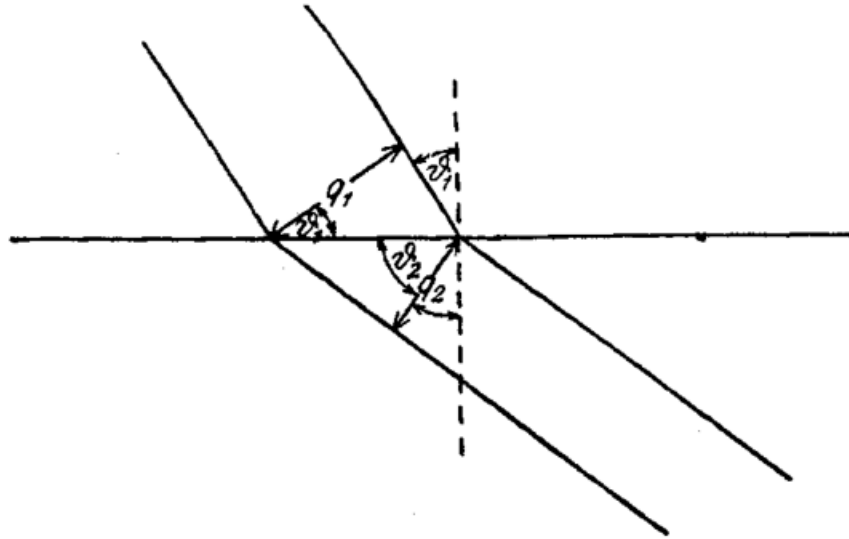
$$s = s_1 - s_2 = \frac{W^2 - 1}{W^2 - \cos^2 \omega H} \cdot S_0 \quad (56c)$$

is independent of the particular choice of the cross section. This is of course already required because otherwise in certain layers of the crystal there would be a permanent inflow, in others a permanent loss of electrons, whereas the entire electron wave field we are considering is supposed to be *stationary*. Yet the currents of the primary and secondary beams individually increase and decrease spatially periodically, both in the same sense, the period of such an intensity beat,

$$\frac{\pi}{\omega} = \frac{2\pi \cos \vartheta_2}{w\sqrt{W^2 - 1}}$$

amounting to a few atomic layers. This finding is analogous to Ewald's *Pendellösung*¹⁾ in the case of real resonance errors. In connection with the periodic intensity fluctuations of the electron beams in the crystal, the intensity of the observable secondary beam reflected back into the vacuum,

$$S_2 = S_0 \frac{\sin^2 \omega H}{W^2 - \cos^2 \omega H} \quad (57)$$



Querschnitt der Strahlenbündel und Richtungscosinus: $\frac{q_1}{q_2} = \frac{\cos \vartheta_1}{\cos \vartheta_2}$

Fig. 4

Fig. 4. Cross section of the beam bundles and direction cosines: $q_1/q_2 = \cos \vartheta_1 / \cos \vartheta_2$.

just as the intensity emerging from the surface $z = H$ back into the vacuum in the direction of the continuation of the primary beam,²⁾

$$S_1 = S_0 \frac{W^2 - 1}{W^2 - \cos^2 \omega H} \quad (57a)$$

depends on the *thickness* of the crystal. The *total* current of the primary and secondary beams emerging from the crystal is of course equal to the incident electron current:

$$S = S_1 + S_2 = S_0. \quad (57b)$$

Footnotes: 1) Ewald III, loc. cit., p. 586. 2) This is the vacuum wave belonging to the primary beam on the exit side, which we had not listed separately so far.

— p. 88 —

The intensity of the diffracted as well as of the transmitted primary beam according to (57) and (57a) is so sensitive to a change of the crystal thickness that it passes from maximum to minimum already upon a change of a few atomic layers. It will therefore not be observable on account of the slightest irregularities of the crystal. Consequently we must disregard these “interferences at the plane-parallel plate” and average over the thickness of the crystal. Then we finally obtain as expression for the reflected electron current

$$S_2 = S_0 \left(1 - \frac{\sqrt{W^2 - 1}}{W} \right) \quad (58)$$

and for the one passing through the crystal

$$S_1 = S_0 \frac{\sqrt{W^2 - 1}}{W}. \quad (58a)$$

2. In the region of selective reflection the selection error is $W < 1$ and accordingly η and ω imaginary. The exact expression for the intensity of the primary wave then reads

$$s_1 = S_0 \cdot \frac{e^{2|\omega|(H-z)} + 2 - 4W^2 + e^{-2|\omega|(H-z)}}{e^{2|\omega|H} + 2 - 4W^2 + e^{-2|\omega|H}}$$

But for a crystal of even moderately considerable thickness (because of the still considerable magnitude of ω already at about 100 atomic layers) the terms $e^{2|\omega|(H-z)}$ and $e^{2|\omega|H}$ are by far dominant and we get

$$s_1 = S_0 e^{-2|\omega|z}. \quad (59)$$

Likewise the intensity of the secondary beam,

$$s_2 = S_0 e^{-2|\omega|z} \quad (59a)$$

also decreases exponentially toward the interior of the crystal; the current running in the direction of the primary beam thus becomes zero (strictly speaking $4(1 - W^2)e^{-2|\omega|H}$) and only the small intensity

$$S_1 = \frac{4(1 - W^2)}{e^{2|\omega|H} + 2 - 4W^2 + e^{-2|\omega|H}} \approx 4(1 - W^2)e^{-2|\omega|H} \quad (60)$$

penetrates through the crystal in the direction of the primary beam. All the remaining incident electrons,

$$S_2 = S_0 \cdot \frac{e^{2|\omega|H} - 2 + e^{-2|\omega|H}}{e^{2|\omega|H} + 2 - 4W^2 + e^{-2|\omega|H}} \quad (60a)$$

— p. 89 —

enter into the reflected beam, and for sufficiently large thickness of the crystal we get practically

$$S_2 = S_0, \quad (61)$$

i.e.: In the region of selective reflection the entire incident electron current passes over into the reflected beam.

The formulas so far refer to fixed given wavelength and direction of incidence. If a whole wavelength region falls on the crystal,

$$\psi = S_0 \int e^{i(\mathfrak{K}r)} d\mathfrak{K},$$

as is usually the case in practice, then a definite lattice plane h_1, h_2, h_3 of the crystal reflects altogether the intensity

$$\left\{ \begin{aligned} \int S_2 d\mathfrak{K} &= S_0 \int_0^{\mathfrak{K}_m - \Delta\mathfrak{K}} \left(1 - \frac{\sqrt{W^2 - 1}}{W} \right) d\mathfrak{K} + S_0 \int_{\mathfrak{K}_m - \Delta\mathfrak{K}}^{\mathfrak{K}_m + \Delta\mathfrak{K}} d\mathfrak{K} \\ &+ S_0 \int_{\mathfrak{K}_m + \Delta\mathfrak{K}}^{\infty} \left(1 - \frac{\sqrt{W^2 - 1}}{W} \right) d\mathfrak{K} = \frac{\pi S_0 w}{1 + \cos \vartheta} \end{aligned} \right. \quad (62)$$

reflected. Of this,

$$\frac{2S_0 w}{1 + \cos \vartheta}$$

falls on the region of selective reflection, the rest is reflected in the form of wavelengths on both sides of this region. The variation of the intensity of the reflected beam as a function of the quantity W linked to the wave number κ by equation (45) is shown in Fig. 4. According to this curve the intensity falls off very rapidly on both sides of the region of selective reflection, so that the “half-width” of a wavelength range diffracted by the crystal,

$$B_{1/2} = 2 \Delta\mathfrak{K} \sqrt{\frac{4}{3}} = \frac{2w}{1 + \cos \vartheta} \sqrt{\frac{4}{3}} = \frac{2|V_{12}|}{\kappa} \frac{\sqrt{\cos \vartheta}}{1 + \cos \vartheta} \sqrt{\frac{4}{3}} \quad (63)$$

is not substantially wider than the region of selective reflection. The *resolving power* of the crystal will therefore, for an absolutely sharply defined direction of incidence of the primary beam,¹⁾

Footnotes: 1) The formula holds for normal incidence of the primary beam. Otherwise $\sqrt{\frac{-\cos \vartheta_2}{\cos \vartheta}}$ takes the place of $\sqrt{\cos \vartheta}$, where ϑ_1 and ϑ_2 denote the angles between the incident and the reflected beam, respectively, and the crystal normal.

— p. 90 —

$$\frac{\lambda}{\Delta\lambda} = \frac{\kappa_0}{2\Delta\kappa} \sqrt{\frac{3}{4}} = \frac{\kappa_0^2}{2V_{12}} \sqrt{\frac{3}{4}} \cdot \frac{1 + \cos \vartheta}{\sqrt{\cos \vartheta}} \quad (64)$$

and has according to § 11 a numerical value of about 100 for wavelengths Λ = about 1 Å. For the rest, the intensity variation of the reflected beam corresponds to the curve calculated by Ewald¹⁾ (cf. curve 4).

The intensity formulas set up here may of course in no way be taken as *exact*. For one thing the [Figure 5: plot of the reflected intensity ratio S_2/S_0 (vertical axis marked 0.2, 0.4, 0.6, 0.8, 1.0) against the selection error W (horizontal axis marked $-3, -2, -1, 0, 1, 2, 3$); the curve is flat at 1.0 for $|W| < 1$ and falls off symmetrically to zero on both sides.] Reflected intensity as a function of the selection error W . Fig. 5

quantities V_{11} , V_{12} , V_{22} are by no means independent of the wave number κ , as assumed here. Besides, a large part of the intensity is certainly lost through the “weak beams” of the dynamical wave system neglected here, and a still more considerable part drops out of the number of electrons observed at all through inelastic collisions. In addition, the incident electron beam by no means consists exactly of electrons of a definite direction. This circumstance of course broadens considerably the wavelength region to which the crystal responds with a definite lattice plane, and at the same time diminishes the intensity of the reflected beam relative to the incident one. Furthermore, the influence of temperature (Debye factor) has remained unconsidered, which likewise weakens the reflected beam.

Footnotes: 1) Ewald III, loc. cit., p. 593, Fig. 19 (Bragg case).

— p. 91 —

§ 7. Sketch of a third approximation

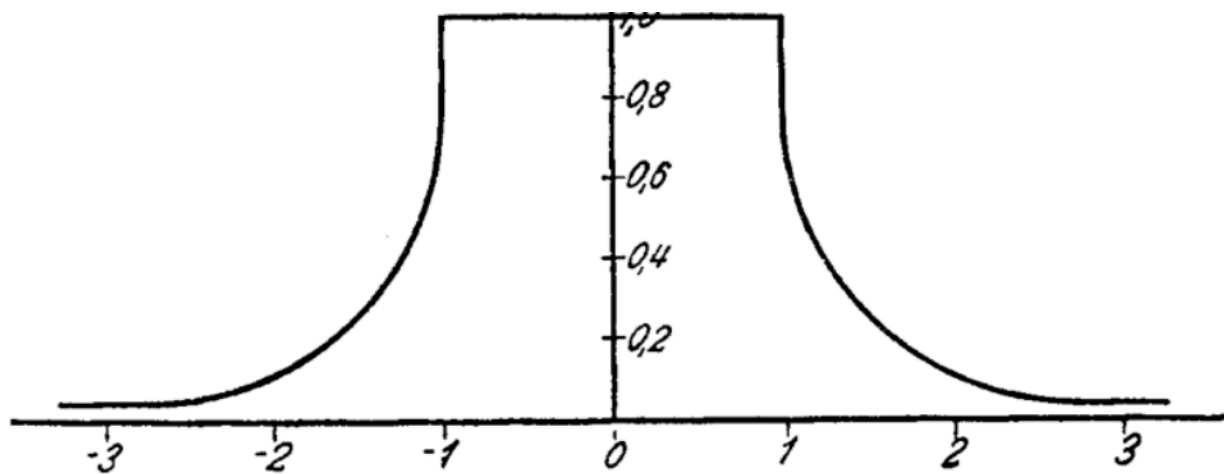
In § 5 we solved the dispersion equation

$$\left(\kappa d \cos \vartheta_1 - \frac{V_{11}}{2}\right) \left(\kappa d \cos \vartheta_2 + \zeta \kappa - \frac{V_{22}}{2}\right) = \left(\frac{V_{12}}{2}\right)^2 \quad (36b)$$

and in § 6 obtained a picture of the intensity variation by setting the “dynamical potential quantities” constant. But, e.g., [Tr.: "(36b)" misprints "(38)"]

$$V_{11} = \sum \frac{v_g^2}{\kappa^2 - k_g^2}$$

depends very strongly on the wave number κ because of the resonance denominators occurring in every term,



Reflektierte Intensität in Abhängigkeit vom Selektionsfehler W

Fig. 5

Fig. 5. Reflected intensity as a function of the selection error W .

$$\begin{cases} \kappa^2 - k_{\mathbf{g}}^2 = \kappa^2 - |\mathbf{f}_0 + 2\pi\mathbf{g}|^2 = \kappa^2 - k_0^2 - 2(\mathbf{f}_0, 2\pi\mathbf{g}) \\ - |2\pi\mathbf{g}|^2 \end{cases} \quad (65)$$

κ being known to be computed from the (given) wave number $\mathfrak{K}$ of the incident beam in vacuum by means of the simple refraction law:

$$\kappa = \mathfrak{K}\mu = \sqrt{\mathfrak{K}^2 + v_0}.$$

A further approximation, which at least takes into account the linear term of the dependence of the dynamical potential quantities on the wave number, can be carried out without difficulty.

We assume again — corresponding to the experiments of Davisson and Germer — that the direction of the primary beam in the crystal is invariably given, and that $\mathfrak{C}_0$ is the propagation vector of that primary beam running in this prescribed direction which would satisfy the Laue equations exactly [cf. derivation of Eq. (33)]. If $|\mathfrak{C}_0| = \kappa_0$, then the propagation vector of the primary beam belonging to the wave number κ is accordingly

$$\mathbf{f}_0 = \mathfrak{C} - d\mathbf{n} = \frac{\kappa}{\kappa_0}\mathfrak{C}_0 - d\mathbf{n}, \quad (66)$$

where $\mathbf{n}$ is a unit vector in the normal direction and d is precisely the quantity to be determined by the “dispersion theory”. With regard to (30), (33), (34) and (34a) the resonance denominator (65) then reads

$$\begin{aligned} \kappa^2 - k_{\mathbf{g}}^2 &= -4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g}) \\ &\quad - 4\pi(\mathbf{g}, \mathfrak{C} - \mathfrak{C}_0) + 4\pi(\mathbf{g}, \mathfrak{C} - \mathbf{f}_0) + 2\kappa\varepsilon \end{aligned}$$

— p. 92 —

or also

$$\begin{cases} \kappa^2 - k_{\mathbf{g}}^2 = -4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g}) \\ - \frac{4\pi}{1 + \cos\vartheta} \zeta g_{\kappa} + 4\pi d g_{\mathbf{n}} + 2\kappa d \cos\vartheta_1. \end{cases} \quad (67)$$

Here g_{κ} , $g_{\mathbf{n}}$ are the components of the vector $\mathbf{g}$ in the direction of the primary beam $\mathfrak{C}$ and of the crystal normal $\mathbf{n}$, respectively. We now expand the dynamical potential quantities in the small quantities ζ and d and break off after the first term of the expansion. Because of

$$\kappa \cos\vartheta_1 + 2\pi g_{\mathbf{n}} = (\mathfrak{C} + 2\pi\mathbf{g}, \mathbf{n}) = (\mathfrak{C}_{\mathbf{g}}\mathbf{n}) = \mathfrak{C}_{\mathbf{g}}^{\mathbf{n}},$$

where $\mathfrak{C}_{\mathbf{g}}$ denotes the propagation vector of the “weak” beam $\mathbf{g}$, this gives

$$\begin{cases} \frac{1}{2}V_{11} = \frac{1}{2} \sum_{\mathbf{g}} \frac{v_{\mathbf{g}}^2}{-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})} + \frac{\zeta}{1 + \cos\vartheta} \sum_{\mathbf{g}} \frac{v_{\mathbf{g}}^2 g_{\kappa}}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} \\ - d \sum_{\mathbf{g}} \frac{v_{\mathbf{g}}^2 \mathfrak{C}_{\mathbf{g}}^{\mathbf{n}}}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} \end{cases} \quad (68)$$

and the dispersion equation (38b) goes over into

$$(\sigma_{11}d - \tau_{11})(\sigma_{22}d - \tau_{22}) = (\sigma_{12}d - \tau_{12})^2 \quad (69)$$

with the abbreviations

$$\left\{ \begin{aligned} \sigma_{11} &= \sum_{\mathbf{g}} \frac{v_{\mathbf{g}}^2 \mathfrak{C}_{\mathbf{g}}^n}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} + \varkappa \cos \vartheta_1 \\ \sigma_{12} &= \sum_{\mathbf{g}} \frac{v_{\mathbf{g}} v_{\mathbf{h}-\mathbf{g}} \mathfrak{C}_{\mathbf{g}}^n}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} \\ \sigma_{22} &= \sum_{\mathbf{g}} \frac{v_{\mathbf{h}-\mathbf{g}}^2 \mathfrak{C}_{\mathbf{g}}^n}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} + \varkappa \cos \vartheta_2 \\ \tau_{11} &= \frac{1}{2} \sum_{\mathbf{g}} \frac{v_{\mathbf{g}}^2}{-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})} + \frac{\zeta}{1 + \cos \vartheta} \sum_{\mathbf{g}} \frac{v_{\mathbf{g}}^2 g_{\varkappa}}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} \\ \tau_{12} &= -\frac{1}{2} v_{\mathbf{h}} + \frac{1}{2} \sum_{\mathbf{g}} \frac{v_{\mathbf{g}} v_{\mathbf{h}-\mathbf{g}}}{-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})} \\ &\quad + \frac{\zeta}{1 + \cos \vartheta} \sum_{\mathbf{g}} \frac{v_{\mathbf{g}} v_{\mathbf{h}-\mathbf{g}} g_{\varkappa}}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} \\ \tau_{22} &= \frac{1}{2} \sum_{\mathbf{g}} \frac{v_{\mathbf{h}-\mathbf{g}}^2}{-4\pi(\mathbf{g}, \mathfrak{C}_0 - \pi\mathbf{g})} \\ &\quad + \zeta \left(\frac{1}{1 + \cos \vartheta} \sum_{\mathbf{g}} \frac{v_{\mathbf{h}-\mathbf{g}}^2 g_{\varkappa}}{[-4\pi(\mathbf{g}, \mathfrak{C}_0 + \pi\mathbf{g})]^2} - \varkappa \right). \end{aligned} \right. \quad (70)$$

— p. 93 —

In the “second approximation” the values of these quantities were

$$\begin{aligned} \sigma_{11} &= \varkappa \cos \vartheta_1 & \sigma_{12} &= 0 & \sigma_{22} &= \varkappa \cos \vartheta_2 \\ \tau_{11} &= \tfrac{1}{2} V_{11} & \tau_{12} &= \tfrac{1}{2} V_{12} & \tau_{22} &= \tfrac{1}{2} V_{22} - \varkappa \zeta \end{aligned}$$

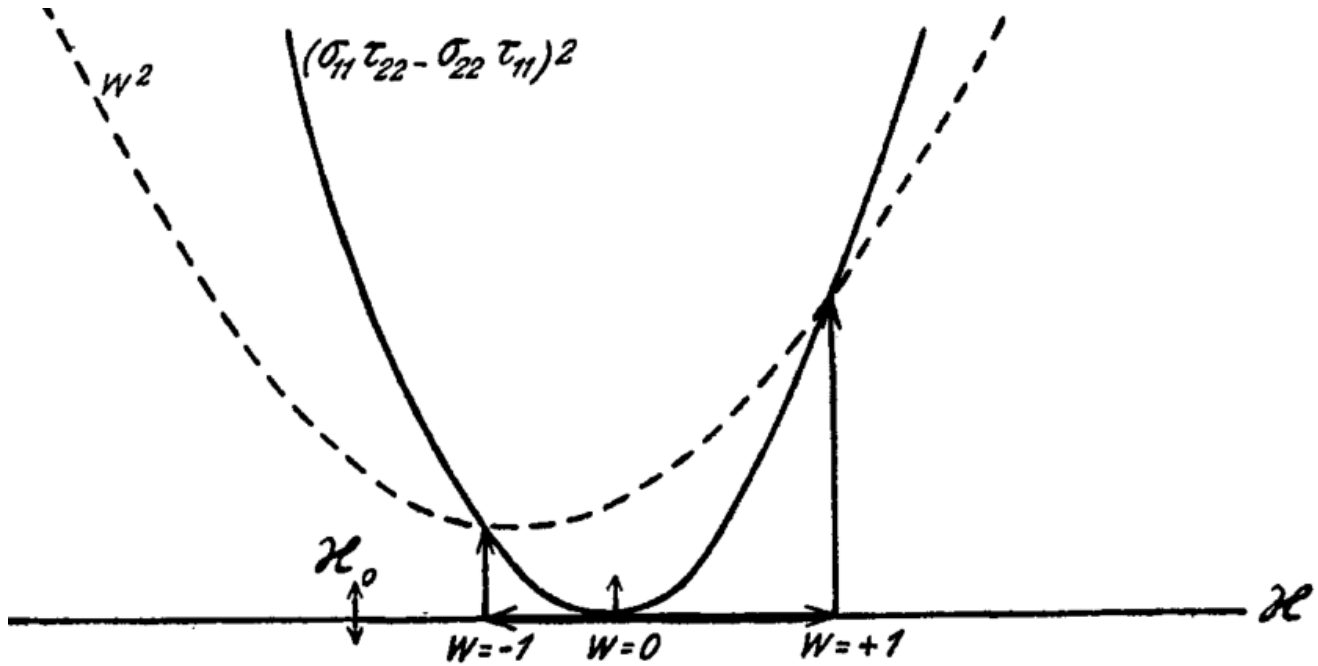
If we further denote, in analogy to the second approximation,

$$w^2 = -(\sigma_{11}\tau_{12} - \sigma_{12}\tau_{11})(\sigma_{22}\tau_{12} - \sigma_{12}\tau_{22}) \quad (71)$$

and

$$W = \frac{\sigma_{11}\tau_{22} - \sigma_{12}\tau_{12}}{w}, \quad (72)$$

[Figure 6: plot of $(\sigma_{11}\tau_{22} - \sigma_{22}\tau_{11})^2$ (solid parabola touching zero at $W = 0$) and w^2 (dashed parabola with elevated minimum) as functions of the wave number $\varkappa$ on the horizontal axis; marked on the axis are $\varkappa_0$ (with a vertical double arrow), $W = -1$, $W = 0$ and $W = +1$, the latter pointing up to the intersection point of the two curves; the region of selective reflection is emphasized as a thick segment of the $\varkappa$ -axis between the two intersection points of the curves (where $W^2 = 1$, asymmetric about $W = 0$).]



Das Gebiet der Selektivreflexion in der dritten Näherung:

$$\left. \begin{aligned} &(\sigma_{11}\tau_{22} - \sigma_{22}\tau_{11})^2 \\ &w^2 = -(\sigma_{11}\tau_{12} - \sigma_{12}\tau_{11})(\sigma_{22}\tau_{12} - \sigma_{12}\tau_{22}) \end{aligned} \right\} \begin{array}{l} \text{als Funktion der} \\ \text{Wellenzahl } \kappa. \end{array}$$

Zwischen den beiden Schnittpunkten der Kurven ($W^2 = 1$) liegt (asymmetrisch zu $W = 0$) das Gebiet der Selektivreflexion (stark ausgezogene κ -Achse)

Fig. 6

Fig. 6. The region of selective reflection in the third approximation: $(\sigma_{11}\tau_{22} - \sigma_{22}\tau_{11})^2$ and $w^2 = -(\sigma_{11}\tau_{12} - \sigma_{12}\tau_{11})(\sigma_{22}\tau_{12} - \sigma_{12}\tau_{22})$ as functions of the wave number κ . Between the two intersection points of the curves ($W^2 = 1$) lies the region of selective reflection (heavily drawn κ -axis), symmetric about $W = 0$.

The region of selective reflection in the third approximation: $w^2 = -(\sigma_{11}\tau_{12} - \sigma_{12}\tau_{11})(\sigma_{22}\tau_{12} - \sigma_{12}\tau_{22}) \left. \vphantom{w^2} \right\}$
as functions of the wave number $\varkappa$. Between the two intersection points of the curves ($W^2 = 1$)
lies (asymmetric about $W = 0$) the region of selective reflection (thick $\varkappa$ -axis). Fig. 6
then we obtain the solution of the dispersion equation in the form analogous to (46):

$$d = \frac{\sigma_{12}\tau_{12} - \sigma_{22}\tau_{11}}{\sigma_{11}\sigma_{22} - \sigma_{12}^2} + \frac{w}{\sigma_{11}\sigma_{22} - \sigma_{12}^2} (W \pm \sqrt{W^2 - 1}). \quad (73)$$

It can be shown that w is in general real for the Bragg case, and has approximately the variation shown by Fig. 6. Then there is also in the third approximation a region of “selective reflection” ($W < 1$). It is bounded by two wave numbers $\varkappa_1$ and $\varkappa_2$, which result as roots of the equation $W^2 = 1$, quadratic in $\varkappa$. The intensity formulas of the previous para-

— p. 94 —

graphs retain their validity, only w and W change their meaning.

But w is sometimes also imaginary. Then it could under certain circumstances happen that w^2 and $(\sigma_{11}\tau_{22} - \sigma_{12}\tau_{12})^2$ have the variation shown in Fig. 7. Then the region of selective reflection ($W < 1$) is split into two separate re-

[Figure 7: like Fig. 6, but with the dashed w^2 parabola dipping below the solid $(\sigma_{11}\tau_{22} - \sigma_{22}\tau_{11})^2$ curve in the middle, producing four intersection points and hence two separate regions of selective reflection on the $\varkappa$ -axis (shown thick); marks $W = -1$, $W = 0$, $W = +1$; the dashed curve labeled w^2 , the solid curve labeled $(\sigma_{11}\tau_{22} - \sigma_{22}\tau_{11})^2$.]

Anomalous variation of the curves shown in Fig. 6 (for notation see there). Splitting of the region of selective reflection (thick $\varkappa$ -axis). Fig. 7

gions. It would be conceivable that the doubly observed reflection (642) in the {110} azimuth at 170 and 188 volts could be traced back to such a splitting.

§ 8. Relation to the continuum theories of X-ray reflection. Extinction laws

Our theory of electron diffraction by crystals developed here stands in relation to the so-called “continuum theories” of X-ray reflections, which use the Fourier representation of the electron density (number of electrons per unit volume) in the crystal:

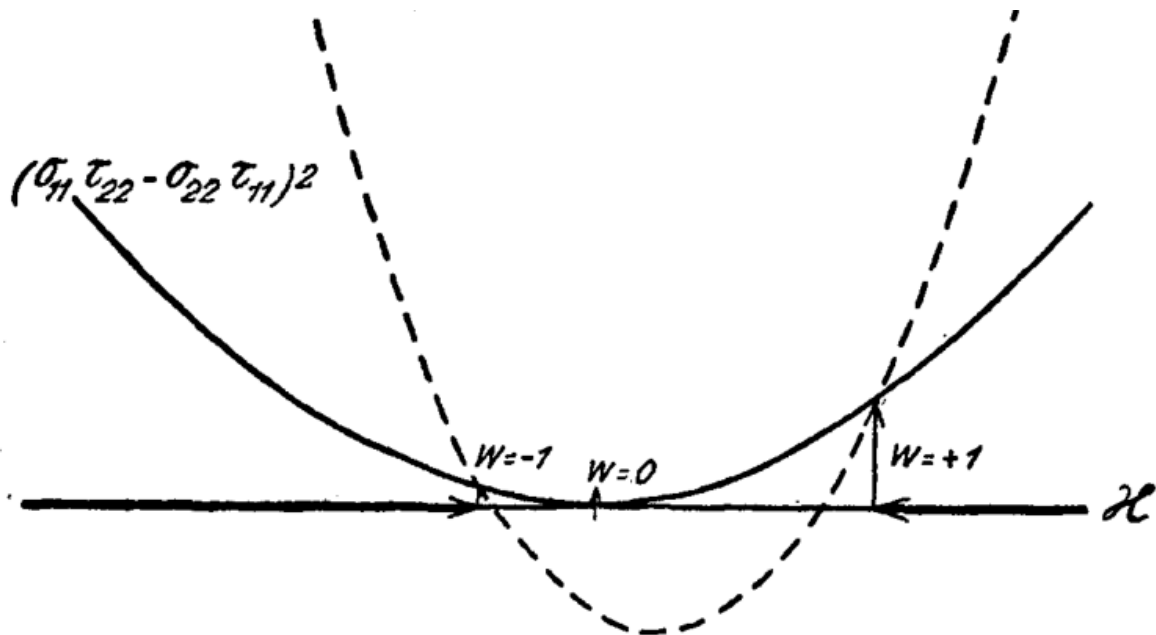
$$\varrho^E(\mathbf{r}) = \sum_{\mathbf{g}} \varrho_{\mathbf{g}}^E e^{2\pi i(\mathbf{g}\mathbf{r})} \quad (74)$$

for the computation of the atomic form factors important for the intensity of the diffracted X-rays. W. H. Bragg¹⁾

Footnotes: 1) W. H. Bragg, Phil. Trans. Roy. Soc. A. 215. p. 253. 1925.

— p. 95 —

first showed that the coefficient $\varrho_{\mathbf{g}}$ of the Fourier representation of the electron density directly gives the atomic form factor for the X-ray reflection ($h_1 h_2 h_3$). Wentzel¹⁾ arrived at the same result by wave-mechanical methods. On the basis of this fact the distribution of the electrons in crystals, particularly in alkali halides, has been empirically determined by various authors²⁾ from the measured intensities of the X-ray pattern. Recently James, Waller and Hartree³⁾



Anomaler Verlauf der in Fig. 6 dargestellten Kurven (Bezeichnungen siehe dort). Aufspaltung des Gebiets der Selektivreflexion (stark ausgezogene $\mathcal{H}$ -Achse)

Fig. 7

Fig. 7. Anomalous behavior of the curves shown in Fig. 6 (for notation see there). Splitting of the region of selective reflection (heavily drawn $\mathcal{H}$ -axis).

have established that the electron distribution in NaCl crystals determined empirically in this way is in good agreement with the distribution which Hartree⁴⁾ has computed theoretically by a wave-mechanical method developed by him. Furthermore a dynamical continuum theory of X-ray reflections has been worked out by Lohr⁵⁾ and Schlapp⁶⁾.

From the Fourier coefficients of the electron density, which determine the atomic form factor, one can now conclude directly to the Fourier coefficients of the potential decisive for electron diffraction. We have set

$$U(\mathbf{r}) = \frac{8\pi^2 me}{h^2} V(\mathbf{r}) = \sum_{\mathbf{g}} v_{\mathbf{g}} e^{2\pi i(\mathbf{g}\mathbf{r})} \quad (7)$$

or also

$$\mathcal{V}(\mathbf{r}) = \sum \mathcal{V}_{\mathbf{g}} e^{2\pi i(\mathbf{g}\mathbf{r})}, \quad \text{where} \quad \mathcal{V}_{\mathbf{g}} = \frac{h^2}{8\pi^2 me} v_{\mathbf{g}}. \quad (75)$$

But electrostatic potential and charge density are now linked by the well-known relation

$$\Delta V = 4\pi e(\varrho^E - \varrho^K), \quad (76)$$

where ϱ^E and ϱ^K denote the spatial charge density of the electrons and of the atomic nuclei. The charge is to be expressed in multiples of the elementary quantum. In a crystal

Footnotes: 1) G. Wentzel, Zeitschr. f. Phys. 43. p. 1. 1927. 2) Among others: Duane, Proc. Nat. Acad. Sci. 11. p. 489. 1925; Havighurst, ibid. p. 502, 507 and Phys. Rev. 29. p. 1. 1927; James and Miss Firth, Proc. Roy. Soc. A. 117. p. 62. 1927. 3) James, Waller and Hartree, Proc. Roy. Soc. A. 118. p. 334. 1928. 4) Hartree, Cambridge Phil. Soc. Proc. 24. p. 89, 111. 1928. 5) E. Lohr, Wiener Ber. 133. p. 517. 1924; 135. p. 655. 1926. 6) L. Schlapp, Phil. Mag. 1. p. 1009. 1926.

— p. 96 —

V , ϱ^E and ϱ^K can be expanded in Fourier series; the Fourier coefficient of the nuclear charge $\varrho_{\mathbf{g}}^K$ is thereby computed from the nuclear charge numbers Z_i and the coordinates $\mathfrak{R}_i$ of the basis atoms according to the equation

$$\varrho_{\mathbf{g}}^K = \frac{1}{(\mathfrak{a}_1[\mathfrak{a}_2\mathfrak{a}_3])} \sum_i Z_i e^{-2\pi i(\mathfrak{R}_i\mathbf{g})}.$$

The Fourier coefficient $\varrho_{\mathbf{g}}$ shall be known*, e.g. through intensity measurement of the X-ray reflection $(g_1 g_2 g_3)$. Then for the corresponding Fourier coefficient of the potential there simply follows

$$\mathcal{V}_{\mathbf{g}} = \frac{e}{\pi g^2} (\varrho_{\mathbf{g}}^K - \varrho_{\mathbf{g}}^E) \quad (77)$$

g is the absolute value of the vector $\mathbf{g}$ in the reciprocal lattice.

Despite the far-reaching analogy between the results of the dynamical theory of X-ray reflections and our theory of electron diffraction, and despite the relation just discussed between the fundamental

* Tr.: The source prints plain $\cdot g$; from context (X-ray intensity measurement) the electron-density coefficient $\cdot E_g$ is meant, as distinguished from $\cdot K_g$ above

quantities of both processes, there nevertheless exist far-reaching differences. The reason for this is actually only a quantitative one: The influence of the crystal fields on the electron waves is, at equal wavelength, much greater than that of the “oscillating dipoles” of the crystal on the X-rays. This already shows in the refractive index, which deviates from unity by some hundredths for electrons, by only millionths for X-rays. Just as v_0 is not negligible compared to $\mathfrak{K}^2$ in the expression for the refractive index

$$\mu = \sqrt{1 + \frac{V_0}{E}} = \sqrt{1 + \frac{v_0}{\mathfrak{K}^2}}$$

so the remaining Fourier coefficients $v_{\mathfrak{g}}$ also play a role compared with the resonance denominators $\mathfrak{K}^2 - k_{\mathfrak{g}}^2$ of the “weak” beams. This has the consequence that the de Broglie waves of the field in the crystal for which the resonance denominators are *not* small nevertheless still possess a not negligible intensity and can react back on the strong beams. This reaction is expressed in the “dynamical potential quantities” V_{11} , V_{12} , V_{22} . In the X-ray interference theory, on the other hand, the intensity of the waves $f_{\mathfrak{g}}$ decreases so rapidly with the distance of the corresponding reciprocal lattice point from the sphere of propagation that the intensity of the “weak beams”

— p. 97 —

becomes vanishingly small, no “reaction” need be taken into account. The same occurs for electrons of much smaller wavelengths, i.e. about in the region investigated by G. P. Thomson of $E = 15$ to 60 kV corresponding to $\lambda = 0.1$ to 0.05 Å.E. The sums in the potential quantities disappear, and in the case of a reflected beam $V_{11} = V_{22} = 0$ and $V_{12} = -v_{\mathfrak{h}}$.

The “reaction” of the weak beams on primary and secondary beam has first of all the consequence that the magnitude of the selectively reflectable wavelength is determined not alone by the refractive index, but in addition also by all Fourier coefficients of the potential $v_{\mathfrak{g}}$, i.e. by the whole potential variation. For X-rays and for electrons of very high velocity these quantities are already determined by the refractive index.

Loosely connected with the reaction is the small resolving power and large reflecting power of the crystal for de Broglie waves.

Furthermore, for electrons the expressions for the integrated reflecting power of a lattice plane $(h_1 h_2 h_3)$, i.e. for the reflected intensity on bombardment with electrons of a larger velocity range, become considerably more complicated. We found for this intensity in § 6

$$\int S_2 d\mathfrak{K} = \frac{\pi S_0 V_{12} \sqrt{\cos \vartheta}}{\mathfrak{K}(1 + \cos \vartheta)} = \frac{\pi S_0 \sqrt{\cos \vartheta}}{\mathfrak{K}(1 + \cos \vartheta)} \left(\sum_{\mathfrak{g}} \frac{v_{\mathfrak{g}} v_{\mathfrak{h}-\mathfrak{g}}}{\mathfrak{K}^2 - k_{\mathfrak{g}}^2} - v_{\mathfrak{h}} \right). \quad (62)$$

Here the “reaction” of the weak beams enters again. For X-rays, as mentioned above, the “integrated reflecting power” is simply proportional to $\varrho_{\mathfrak{h}}^E$. The same holds again for very fast electrons: V_{12} becomes there equal to $-v_{\mathfrak{h}}$.

The extinction laws too no longer agree under certain circumstances for electron and X-ray diffraction. The X-ray reflection $(h_1 h_2 h_3)$ is known to be extinguished when the corresponding point of the reciprocal lattice has weight zero (Ewald IV), i.e. when

$$\varrho_{\mathfrak{h}}^E = \varrho_{\mathfrak{h}}^K = v_{\mathfrak{h}} = 0$$

holds. Then of course the electron wave $\psi_{\mathfrak{h}}$

— p. 98 —

can no longer couple directly to the primary wave. For in the dynamical equation

$$\psi_{\mathfrak{h}}(\kappa^2 - k_{\mathfrak{h}}^2) = - \sum \psi_{\mathfrak{g}} v_{\mathfrak{h}-\mathfrak{g}}$$

the amplitude ψ_0 of the primary wave is afflicted with the factor $v_{\mathfrak{h}}$. But it can happen under certain circumstances that in the sum on the right-hand side a non-vanishing amplitude $\psi_{\mathfrak{g}}$ occurs with a coefficient $v_{\mathfrak{h}-\mathfrak{g}}$ different from zero, and that thereby $\psi_{\mathfrak{h}}$ becomes indirectly of finite magnitude. The electron reflection ($h_1 h_2 h_3$) will then not be extinguished even for vanishing Fourier coefficient $v_{\mathfrak{h}}$, if two coefficients $v_{\mathfrak{g}}$ and $v_{\mathfrak{h}-\mathfrak{g}}$ are simultaneously different from zero.

For the very simplest structures such a breaking of the extinction laws cannot of course come into question. For the base-centered lattice, e.g., X-ray reflections are extinguished when $h_1 + h_2$ is odd. Thus, in order that $v_{\mathfrak{g}} \neq 0$, $g_1 + g_2 \equiv 0 \pmod{2}$ must hold, and in order that $v_{\mathfrak{h}-\mathfrak{g}} \neq 0$, $(h-g)_1 + (h-g)_2 \equiv 0 \pmod{2}$ must hold. From this, however, follows $h_1 + h_2 \equiv 0 \pmod{2}$, hence $v_{\mathfrak{h}} \neq 0$. Here therefore no electron reflection can occur through the “reaction” of weak beams when the corresponding X-ray reflection is extinguished. The same holds e.g. for the face-centered and for the body-centered lattice.

If, on the other hand, extinction is observed for X-rays when a definite index or an index sum (or difference) is divisible by a certain number p , then this extinction law is in general broken for electron diffraction. Let e.g. ($h_1 h_2 h_3$) be extinguished when

$$h_1 \equiv rp \pmod{np},$$

where p denotes an integer > 1 , and r can run through one or several numbers in the series $0, 1 \dots n-1$.¹⁾ Let, on the other hand,

$$h_1 \not\equiv 0 \pmod{p}$$

Footnotes: 1) The two other indices h_2 and h_3 may thereby still be subject to fairly arbitrary conditions which do not interest us here. Exactly the same holds of course when e.g. the extinction law reads $h_1 + h_2 \equiv rp \pmod{np}$, etc.

— p. 99 —

— under the same conditions for h_2, h_3 — not extinguished. Then always

$$\begin{aligned} v_{\mathfrak{g}} &\neq 0, & \text{if } g_1 &\not\equiv 0 \pmod{r}, \\ v_{\mathfrak{h}-\mathfrak{g}} &\neq 0, & \text{if } (h-g)_1 &\not\equiv 0 \pmod{p}, \end{aligned}$$

i.e. for instance

$$(h-g)_1 \equiv rp - g_1 \pmod{np}.$$

The electron reflection $\mathfrak{h}$ thus occurs although $h_1 \equiv rp \pmod{np}$, i.e. $v_{\mathfrak{h}} = 0$.

An example of this is e.g. diamond. There are extinguished 1. all reflections with “mixed” (even and odd) indices, 2. all reflections for which $h_1 + h_2 + h_3 \equiv 2 \pmod{4}$.

The second law is broken for electrons, for e.g.

$$v_{\mathbf{g}} \neq 0, \quad \text{if} \quad g_1 + g_2 + g_3 \equiv 1 \pmod{4}$$

and

$$v_{\mathbf{h}-\mathbf{g}} \neq 0, \quad \text{if} \quad (h-g)_1 + (h-g)_2 + (h-g)_3 \equiv 1 \pmod{4}.$$

$\psi_{\mathbf{h}}$ thus occurs even when $h_1 + h_2 + h_3 \equiv 2 \pmod{4}$.¹⁾

SiC (carborundum II) behaves similarly. Here the X-ray diffraction $h_1 - h_2 \equiv 0 \pmod{3}$ is extinguished [unless $h_3 \equiv 0 \pmod{6}$, which is however irrelevant]. This extinction too no longer takes place in electron diffraction.

On the other hand all extinction laws

$$h_1 \not\equiv 0 \pmod{p}$$

and similar ones remain in force. For here $v_{\mathbf{g}}$ and $v_{\mathbf{h}-\mathbf{g}}$ are non-zero only when $g_1, (h-g)_1$ and consequently also h_1 are divisible by p . In this case electron interference beams can only arise when $v_{\mathbf{h}} \neq 0$, i.e. when they already arise directly from the primary beam.

For high electron velocities the “indirect reflection” due to weak-beam back-coupling

Footnotes: 1) It is not inconceivable that exactly the corresponding phenomenon is partly responsible for the imperfect extinction of the X-ray reflection (222) in diamond. Since diamond is an ideal crystal, the small influence of the “weak” beams e.g. (111), ($\bar{1}\bar{1}1$), etc., may suffice to produce the reflection (222) “indirectly”. *Translator’s notes on symbols: (i) The symbol printed as a swash italic z (e.g. in $\xi = 2rz na_0$ and in $(3Nz \ 8\pi)$) is transcribed exactly as printed; in context it plays the role of the effective nuclear charge / number of conduction electrons. (ii) In eq. (81) the subscripts of the last terms of G^2 and $\bar{g}$ are barely legible at this scan resolution (they look like “2”); they are transcribed as g_3 in agreement with the parallel definitions (80) and (80a) and the stated permutation symmetry of the three components. (iii) In the W_i equation the original prints z in $(3Nz \ 8\pi)$ but a capital $Z^{2/3}$ in the second half; the surrounding text denotes the number of conduction electrons per atom by z .*

— p. 100 —

is of course no longer relevant. At low velocities it may be observed, provided a conducting crystal with a suitable structure is found. However, the inelastic collisions strongly attenuate all secondary beams $\psi_{\mathbf{g}}$, and this attenuation occurs twice for the “indirectly reflected” beams, since (in Bragg’s picture) they must be thought of as arising from twofold lattice-plane reflection of the primary beam. Their intensity will therefore in practice be considerably smaller than that of the directly reflected beams, although according to the theory developed here it should be of about the same intensity as these.

§ 9. Symmetric Reflection. Coordinate Transformation

In the experiments of Davisson and Germer the primary beam strikes an octahedral face of the face-centered cubic nickel crystal perpendicularly, i.e. in the direction of a *threefold axis*. Moreover, three mirror planes rotated by 60° with respect to one another are determined by the same crystal normal as well as by the altitudes of the equilateral triangles formed by the atoms lying in the octahedral face. As a consequence of these symmetry properties of the crystal with respect to the primary beam, reflected beams in the experiments occur not singly, but in *three* or *six* mutually symmetric directions.

Three symmetric reflection directions come into question for the reflections in the (111) and (100) azimuths. They arise from one another by rotation through 120° about the primary beam as axis. In this case the beams lie in the aforementioned *mirror planes*. In all other azimuths six beams always occur simultaneously, three of which arise from one another by rotation through multiples of 120° , while one triple is obtained from the other by reflection in one of the mirror planes. The reflection in the (110) azimuth, observed by Davisson and Germer alone besides those in the (111) and (100) azimuths, is distinguished by the fact that all beams run *perpendicular* to one mirror plane each; consequently all six symmetric beams go over into one another by rotations through multiples of 60° about the primary beam.

The intuitively clear fact of reflection in several symmetric directions for the case that the primary beam

— p. 101 —

falls along a symmetry axis can also be understood computationally from the theory developed here. For this one uses the fact that

$$U(\mathbf{r}) = \frac{8\pi^2 m e}{h^2} V(\mathbf{r}) = \sum_g v_g e^{2\pi i(\mathbf{g}\mathbf{r})} \quad (7)$$

goes over into itself when the position vector $\mathbf{r}$ is rotated through the angle $2\pi/n$ about an n -fold rotation axis. For then the two coefficients v_g and v_h must be equal, if g arises from h by a rotation about the corresponding symmetry axis of the reciprocal lattice, likewise through $2\pi/n$ or a multiple thereof. Likewise a symmetry plane involves the equality of two coefficients v_g and v_h each, when the vector g is carried over into h by reflection in the symmetry plane (in the reciprocal lattice).

On the basis of these symmetry properties of the Fourier coefficients it is easy to show that the dynamical potential quantities for all symmetrically reflected beams are equal. In the derivation of equations (36) of § 4 we must now assume the $n + 1$ amplitudes

$$c_1, c_2^1, c_2^2 \dots c_2^n$$

of the primary and the n secondary beams to be large, instead of only two as there. Then for the amplitude of a weak beam we obtain from the dynamical equations (13), in place of the expression (35), now:

$$\psi_g = - \frac{c_1 v_g + \sum_i c_2^i v_{g-h_i}}{\kappa^2 - k_g^2} \quad (35a)$$

and if we insert this into the dynamical equations for the primary beam and the secondary beams

$$\begin{aligned} c_1(\kappa^2 - k_0^2) &= - \sum \psi_g v_{-g} \\ c_2^i(\kappa^2 - k_{h^i}^2) &= - \sum \psi_g v_{h^i-g} \end{aligned}$$

we obtain

$$\begin{cases} c_1(2\kappa\varepsilon_1 - \mathcal{V}_{11}) - \sum_{i=1}^n c_2^i \mathcal{V}_{12}^i = 0 \\ -c_1 \mathcal{V}_{12}^i + c_2^i 2\kappa\varepsilon_2^i - \sum_{j=1}^n \mathcal{V}_{22}^{ij} c_j = 0 \quad \text{for all } i(1, 2, \dots, n). \end{cases} \quad (36b)$$

— p. 102 —

Here — under the assumption of a center of symmetry —

$$\begin{cases} \mathcal{V}_{11} = \sum \frac{v_g^2}{\kappa^2 - k_g^2} \\ \mathcal{V}_{12}^i = \sum \frac{v_g v_{h^i-g}}{\kappa^2 - k_g^2} - v_{h^i} \\ \mathcal{V}_{22}^{ij} = \sum \frac{v_{g-h^i} v_{g-h^j}}{2\kappa^2 - k_g^2} - v_{h^i-h^j} \quad \text{for } i \neq j \\ \mathcal{V}_{22}^{ii} = \sum \frac{v_{g-h^i}}{\kappa^2 - k_g^2} \end{cases} \quad (37b)$$

It then follows from the symmetry of the problem and can also be shown computationally that $\mathcal{V}_{12}^i$ as well as $\sum_j \mathcal{V}_{22}^{ij}$ are equal for all i . The secondary waves all have the same amplitude. If we further set

$$\begin{cases} \varepsilon_2 = \varepsilon_2^i & c = c_2^i \sqrt{n} \\ \mathcal{V}_{12} = \mathcal{V}_{12}^i \sqrt{n} & \mathcal{V}_{22} = \sum_{j=1}^n \mathcal{V}_{22}^{ij} \end{cases} \quad (78)$$

the equations (36b) go over exactly into the equations

$$\begin{cases} c_1(2\kappa\varepsilon_1 - \mathcal{V}_{11}) - c_2 \mathcal{V}_{12} = 0 \\ -c_1 \mathcal{V}_{12} + c_2(2\kappa\varepsilon_2 - \mathcal{V}_{22}) = 0 \end{cases} \quad (36)$$

as we derived them earlier for *one* reflected beam.¹⁾ All the considerations put forward there about the dispersion equation etc. thus remain the same,

Footnotes:

- 1) The corresponding considerations no longer hold when the Laue equations are accidentally fulfilled simultaneously for two reflected beams. Belonging here are e.g. the third-order reflections in the $\{111\}$ and $\{100\}$ azimuths, which according to the elementary diffraction theory should always occur pairwise at the same wavelength, but have been observed for electrons

at different wavelengths (Davisson and Germer II, p. 733). There the potential quantities $\mathcal{V}_{12}$ and $\mathcal{V}_{22}$ are different. The following example shows this: The reflections $\mathfrak{h}' = (660)$ and $\mathfrak{h}'' = (822)$ are supposed to occur simultaneously. It is also true that $v_{h'} = v_{h''}$, since the absolute values of h , which for spherically symmetric atoms alone determine the coefficients v_h (§ 11), are both equal

$$\sqrt{\frac{36}{a^2} + \frac{36}{a^2}} = \sqrt{\frac{64}{a^2} + \frac{4}{a^2} + \frac{4}{a^2}} = \frac{1}{a}\sqrt{72}$$

But e.g. for $\mathfrak{g} = (911)$ [we have]

$$\begin{aligned} \mathfrak{h}'_1 - \mathfrak{g} &= (6 - 9, 6 - 1, 0 - 1) = (\bar{3}51) & \mathfrak{h}'_2 - \mathfrak{g} &= (5\bar{3}1) & \mathfrak{h}'_3 - \mathfrak{g} &= (559) \\ \mathfrak{h}''_1 - \mathfrak{g} &= (\bar{1}11) & \mathfrak{h}''_2 - \mathfrak{g} &= (\bar{7}71) & \mathfrak{h}''_3 - \mathfrak{g} &= (7\bar{1}7) \end{aligned}$$

The potential quantities $\mathcal{V}_{12}$ and $\mathcal{V}_{22}$ are different for the two reflected beams. It is therefore understandable that the two diffraction beams no longer coincide for electrons.

— p. 103 —

the considerations put forward remain the same, and the intensity of all reflected beams together also remains the same as before, while of course only $1/n$ of the total reflected intensity now falls on each single one according to (78). Only the expressions for the dynamical potential quantities are changed substantially, which we write symmetrically:

$$\begin{aligned} \mathcal{V}_{11} &= \sum_g \frac{v_g^2}{\kappa^2 - k_g^2} \\ \mathcal{V}_{12} &= \frac{1}{\sqrt{n}} \sum_g \frac{v_g \sum_{i=1}^n v_{h^i - g}}{\kappa^2 - k_g^2} - \frac{1}{\sqrt{n}} \sum_{i=1}^n v_{h^i} \\ \mathcal{V}_{22} &= \frac{1}{n} \sum_g \frac{(\sum_{i=1}^n v_{h^i - g})^2}{\kappa^2 - k_g^2} - \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n v_{h^i - h^j} . \end{aligned} \tag{79}$$

The prime means that $i = j$ (v_0) is to be assumed.

We can now calculate the dynamical potential quantities in the special case of the Davisson and Germer experiments, but must note for this that the cubic axes of the nickel crystal¹⁾ are directed out of the crystal at equal angles to the (111) surface of the crystal. The symmetrically reflected beams then arise from one another by permutation of the three reflection indices. In general this yields 6 symmetrically reflected beams; only when the reflection occurs in the (100) or (111) azimuth do 2 reflection indices become equal and one obtains only 3 reflected beams by permutation (cf. beginning of this section). The primary wave is directed into the crystal perpendicularly, its propagation vector $\mathfrak{f}_0$ therefore has the component $-k_0/\sqrt{3}$ along each axis. For the case of exact fulfillment of the Laue equations one therefore obtains for the resonance denominators (cf. § 7, eq. 70)

Footnotes:

- 1) In the first preliminary communication (Naturwissenschaften 15, p. 787, 1927) I used other coordinates. — p. 104 —

$$\begin{cases} -4\pi(\mathbf{g}, \mathbf{e}_0 + \pi\mathbf{g}) = \frac{4\pi}{a\sqrt{3}} \varkappa_0 (g_1 + g_2 + g_3) \\ -\frac{4\pi^2}{a^2} (g_1^2 + g_2^2 + g_3^2). \end{cases} \quad (80)$$

Here a is the side of the elementary cube, and $\varkappa_0$ is determined by the vanishing of the resonance error for the secondary beam

$$\frac{4\pi}{a\sqrt{3}} \varkappa_0 (h_1 + h_2 + h_3) = \frac{4\pi^2}{a^2} (h_1^2 + h_2^2 + h_3^2).$$

Thus:

$$\varkappa_0 = \frac{\pi\sqrt{3}}{a} \frac{h_1^2 + h_2^2 + h_3^2}{h_1 + h_2 + h_3} \quad (80a)$$

in agreement with eq. (6) of the first section. We further introduce the abbreviations:

$$\begin{cases} G^2 = g_1^2 + g_2^2 + g_3^2 \\ \bar{g} = g_1 + g_2 + g_3 \\ \varkappa = \frac{h_1^2 + h_2^2 + h_3^2}{h_1 + h_2 + h_3} = \frac{H^2}{\bar{h}} \\ \bar{v}_{g-h} = \text{sum of all } v_{g-h} \text{ arising by permutation of the three components of } g \text{ among themselves.} \end{cases} \quad (81)$$

Then

$$\begin{cases} \sigma_{11} = \frac{a^3}{16\pi^3\sqrt{3}} \sum_g \frac{v_g^2(3\varkappa - 2\bar{g})}{(\varkappa\bar{g} - G^2)^2} \\ \sigma_{12} = \frac{a^3\sqrt{n}}{16\pi^3\sqrt{3}} \sum_g \frac{v_g \bar{v}_{h-g}(3\varkappa - 2\bar{g})}{(\varkappa\bar{g} - G^2)^2} \\ \sigma_{22} = \frac{a^3n}{16\pi^3\sqrt{3}} \sum_g \frac{(\bar{v}_{h-g})^2(3\varkappa - 2\bar{g})}{(\varkappa\bar{g} - G^2)^2} \\ \tau_{11} = \frac{a^2}{8\pi^2} \sum_g \frac{v_g^2}{\varkappa\bar{g} - G^2} - \frac{\zeta}{1 + \cos\vartheta} \frac{a^3}{8\pi^3\sqrt{3}} \sum_g \frac{v_g^2 \bar{g}}{(\varkappa\bar{g} - G^2)^2} \\ \tau_{12} = -\frac{\sqrt{n}}{2} v_h + \frac{a^2\sqrt{n}}{8\pi^2} \sum_g \frac{v_g \bar{v}_{h-g}}{\varkappa\bar{g} - G^2} - \frac{\zeta}{1 + \cos\vartheta} \frac{a^3\sqrt{n}}{8\pi^3\sqrt{3}} \sum_g \frac{v_g \bar{v}_{h-g} \bar{g}}{(\varkappa\bar{g} - G^2)^2} \\ \tau_{22} = -\sum_{i=2}^n v_{h^1-h^i} + \frac{a^2n}{8\pi^2} \sum_g \frac{\bar{v}_{h-g}^2}{\varkappa\bar{g} - G^2} - \zeta \left(\frac{1}{1 + \cos\vartheta} \frac{a^3n}{8\pi^3\sqrt{3}} \sum_g \frac{\bar{v}_{h-g}^2 \bar{g}}{(\varkappa\bar{g} - G^2)^2} + \varkappa \right). \end{cases} \quad (82)$$

§ 10. Calculation and Discussion of the Work Function

We have developed the theory of crystal diffraction of electrons up to this point without being able to state the numerical magnitude of the effects to be expected. In doing so we have advanced from the purely wave-kinematical theory of § 1 to a dynamical theory and have seen that the *mean potential* V_0 in the crystal is responsible for the crystal possessing a *refractive index* for de Broglie waves

$$\mu = \sqrt{1 + \frac{V_0}{E}}$$

whereas the detailed course of the potential causes all the finer phenomena. We shall now remedy this omission, and on the basis of a wave-mechanical picture of the fields of the crystal atoms arrive first at an estimate of the mean potential V_0 , later also at a determination of the higher Fourier coefficients V_h of the potential.

The *mean potential* V_0 is the absolute term of the Fourier expansion of the electrostatic potential V in the crystal¹⁾ and is calculated as the space integral over the potential $V(\mathbf{r})$ prevailing at each point of the unit cell of the crystal, divided by the volume of the unit cell:

$$\bar{V}_0 = \frac{1}{(a_1[a_2a_3])} \int_{\text{El.-Z.}} V(\mathbf{r}) d\tau. \quad (83)$$

To the potential at the field point $\mathbf{r}$ contribute -- at least for the neutral atoms with which we have to deal -- the greatest part from the unit cell in which the field point lies. Nevertheless, for generality's sake, we also want to take into account the contributions of the surrounding cells to the potential of the one we consider. Now all unit cells are built alike. If we denote the unit cell we consider by 0, and by $\mathbf{R}$ the vector by which they arise from cell 0, then it is clear that the charge distribution of cell 0 produces exactly the same potential in cell $\mathbf{R}$ as cell $-\mathbf{R}$ does in cell 0. We can thus replace the integration

Footnotes:

- 1) The Fourier expansion of V differs from our expansion (7) by the factor $\hbar^2/8\pi^2me$.

— p. 106 —

over the potential exerted by *all* the atoms of the crystal on *one* unit cell by an integration over the potential produced by the atoms of *one* unit cell in the *whole* crystal. I.e., we take into account the influence of all crystal atoms by using only the atoms of one unit cell for the calculation of the potential $V(\mathbf{r})$, and integrating $V(\mathbf{r})$ over infinite space instead. If the crystal consists of neutral atoms, this integral can again be conceived as the sum of the integrals over the potentials arising from the individual atoms¹⁾:

$$\bar{V}_0 = \sum_i \int_0^\infty V_i(\mathbf{r}) d\tau.$$

Here $V_i(\mathbf{r})$ denotes the potential exerted by the i th atom of the basis on a field point whose position vector, referred to the nucleus of this very atom, is $\mathbf{r}$.

We have already demanded in § 1, from the consideration that the metal crystal must not become depleted of electrons, $V_0 > 0$. We now see the theoretical reason for this: For the field at a point $\mathbf{r}$ lying inside the charge cloud of a neutral atom²⁾, all those electrons drop out which are outside a sphere about the nucleus of radius $r = |\mathbf{r}|$. The atomic residue determining the field is then always positively charged, because the atomic nucleus is not completely neutralized by the part of the electron cloud inside the sphere r . An atom that is electrically neutral outward behaves, with respect to the electric field at a field point in its interior, like a point charge concentrated in the nucleus, whose magnitude decreases with increasing distance of the field point from the nucleus, but always remains positive. The electrostatic potential in the interior of a neutral atom, and likewise the mean potential V_0 in a crystal composed of neutral atoms, is therefore always positive, in agreement with our earlier consideration.

Footnotes:

- 1) For *ions* the space integrals over the potentials of the individual ions no longer converge, but only the integral over the potential arising from the whole — neutral — unit cell.
- 2) The following considerations refer, strictly speaking, only to spherically symmetric atoms (cf. the simplifying assumptions further below); if one goes far enough into the interior, they of course hold generally, so that the conclusion for V_0 probably holds for every crystal of neutral atoms.

— p. 107 —

In order to actually calculate the mean potential V_0 in a crystal, we must now make rather far-reaching idealizations:

1. We assume the individual atoms of the crystal to be *spherically symmetric*, as we have already done in our qualitative considerations above. For the closed shells of free¹⁾ atoms this assumption holds exactly according to Unsöld²⁾, and it will still conform to the actual conditions very extensively for the inner, tightly bound electrons even under the influence of the (not spherically symmetric) fields of the remaining crystal atoms. The *outer* electrons will certainly be deformed more strongly by these fields, but because of the rapid falloff of the fields of neutral atoms one will be able to justify the assumption of spherically symmetric distribution even here, especially for very symmetric crystals.
2. We assume *hydrogen eigenfunctions* for the electrons. We take into account the core electrons inside a given closed electron shell by *screening numbers*, which we take from the work of L. Pauling³⁾ “Electron distribution in many electron atoms”. The use of a continuous distribution of the electrons over the atom does not have the defects of the method of concentrating the charge of each closed shell on a spherical shell of suitably chosen radius (entirely obscure values of the Fourier coefficients, which are extraordinarily sensitive to the choice of the radii of the spherical shells). However, the distribution of the electrons precisely according to hydrogen eigenfunctions certainly does not correspond to the true conditions in complicated atoms, because it assumes that the electron moves in the potential field of a point charge, i.e. that the “effective nuclear charge” is a constant throughout the region over which the charge of the electron is distributed. In reality, however, the “effective nuclear charge” decreases strongly and continuously from the nucleus outward. We can therefore expect here only

Footnotes:

- 1) What matters is that the *field* to which the electrons are subject is a *central field*.
- 2) A. Unsöld, Ann. d. Phys. 82, p. 355, 1927.
- 3) L. Pauling, Proc. Roy. Soc. 114, p. 181, 1927.

— p. 108 —

an approximate estimate of the mean potential V_0 .¹⁾

3. For lack of more precise knowledge of the distribution of the *conduction electrons* not bound to the individual atoms, we assume them to be distributed uniformly over the whole crystal. We ascribe to each atom as many conduction electrons as neutralize the nuclear charge together with the bound electrons. In doing so we further replace the parallelepiped which we can delimit as the “domain of a single atom” by the sphere of equal volume, and treat the conduction electrons to be counted to the atom as distributed uniformly over this sphere.

We first use assumption 1 (spherically symmetric distribution of the electrons about the nucleus) without going into the details of this distribution for the time being. If $\varrho(r)r^2dr$ is the number of electrons contained in a spherical shell of radius r , and Z the nuclear charge, then according to known formulas of electrostatics the potential at distance R from the nucleus is:

$$V(R) = \frac{Ze}{R} - \frac{e}{R} \int_0^R \varrho(r)r^2 dr - e \int_R^\infty \frac{\varrho(r)r^2 dr}{r}.$$

Now the atom as a whole shall be neutral, i.e. the total number of electrons shall equal the nuclear charge:

$$\int_0^\infty \varrho(r)r^2 dr = Z.$$

With this

$$V(R) = e \int_R^\infty \varrho(r)r^2 dr \left(\frac{1}{R} - \frac{1}{r} \right). \quad (84)$$

If now the crystal consists of entirely equal atoms, of which ν are contained in the unit cell, then from formula (83a) for the mean potential, with regard to the spherically symmetric potential distribution:

Footnotes:

- 1) I hope to be able to report soon on more precise calculations of the mean potential according to a method developed by Hartree (Cambridge Phil. Soc. Proc. 24, p. 89) (111), which for the calculation of the electron distribution takes into account the actual field arising from the interplay of nuclear and electron charge.

$$\bar{V}_0 = \frac{\nu}{(a_1[a_2a_3])} \int_0^\infty 4\pi R^2 V(R) dR. \quad (83b)$$

Here $\nu/(a_1[a_2a_3])$ denotes the number N of atoms per unit volume. If we now insert the expression (84) for the potential into (83b):

$$\bar{V}_0 = 4\pi N \left[\int_0^\infty R dR \int_R^\infty \varrho(r) r^2 dr - \int_0^\infty R^2 dR \int_R^\infty \varrho(r) r dr \right],$$

we can interchange the integration variables:

$$\bar{V}_0 = 4\pi N \left[\int_0^\infty \varrho(r) r^2 dr \int_0^r R dR - \int_0^\infty \varrho(r) r dr \int_0^r R^2 dR \right]$$

and obtain as the result:

$$\bar{V}_0 = \frac{2\pi}{3} N \int_0^\infty \varrho(r) r^4 dr. \quad (85)$$

Since $\varrho(r)r^2dr$ is the number of electrons at distance r from the nucleus, the mean potential V_0 represents something similar to the *moment of inertia* of the electrons of the atom.

We now insert the actual electron density $\varrho(r)/4\pi$ in the atom. It is equal to the sum of the densities which the individual electrons have at distance r from the nucleus. The integral (85) can thus be calculated as a sum of similarly constructed integrals over the densities of the individual electrons. For an electron with principal quantum number n and azimuthal quantum number l , the 4π -fold density is now however simply given by the square of the r -dependent part of the Schrödinger hydrogen eigenfunction:

$$\varrho_{nl}(r) = X_{nl}^2 = C_{nl} e^{-\xi} \xi^{2l} [L_{n+l}^{2l+1}(\xi)]^2.$$

Here

$$\xi = \frac{2rz}{na_0}$$

and

$$a_0 = \frac{h^2}{4\pi^2 m e^2}$$

is the radius of the “innermost orbit” of the Bohr hydrogen model. For the contribution of the n, l -electron to the mean potential V_0 ,

$$\bar{V}_0^{nl} = \frac{2\pi}{3} N \int_0^\infty \varrho_{nl} r^4 dr$$

— p. 110 —

we make use of the method given by Waller¹⁾ for the evaluation of such integrals, which rests on the representation of the Laguerre polynomials $L_{n+l}^{2l+1}(\xi)$ by a *generating function*:

$$\sum \frac{L_p^r(\xi)}{p!} t^p = (-1)^r \frac{t^r}{(1-t)^{r+1}} e^{-\frac{\xi t}{1-t}}. \quad (86)$$

Two such representations multiplied and integrated give, with $r = 2l + 1$,

$$\left\{ \begin{aligned} & \sum \frac{t^{p_1} u^{p_2}}{p_1! p_2!} \int_0^\infty \xi^{s+2l+2} e^{-\xi} L_{p_1}^{2l+1}(\xi) L_{p_2}^{2l+1}(\xi) d\xi \\ &= \frac{(ut)^{2l+1}}{[(1-u)(1-t)]^{2l+2}} \int_0^\infty \xi^{s+2l+2} e^{-\xi \frac{1-ut}{(1-u)(1-t)}} d\xi \\ &= (s+2l+2)! \frac{[(1-u)(1-t)]^{s+1}}{(1-ut)^{2l+s+3}} (ut)^{2l+1}. \end{aligned} \right. \quad (86a)$$

The integral

$$\int \xi^{s+2l+2} e^{-\xi} [L_{n+l}^{2l+1}(\xi)]^2 d\xi$$

is therefore equal to the coefficient of $(ut)^{n-l-1}$ in the expansion of the expression

$$(n+l)!^2 (s+2l+2)! \frac{[(1-u)(1-t)]^{s+1}}{(1-ut)^{s+2l+3}}.$$

From this Waller obtains for $s = 0$ the *normalization factors* for the Schrödinger eigenfunctions

$$C_{nl}^2 = \left(\frac{z}{na_0} \right)^3 \frac{(n-l-1)!}{(n+l)!^3 \cdot 2n}.$$

For our expression (85a) we obtain with $s = 2$:

$$\begin{aligned} \frac{2\pi}{3} N \int_0^\infty r^4 \varrho(r) dr &= \frac{2\pi}{3} N \left(\frac{na_0}{z} \right)^2 \frac{1}{2n} \\ &\quad \cdot \frac{(n-l-1)!}{(n+l)!^3} \int_0^\infty e^{-\xi} \xi^{2l+4} [L_{n+l}^{2l+1}(\xi)]^2 d\xi \\ &= \frac{\pi N}{3n} \left(\frac{na_0}{z} \right)^2 \frac{(n-l-1)!(2l+4)!}{(n+l)!} P_{nl}, \end{aligned}$$

where P_{nl} is the coefficient of $(ut)^{n-l-1}$ in the expansion

Footnotes:

1) I. Waller, Ztschr. f. Phys. 38, p. 635 (especially p. 644). — p. 111 —

$$\frac{[(1-u)(1-t)]^3}{(1-ut)^{2l+5}} = (1+ut-(u+t))^3 \sum_k (ut)^k \binom{2l+4+k}{k}$$

is. Since we need only pay attention to terms containing u and t in the same power, we may just as well consider the expansion

$$\begin{aligned} & [(1+ut)^3 + 3(1+ut) \cdot 2ut] \sum_k (ut)^k \binom{2l+4+k}{k} \\ &= (1+9ut+9u^2t^2+u^3t^3) \sum_k (ut)^k \binom{2l+4+k}{k} \end{aligned}$$

and then obtain for the coefficient sought:

$$P_{nl} = \binom{n+l+3}{n-l-1} + 9 \binom{n+l+2}{n-l-2} + 9 \binom{n+l+1}{n-l-3} + \binom{n+l}{n-l-4}.$$

By evaluating the binomial coefficients the contribution of an n, l -electron to the mean potential finally becomes:

$$\bar{V}_{nl}^0 = \pi N e \left(\frac{na_0}{z} \right)^2 \left(\frac{5n^2+1}{3} - l(l+1) \right). \quad (87)$$

The *conduction electrons* shall, according to our assumption, be distributed *uniformly* over a sphere whose volume is equal to the volume falling on each atom in the crystal:

$$\frac{4\pi}{3} R_0^3 = \frac{1}{N}.$$

The charge density of the conduction electron is:

$$\frac{\rho}{4\pi} = \frac{e}{\frac{4\pi}{3} R_0^3}.$$

Accordingly the contribution of each conduction electron to the mean potential becomes

$$\bar{V}_L^0 = \frac{\frac{2\pi}{3} N e}{\frac{4}{3} R_0^3} \int_0^{R_0} r^4 dr = \frac{2\pi}{5} N e R_0^2 = \frac{3}{10} \frac{e}{R_0}. \quad (88)$$

To calculate V_0 the contributions of all — bound and free — electrons are to be summed.

In the example of the nickel used by Davisson and Germer we want to carry out the calculation. Free Ni atoms contain, according to the evidence of the optical and X-ray spectra, besides the closed K, L, M I and M II shells, 8 M III electrons and two N I electrons. The orbit of the latter is, however, so large that it would no longer fit into the lattice

at all. We therefore assume that these two N electrons are split off from the atomic association and are freely mobile in the crystal as conduction electrons. The number N of atoms per cubic centimeter is for Ni $0.92 \cdot 10^{23}$, so that

$$\pi N e a_0^2 = 3.14 \cdot 0.92 \cdot 10^{23} \cdot 4.77 \cdot 10^{-10} \cdot (0.532)^2 \cdot 10^{-16} = 0.00390.$$

With this constant we would obtain V_0 in electrostatic units; by multiplication by 300 we obtain it in volts:

$$V_{nl}^0 = 1.17 \cdot \frac{n^2}{z} \left(\frac{5n^2 + 1}{3} - l(l+1) \right).$$

The radius R_0 determining the contribution of the conduction electrons follows from

$$\frac{4\pi}{3} R_0^3 = 1.09 \cdot 10^{-23}$$

as $R_0 = 1.38 \cdot 10^{-8}$ cm.

With these values and the screening numbers of L. Pauling¹⁾ the mean potential V_0 is calculated according to the following scheme:

Quantum numbers		Number of electrons	Effective nuclear charge	Contribution to V_0
n	l			
1	0	2	28	$2 \cdot 1.17 \cdot \frac{1}{28^2} \cdot 2 = 0.006$
2	0	2	23.5	$2 \cdot 1.17 \cdot \frac{4}{23.5^2} \cdot 7 = 0.12$
2	1	6	23.5	$6 \cdot 1.17 \cdot \frac{4}{23.5^2} \cdot 5 = 0.25$
3	0	2	17.1	$2 \cdot 1.17 \cdot \frac{9}{17.1^2} \cdot 15\frac{1}{3} = 1.10$
3	1	6	14.85	$6 \cdot 1.17 \cdot \frac{9}{14.85^2} \cdot 13\frac{1}{3} = 3.83$
3	2	8	10.3	$8 \cdot 1.17 \cdot \frac{9}{10.3^2} \cdot 9\frac{1}{3} = 7.40$
Conduction		2		$2 \cdot 0.3 \cdot \frac{300 \cdot 4.77 \cdot 10^{-10}}{1.38 \cdot 10^{-8}} = 6.22$
				$V_0 = 18.93$ volts

Regarding this value of V_0 it is first to be remarked that according to the investigations of Hartree²⁾ the screening numbers given by L. Pauling

Footnotes:

1) L. Pauling, l. c.

2) Hartree, l. c.

— p. 113 —

yield too large extensions of the electron shells. Also the *conduction electrons* will presumably not, as we assumed, be distributed uniformly over the whole crystal, but be more strongly concentrated in the neighborhood of the crystal atoms. But since our mean potential V_0 is proportional to the mean value of the square of the distance r^2 , averaged over the whole electron cloud of the atom, we have certainly obtained *too large* a value for V_0 . On the other hand we have neglected the fact that the Davisson-Germer electron penetrating the crystal from outside polarizes the crystal atoms and thereby somewhat increases the potential $V(\mathbf{r})$ at each point of the crystal. But this should hardly carry weight against the first-mentioned fact. It is therefore *probable that the exactly calculated theoretical value of the mean potential lies near the experimental mean value of 14.8 volts.*¹⁾

It has already been pointed out earlier by Rosenfeld and Witmer²⁾ and independently by me³⁾ that the mean electrostatic potential V_0 in the crystal must *agree* with the *work function* W_a of the Richardson effect in Sommerfeld's notation.⁴⁾ Perhaps only one more explanation is needed as to why in the electron diffraction experiments W_a itself enters and not, as in the *Richardson effect*, the *difference* between W_a and the energy W_i of the conduction electrons moving with the limiting velocity of Fermi statistics: In the experiments of Davisson and Germer, Thomson, Rupp the energy E of the incident electrons has nothing at all to do with the energy of the conduction electrons. E is simply increased by the potential difference $V_0 = W_a$ upon entering the crystal. In the Richardson effect, on the other hand, W_i represents, so to speak, the kinetic energy of the conduction electrons in the crystal, and what matters is the *difference* of this energy from the potential threshold to be overcome upon exit, i.e. $W_a - W_i$.

Footnotes:

- 1) L. Rosenfeld and E. Witmer (Ztschr. f. Phys. 49, p. 736, 1928) also used newer experiments by Davisson and Germer, as well as the experiments by Rupp, and find the value 18 volts for V_0 .
- 2) L. Rosenfeld and E. Witmer, l. c.
- 3) H. Bethe II, l. c.
- 4) A. Sommerfeld, Ztschr. f. Phys. 47, p. 1 (especially p. 27ff.).

[Printer's signature at foot of page: Annalen der Physik. IV. Folge. 87. — 8]

— p. 114 —

By means of the work function W_a to be taken from electron diffraction experiments and the constant of the Richardson effect $W_a - W_i$, which always lies around 4 volts, the energy W_i of the conduction electrons can be calculated, which stands in close relation to the number z of conduction electrons split off per atom:

$$W_i = \frac{h^2}{2m} \left(\frac{3Nz}{8\pi} \right)^{2/3} = 7.4 \cdot Z^{2/3} \text{ volts (for Ni).}$$

On the basis of this relation Rosenfeld and Witmer have calculated the number of conduction electrons per atom for the metals investigated by Davisson and Germer as well as by Rupp.¹⁾ From our value $W_a = 14.8$ volts and the Richardson constant $W_a - W_i = 4$ volts follows $W_i = 10.8$

volts and with it about $z = 2$ conduction electrons per atom²⁾ in agreement with our consideration that probably the two N electrons of the free Ni atom are split off in the crystal. The complete breakdown of all unclosed shells in the crystal association, put forward by Houston³⁾ as possible to explain the conductivity of metals, would mean for Ni the splitting off of *ten* (8 MIII and 2 NI) electrons. From this, however, the extraordinarily high amount of 34 volts would result for W_i , which cannot be reconciled with the experimental results of Davisson and Germer.

The value of W_a calculated in this section from the *atomic fields* of course also depends on the assumed number of conduction electrons per atom. For nickel, for example, it would increase by the *difference* of the contributions of a conduction electron (3.11 volts) and one of the most weakly bound (MIII) electrons (0.83 volts) to V_0 for each further split-off electron, i.e. each time by 2.28 volts. The kinetic energy of the [electrons] moving with the “limiting velocity”

Footnotes:

- 1) Cf. the Table IV given in the work of Rosenfeld and Witmer, l. c. Remarkable is the large number of conduction electrons for Ag (3), but also that Cu and Au split off not merely *one* conduction electron per atom.
- 2) Rosenfeld and Witmer obtain with $W_a = 18$ volts two or three conduction electrons per Ni atom (cf. footnote 1 of the previous page).
- 3) W. Houston, Ztschr. f. Phys. 48, p. 449, 1928. — p. 115 —

... W_i of the moving conduction electrons increases considerably *more rapidly* with the number of conduction electrons, at least as long as the latter is not too large. It would thus in principle be *conceivable* to determine, by a correspondingly *improved calculation* of the work function W_a , in a *purely theoretical* manner, that number of conduction electrons per atom for which precisely the difference $W_a - W_i$ falls into the order of magnitude of Richardson’s constant. For the present, however, the accuracy of our calculation of W_a does not suffice for this purpose. But it is already a fine result that the number of conduction electrons per atom can be determined by a *combination of electron-diffraction and Richardson experiments*. As far as can be judged from the material furnished by Rupp¹⁾ and evaluated by Rosenfeld and Witmer²⁾, this number, like the work function W_a itself, is nearly constant (equal to 2–3) for different metals. One would actually expect the electrons outside the closed shells to split off, and its relation to chemical valence. But the material still appears too scanty to draw conclusive inferences from it.

§ 11. Calculation of the Fourier coefficients of the potential

We now proceed to calculate the Fourier coefficients determining the potential *profile* on the basis of the same assumptions about the structure of the atoms that we used in the preceding section for the calculation of the mean potential. The coefficients of the expansion³⁾

$$V = \sum V_g e^{2\pi i(\mathbf{g}\mathbf{r})}$$

we reduce, by means of the relation derived in § 8 from the electrostatic equation

$$\Delta V = 4\pi e(\varrho_{\text{El}} - \varrho_{\text{Kern}})$$

$$V_g = \frac{e}{\pi g^2}(\varrho_g^{\text{K}} - \varrho_g^{\text{El}}) \quad (77a)$$

to the Fourier coefficients ϱ_g^K of the nuclear density and ϱ_g^{El} of the electron density. Here g denotes the absolute value of the

Footnotes: 1) E. Rupp, loc. cit. 2) Rosenfeld and Witmer, loc. cit. 3) This expansion differs from the expansion (7) generally used in this paper by the factor $\frac{h^2}{8\pi^2 m e}$.

— p. 116 —

vector $\mathbf{g}$, which stands as the index of the Fourier coefficients; i.e., in orthogonal coordinates

$$g^2 = |\mathbf{g}|^2 = \frac{g_1^2}{a_1^2} + \frac{g_2^2}{a_2^2} + \frac{g_3^2}{a_3^2}.$$

The Fourier coefficients of the electron density are now, as is well known, computed from

$$\varrho_g^{\text{El}} = \frac{1}{(\mathbf{a}_1[\mathbf{a}_2\mathbf{a}_3])} \int \frac{\varrho(\mathbf{r})}{4\pi} e^{-2\pi i(\mathbf{g}\mathbf{r})} d\tau \quad (89)$$

where $\varrho(\mathbf{r})$ denotes the 4π -fold electron density at the field point $\mathbf{r}^{1)}$ and the integration is to be extended over one unit cell of the crystal. To take into account the fact that the charge clouds of the atoms extend — albeit by quite small amounts — into foreign unit cells, we again extend the region of integration to infinity²⁾ and consider instead only the electron density originating from the atoms of one unit cell. We split the integral into a sum of integrals of identical structure over the electron densities of the individual atoms of the unit cell, which are located at the field point $\mathbf{r}$, $\varrho_i(\mathbf{r})$, and denote the position of the i th basis atom by $\mathfrak{R}_i$, the position vector of the field point, referred to the nucleus of this atom, simply by $\mathbf{r}$. Then:

$$\varrho_g^{\text{El}} = \frac{1}{(\mathbf{a}_1[\mathbf{a}_2\mathbf{a}_3])} \sum_i e^{-2\pi i(\mathbf{g}\mathfrak{R}_i)} \int_0^\infty \frac{\varrho_i(\mathbf{r})}{4\pi} e^{-2\pi i(\mathbf{g}\mathbf{r})} d\tau. \quad (90)$$

The corresponding coefficient of the nuclear density is, since the nuclei have no extension, with the notation Z_i for the nuclear charge of the i th atom, simply:

$$\varrho_g^K = \frac{1}{(\mathbf{a}_1[\mathbf{a}_2\mathbf{a}_3])} \sum_i e^{-2\pi i(\mathbf{g}\mathfrak{R}_i)} Z_i. \quad (90a)$$

In a crystal consisting entirely of identical atoms, the integrals in (90) all become equal. The sums in (90) and (90a) then simply coincide with the *structure factor* of X-ray

Footnotes: 1) We include the factor 4π in the electron density in order to remain in agreement with the preceding section, and in order to be able afterwards to set ϱ simply equal to the square of the eigenfunction. 2) Cf. the treatment in the preceding section of the potential exerted on the atoms under consideration by foreign unit cells.

— p. 117 —

...agree with the interference theory.¹⁾ For a face-centred lattice, all Fourier coefficients ϱ_g with “mixed indices” thus vanish, whereas for unmixed indices

$$\varrho_g^K = NZ$$

N being again the number of atoms per unit volume.

If we now again assume the electrons to be distributed with spherical symmetry, we obtain

$$\varrho_g^{\text{El}} = N \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{\varrho_{\text{El}}(r)}{4\pi} e^{-2\pi i g r \cos \vartheta} r^2 dr \sin \vartheta d\vartheta d\varphi$$

where ϑ denotes the angle between the fixed direction of the vector $\mathbf{g}$ and the position vector $\mathbf{r}$, which in (90) was the variable of integration. The integrations over φ and ϑ can be carried out:

$$\begin{cases} \varrho_g^{\text{El}} = \frac{N}{2\pi i g} \int_0^\infty \varrho(r) r dr \cdot \frac{1}{2} (e^{2\pi i g r} - e^{-2\pi i g r}) \\ = \frac{N}{2\pi g} \int_0^\infty \varrho(r) r dr \sin 2\pi g r. \end{cases} \quad (91)$$

Here $\varrho(r)$ is again the sum of the 4π -fold densities of the *individual* electrons at distance r from the nucleus; and these in turn are given by the square of the r -dependent part of the Schrödinger eigenfunction:

$$\begin{cases} \varrho_g^{nl} = \frac{N}{4\pi i g n a_0} \frac{(n-l-1)!}{(n+l)!^3 2n} \int_0^\infty e^{-\xi} \xi^{2l+1} [L_{n+l}^{2l+1}(\xi)]^2 d\xi \\ \cdot \left(e^{2\pi i \frac{n a_0 g}{2z} \xi} - e^{-2\pi i \frac{n a_0 g}{2z} \xi} \right). \end{cases} \quad (91a)$$

For the evaluation of the integral in (91a) we use a modification of Waller's method sketched in the preceding section.²⁾ One of the integrals to be calculated³⁾ is

$$J_{nl} = \int_0^\infty e^{-(1+i\alpha)\xi} \xi^{2l+1} [L_{n-l}^{2l+1}(\xi)]^2 d\xi \quad (92)$$

[Translator's note: the subscript of L is printed here as $n-l$; in (91a) above and in the Waller integral on p. 118 it is printed as $n+l$, which is also what the subsequent derivation requires.]

Footnotes: 1) We have already made use in § 8 of the fact that v_h vanishes simultaneously with the structure factor for the reflection (h_1, h_2, h_3) . 2) I. Waller, Ztschr. f. Phys. 38, p. 644, 1926. 3) The other differs from this one only in the sign of i .

— p. 118 —

wherein

$$\alpha = \frac{2\pi a_0 n}{2z} g \quad (92a)$$

differs, namely, from the integral evaluated by Waller,

$$\int_0^\infty e^{-\xi} \xi^{2l+s+2} [L_{n+l}^{2l+1}(\xi)]^2 d\xi$$

with $s = -1$ only in the factor in the exponent of e . Accordingly, the derivation of the preceding section can be transferred to our case, and one obtains, analogously to (86a):

$$\begin{aligned}
& \sum_{p_1} \sum_{p_2} \frac{t^{p_1} u^{p_2}}{p_1! p_2!} \int_0^\infty \xi^{2l+1} e^{-(1+i\alpha)\xi} L_{p_1}^{2l+1} L_{p_2}^{2l+1} d\xi \\
&= \frac{(ut)^{2l+1}}{[(1-u)(1-t)]^{l+2}} \int_0^\infty \xi^{2l+1} e^{-\xi \left(\frac{1-ut}{(1-u)(1-t)} + i\alpha \right)} d\xi \\
&= (2l+1)!(ut)^{2l+1} [1-ut+i\alpha(1-u)(1-t)]^{-(2l+2)}.
\end{aligned}$$

The value of the integral (92) is therefore equal to the coefficient of $(ut)^{n-l-1}$ in the expansion of

$$\begin{aligned}
& (n+l)!^2 (2l+1)! (1+i\alpha - i\alpha(u+t) - (1-i\alpha)ut)^{-(2l+2)} \\
&= (n+l)!^2 (2l+1)! (1+i\alpha)^{-(2l+2)} \sum_{\tau} \binom{2l+1+\tau}{\tau} \left(\frac{1-i\alpha}{1+i\alpha} ut + \frac{i\alpha}{1+i\alpha} (u+t) \right)^{\tau}.
\end{aligned}$$

Now only those terms come into consideration which contain u and t in the same power, i.e., of the entire expansion only the part:

$$(n+l)!^2 (2l+1)! (1+i\alpha)^{-(2l+2)} \sum_{\tau} \binom{2l+1+\tau}{\tau} \sum_{\sigma=0}^{\tau/2} \binom{\tau}{2\sigma} \left(\frac{i\alpha}{1+i\alpha} \right)^{2\sigma} \left(\frac{1-i\alpha}{1+i\alpha} \right)^{\tau-2\sigma} \binom{2\sigma}{\sigma} (ut)^{\tau-\sigma}.$$

Here we have to pick out the terms for which the exponent of ut ,

$$\tau - \sigma = n - l - 1$$

results. Since furthermore $\tau - 2\sigma \geq 0$ must hold, we need only terms with

$$\sigma \leq \frac{\tau}{2} \leq n - l - 1$$

— p. 119 —

to be taken into account, and we obtain

$$\begin{aligned}
J_{nl} &= (n+l)!^2 (2l+1)! (1+i\alpha)^{-(2l+2)} \\
& \sum_{\sigma=0}^{n-l-1} (1+i\alpha)^{-(n-l-1+\sigma)} (i\alpha)^{2\sigma} (1-i\alpha)^{n-l-1-\sigma} \binom{n-l-1+\sigma}{2\sigma} \cdot \binom{2\sigma}{\sigma} \binom{n+l+\sigma}{n-l-1+\sigma} \\
&= (n+l)!^2 (2l+1)! (1+i\alpha)^{-2n} \sum_{\sigma=0}^{n-l-1} (1+\alpha^2)^{n-l-1-\sigma} (-\alpha^2)^{\sigma} \cdot \frac{(n+l+\sigma)!}{(n-l-1-\sigma)!(2l+1)!\sigma!^2} \\
&= (n+l)!^3 (1+i\alpha)^{-2n} \sum_{\varrho=0}^{n-l-1} \alpha^{2\varrho} \sum_{\sigma=0}^{\varrho} (-1)^{\sigma} \binom{n+l+\sigma}{\sigma} \binom{n-l-1-\sigma}{\varrho-\sigma} \cdot \frac{1}{(n-l-1-\sigma)!} \\
&= \frac{(n+l)!^3}{(n-l-1)!} (1+i\alpha)^{-2n} \sum_{\varrho=0}^{n-l-1} \alpha^{2\varrho} \binom{n-l-1}{\varrho} \sum_{\sigma=0}^{\varrho} (-1)^{\sigma} \binom{n+l+\sigma}{\sigma} \binom{\varrho}{\sigma}.
\end{aligned}$$

This goes over, after an uninteresting transformation of the sum over $\sigma^{1)}$, into:

$$J_{nl} = \frac{(n+l)!^3}{(n-l-1)!} (1+i\alpha)^{-2n} \sum_{\varrho=0}^{n-l-1} (-1)^{\varrho} \alpha^{2\varrho} \binom{n-l-1}{\varrho} \binom{n+l}{\varrho}. \quad (92b)$$

Footnotes: 1) Namely,

$$(1+x)^{-(n+l+1)} = \sum_{\sigma} (-1)^{\sigma} \binom{n+l+\sigma}{\sigma} x^{\sigma}$$

$$(1+x)^{\varrho} = \sum_{\tau} \binom{\varrho}{\tau} x^{\tau}$$

By multiplication:

$$(1+x)^{-(n+l-\varrho+1)} = \sum_s (-1)^s \binom{n+l-\varrho+s}{s} x^s$$

$$= \sum_s x^s \sum_{\sigma} (-1)^{\sigma} \binom{n+l+\sigma}{\sigma} \binom{\varrho}{s-\sigma}$$

Hence by comparison of coefficients for $s = \varrho$:

$$\sum_{\sigma} (-1)^{\sigma} \binom{n+l+\sigma}{\sigma} \binom{\varrho}{\sigma} = (-1)^{\varrho} \binom{n+l}{\varrho}.$$

— p. 120 —

If we insert this value for the integral (92) into (91a), we obtain for the desired Fourier coefficient of the electron density:

$$\varrho_g^{nl} = \frac{N}{2i\alpha} \frac{1}{2n} \sum_{\varrho=0}^{n-l-1} (-1)^{\varrho} \alpha^{2\varrho} \binom{n-l-1}{\varrho} \binom{n+l}{\varrho} [(1-i\alpha)^{-2n} - (1+i\alpha)^{-2n}]$$

or

$$\left\{ \begin{aligned} \varrho_g^{nl} &= \frac{N}{2n} \sum_{\sigma=0}^{n-1} \binom{2n}{2\sigma+1} (-1)^{\sigma} \alpha^{2\sigma} \\ &\cdot \sum_{\varrho=0}^{n-l-1} (-1)^{\varrho} \alpha^{2\varrho} \binom{n-l-1}{\varrho} \binom{n+l}{\varrho} \cdot \frac{1}{(1+\alpha^2)^{2n}}. \end{aligned} \right. \quad (93)$$

The expressions (92) read, for the first three principal quantum numbers:

$$\varrho_g^{10} = N \cdot \frac{1}{(1+\alpha^2)^2}$$

$$\varrho_g^{20} = N \cdot \frac{1-\alpha^2}{(1+\alpha^2)^4} (1-2\alpha^2)$$

$$\varrho_g^{21} = N \cdot \frac{1-\alpha^2}{(1+\alpha^2)^4}$$

$$\varrho_g^{30} = N \cdot \frac{1 - \frac{10}{3}\alpha^2 + \alpha^4}{(1 + \alpha^2)^6} (1 - 6\alpha^2 + 3\alpha^4)$$

$$\varrho_g^{31} = N \cdot \frac{1 - \frac{10}{3}\alpha^2 + \alpha^4}{(1 + \alpha^2)^6} (1 - 4\alpha^2)$$

$$\varrho_g^{32} = N \cdot \frac{1 - \frac{10}{3}\alpha^2 + \alpha^4}{(1 + \alpha^2)^6}$$

The auxiliary variable α depends thereby on the indices of the desired Fourier coefficients according to (92a). Written out in the individual indices, the relation (92a) reads:

$$\alpha^2 = \left(\frac{\pi n a_0}{z} \right)^2 \left(\frac{g_1^2}{a_1^2} + \frac{g_2^2}{a_2^2} + \frac{g_3^2}{a_3^2} \right) \quad (92c)$$

and goes over, for cubic axes, into the particularly simple expression

$$\alpha^2 = \left(\frac{\pi n a_0}{z a} \right)^2 (g_1^2 + g_2^2 + g_3^2) \quad (92d)$$

— p. 121 —

...over, which for the lattice constant $a = 3.52 \text{ \AA}$ of nickel takes the value

$$\alpha^2 = 0.227 \left(\frac{n}{z} \right)^2 (g_1^2 + g_2^2 + g_3^2) \quad (92e)$$

From formula (93) the basic properties of the Fourier expansion of the density of an (n, l) electron can be read off.¹⁾ The Fourier coefficients depend, first of all, as a consequence of the assumed spherically symmetric distribution of the electrons, *only* on the absolute value of the vector $\mathbf{g}$ in the reciprocal lattice, but not on the individual indices g_1, g_2, g_3 . Furthermore, for *small* indices g (small α) the first sum in (93) approaches the value $2n$, the second unity, and thus the Fourier coefficient [approaches] the number N of atoms per unit volume. This is quite natural: the exponential function $e^{2\pi i(\mathbf{g}\mathbf{r})}$ becomes nearly unity over the whole region of the atom, and the integral

$$\int_0^\infty \int_0^\pi \int_0^{2\pi} \varrho_{nl}(r) r^2 dr e^{-2\pi i(\mathbf{g}\mathbf{r})} \cdot \frac{1}{4\pi} \sin \vartheta d\vartheta d\varphi,$$

which, apart from the factor N , determines the Fourier coefficient, simply goes over into the space integral over the density of the (nl) electron,

$$\int \varrho(r) r^2 dr,$$

which is normalized to unity. Thus the Fourier coefficient for small indices must become equal to the number N of atoms per unit volume. For *large* indices g , on the other hand, the first sum becomes $2n\alpha^{2(n-1)}$, the second takes the value $(-\alpha^2)^{n-l-1} \binom{n+l}{n-l-1}$ (since the terms with smaller powers of α^2 may be neglected), and the entire Fourier coefficient becomes approximately:

$$\varrho_g^{2l} = (-1)^{n-l-1} \binom{n+l}{n-l-1} \frac{\alpha^{2(2n-l-2)}}{(1+\alpha^2)^{2n}}$$

[Translator's note: printed as ϱ_g^{2l} without the factor N ; cf. (93).]

Footnotes: 1) The following considerations, just like formula (93), would also hold — exactly, in fact — for the coefficients of the Fourier *integral* by which the charge distribution of free hydrogen-like atoms can be represented. They certainly remain qualitatively valid also for atoms for which hydrogen eigenfunctions may no longer be used.

— p. 122 —

and goes to zero as $\alpha^{-2(l+2)}$. This too is easily understood: the oscillations of the exponential function $e^{2\pi i(\mathbf{g}\mathbf{r})}$ become so rapid that the contributions of the individual regions of space to the integral (90c) mutually cancel. It is to be expected that this cancellation is *more complete* for the very smooth density curves of the electrons with large azimuthal quantum number l than for the electrons with small l , whose density exhibits many maxima and minima in the radial direction, and that consequently the Fourier coefficients for the former fall off more rapidly with increasing index ($\alpha^{-2(l+2)}$).

The conduction electrons, as a consequence of their assumed uniform distribution over the entire crystal, naturally contribute nothing to the Fourier coefficients of the electron density (except to ϱ_{000}).

The Fourier coefficient of the nuclear density, which we found to be NZ for crystals of identical atoms, remains constant for all indices g for which it differs from zero at all¹⁾. When the index g is small, it is *cancelled* to a large extent by the Fourier coefficient ϱ_g^{El} of the electron density²⁾; for large indices g , on the other hand, the Fourier coefficient of the electron density no longer plays any role compared to it, according to what was just said. Accordingly, by (77a), the Fourier coefficient V_g of the potential behaves for large indices g as $1/g^2$.

Tab. 3 contains, for nickel, the contributions of the individual electron shells to the Fourier coefficients of the electron density, as long as $G^2 = g_1^2 + g_2^2 + g_3^2 < 30$, furthermore the coefficient $\varrho_g = \varrho_g^K - \varrho_g^{\text{El}}$ itself up to $G^2 = 120$, both omitting the factor N . The second-to-last column gives the Fourier coefficient V_g of the potential in volts, the last the coefficient generally used in this paper,

$$v_g = \frac{8\pi^2 me}{h^2} V_g = \frac{8\pi^2 me^2}{\pi h^2 g^2} \varrho_g = \frac{2\varrho_g}{\pi a_0 g^2}$$

expressed in the unit Ångström⁻².

Footnotes: 1) I.e., for which the structure factor does not vanish. 2) For the index triple 0, 0, 0, $\varrho_{000}^K - \varrho_{000}^{\text{El}} = 0$, which simply means that the crystal as a whole is neutral.

— p. 123 —

Table 3 Fourier coefficients of the potential and of the charge density for nickel

G ²	g ₁ g ₂ g ₃	Nucleus	K	L	M ^I	M ^{II}	M ^{III}	ρ_g in elementary charges per atomic volume	V _g in volts	v_g in Ångström ⁻²
3	111	28	-2	-7.8	-1.45	-4.15	-4.85	7.75	13.4	3.57
4	200	28	-2	-7.7	-1.3	-3.45	-3.85	9.7	12.6	3.31
8	220	28	-2	-7.4	-0.8	-2.15	-1.75	13.9	9.05	2.38
11	311	28	-2	-7.3	-0.55	-1.35	-0.9	15.9	7.5	1.98
12	222	28	-2	-7.2	-0.45	-1.15	-0.7	16.5	7.15	1.88
16	400	28	-2	-7.0	-0.25	-0.6	-0.15	18.0	5.85	1.54
19	331	28	-2	-6.9	-0.15	-0.3	+0.1	18.75	5.15	1.35
20	420	28	-2	-6.8	-0.1	-0.25	+0.15	19.0	4.95	1.30
24	422	28	-1.95	-6.5	-0.05	-0.1	+0.25	19.65	4.25	1.12
27	⁵¹¹ ₃₃₃	28	-1.95	-6.4	0.0	0.0	+0.3	19.95	3.85	1.02

G ²	g ₁ g ₂ g ₃	ρ_g in elementary charges per atomic volume	V _g in volts	v_g in Ångström ⁻²
32	440	20.2	3.29	0.87
35	531	20.4	3.04	0.80
36	⁶⁰⁰ ₄₄₂	20.4	2.96	0.78
40	620	20.6	2.68	0.71
43	533	20.65	2.50	0.66
44	622	20.65	2.45	0.64
48	444	20.8	2.26	0.59
51	⁵⁵¹ ₇₁₁	20.9	2.14	0.56
52	640	20.9	2.09	0.55
56	642	21.0	1.95	0.51
59	⁵⁵³ ₇₃₁	21.1	1.86	0.49
64	800	21.1	1.72	0.45
67	733	21.25	1.65	0.43

G ²	g ₁ g ₂ g ₃	ρ_g in elementary charges per atomic volume	V _g in volts	v_g in Ångström ⁻²
80	840	21.55	1.40	0.37
83	⁷⁵³ ₉₁₁	21.65	1.36	0.36
84	842	21.65	1.34	0.35
88	664	21.75	1.29	0.34
91	931	21.95	1.26	0.33
96	844	22.05	1.19	0.31
99	⁹³³ ₇₇₁ 755	22.10	1.16	0.30
100	^{10,0,0} ₈₆₀	22.20	1.16	0.30
120	10,4,2	22.8	0.99	0.26
150	—	23.5	0.82	0.21
200	^{10,10,0} _{14,2,0} 10,8,6	24.45	0.76	0.17
300	—	25.55	0.44	0.12
500	—	26.45	0.27	0.072

G^2	$g_1g_2g_3$	ϱ_g in elemen- tary charges per atomic volume	V_g in volts	v_g in Ångström	G^2	$g_1g_2g_3$	ϱ_g in elemen- tary charges per atomic volume	V_g in volts	v_g in Ångström ⁻²
68	820 644	21.35	1.63	0.43	1000	—	26.85	0.14	0.037
72	822 660	21.4	1.55	0.41	5000	—	27.6	0.03	0.007
75	751 555	21.4	1.49	0.39	30000	—	28.0	0.005	0.001
76	662	21.4	1.47	0.39					

§ 12. Numerical calculations of the quantities of electron diffraction theory

We now want to use the Fourier coefficients obtained in the previous section in order to actually carry out numerically, for one of the “Laue spots” observed by Davisson and Germer, the calculations of §§ 4—7 and 9.

— p. 124 —

We choose for this purpose the reflection point (622), which was observed in the (111) azimuth at 160 volts electron energy (in vacuum).¹⁾ For this reflection point the exact Laue equation gives, as the wave number in the crystal,

$$\kappa_0 = \frac{\pi\sqrt{3}}{a} \cdot \frac{h_1^2 + h_2^2 + h_3^2}{h_1 + h_2 + h_3} = \frac{3.14 \cdot 1.73 \cdot 44}{3.52 \cdot 10} = 6.80 \text{ Å}^{-1}.$$

To calculate the dynamical potential quantities $\sigma_{11}, \sigma_{12}, \sigma_{22}, \tau_{11}, \tau_{12}, \tau_{22}$ of § 7, we order the sums in (82) according to the magnitude $G^2 = g_1^2 + g_2^2 + g_3^2$, which, according to the discussion of the previous section, determines the potential coefficient v_g , and the summation terms belonging to the same G again according to the sign of the indices g_1, g_2, g_3 . Permutations of the indices among themselves are already taken into account in formulas (82).

The evaluation must be carried out separately for each summation term, until G becomes so large that no small resonance denominators occur any more. For larger G one may replace the summation by an integration, whereby one may write for v_g approximately F/G^2 with a suitably chosen constant F .

For the (622) reflection from nickel, one obtains for the dynamical potential quantities (82) the values:

$$\begin{aligned}\sigma_{11} &= 22.56 \text{ Å}^{-1} \\ \sigma_{12} &= 4.92 \text{ Å}^{-1} \\ \sigma_{22} &= -15.50 \text{ Å}^{-1} \\ \tau_{11} &= 3.84 - 7.57 (\kappa - \kappa_0) \text{ Å}^{-2} \\ \tau_{12} &= 1.04 - 11.92 (\kappa - \kappa_0) \text{ Å}^{-2}\end{aligned}$$

$$\tau_{22} = -5.52 - 49.25 (\varkappa - \varkappa_0) \text{ \AA}^{-2}$$

With these, the quantities defined in § 7 become furthermore

$$\begin{aligned} w^2 &= -50.3 + 717 (\varkappa - \varkappa_0) + 99300 (\varkappa - \varkappa_0)^2 \text{ \AA}^{-3}, \\ \sigma_{11} \sigma_{22} - \sigma_{12}^2 &= -375 \text{ \AA}^{-2}. \end{aligned}$$

Footnotes: 1) The calculation for the most accurately investigated reflection (331) at 54 volts unfortunately presents difficulties for this reason, because in the vicinity (at 65 volts) the especially strong reflection (111) occurs, and the beam in question cannot be regarded as “weak”. The treatment of three strong beams, however, presents great computational difficulties.

— p. 125 —

With this we now have everything we need for the calculation of the quantities of the dynamical theory that interest us: 1. For the *midpoint* ($W = 0$) of the region of selective reflection we obtain from equation (72)

$$22.56 (-5.52 - 49.25 (\varkappa_m - \varkappa_0)) = -15.50 (3.84 - 7.57 (\varkappa - \varkappa_0))$$

the value

$$\varkappa_m - \varkappa_0 = -0.053 \text{ \AA}^{-1}, \quad \varkappa_m = 6.75 \text{ \AA}^{-1}.$$

The corresponding vacuum wave number is

$$K_m = \sqrt{6.75^2 - 4.14} = 6.43 \text{ \AA}^{-1}.$$

Thus the selectively reflectable wave number is shifted by 0.32 through the *refractive index*, and by $0.05 \text{ \AA}^{-1}$ through the influence of the potential variation according to the *dynamical theory*. *The influence of the second and third approximation on the magnitude of the selectively reflectable wavelength is thus about one sixth of that of the first approximation.*

Had one simply sought, according to the second approximation, *that* wave number $\varkappa_\mu$ for which the dynamical potential quantities V_{11} , V_{12} , V_{22} yield precisely $\varkappa_\mu$ as the midpoint of the selective-reflection region, one would have found for $\varkappa_\mu - \varkappa_0$ the somewhat larger value $-0.066 \text{ \AA}^{-1}$.

2. The *boundaries* of the region of selective reflection are given by the roots of the equation $W^2 = 1$ (cf. § 7):

$$\begin{aligned} &[22.56 (-5.52 - 49.25 (\varkappa - \varkappa_0)) + 15.50 (3.84 - 7.57 (\varkappa - \varkappa_0))]^2 \\ &= -50.3 + 717 (\varkappa - \varkappa_0) + 99300 (\varkappa - \varkappa_0)^2. \end{aligned}$$

From this one obtains $\varkappa_1 - \varkappa_0 = -0.039$, $\varkappa_2 - \varkappa_0 = -0.103$. The region of selective reflection has a *width* of $0.064 \text{ \AA}^{-1}$, which is about 1 percent of the wave number. The *resolving power of the crystal for electron diffraction* is

$$\frac{\lambda}{\Delta\lambda} = \frac{\varkappa_0}{\varkappa_1 - \varkappa_2} = \text{about } 100. \quad (94)$$

If one further takes into account here that our calculation neglects the *inelastic collisions*, which in any case *reduce* the resolving power, and that the direction of the incident beam is by no means sharp, then this resolving power stands in *good agreement* with the *width* of the reflection maxima measured by Davisson and Germer of 7 to 25 percent of the wave number, corresponding to velocities

— p. 126 —

...of 300 to 50 volts. The half-width would presumably amount to about half of that, i.e., for 160 volts one may reckon with about 5 percent half-width ($\frac{\lambda}{\Delta\lambda} = 20$).

Zwicky¹⁾ had, from the first approximation by the Born method²⁾, inferred for the resolving power the obviously far too high value 10^8 . From our second approximation $\frac{\lambda}{\Delta\lambda} = 60$ would follow.

3. The *intensity* of the reflected beam is given in Tab. 5 as a function of the wave number $\varkappa$, both for a “thick” crystal and for one which consists only [Tr.: "Tab. 5" misprints "Tab. 4".]

Table 4 Intensity variation of the (622) reflection (Third approximation)

$\varkappa - \varkappa_0$ in $\text{\AA}^{-1}$	w in $\text{\AA}^{-3}$	W	$\varkappa_0 = 6.80$ $\text{\AA}^{-1}$ —		Penetration depth	
			Reflected intensity			
			1 atomic layer	entire crystal	$\text{\AA}$	atomic layers
(+ 0.1)	(31.7)	(2.96)	(0.042)	(0.059)	—	—
0.05	15.1	4.20	0.008	0.029	—	—
0	7.09 <i>i</i>	4.60 <i>i</i>	0.002	0.024	—	—
− 0.03	4.55	3.08	0.001	0.044	—	—
− 0.035	7.07	1.54	0.002	0.240	—	—
− 0.039	8.80	1.00	0.003	1.000	133	65
− 0.045	11.22	0.43	0.005	1.000	66	32
− 0.053	14.00	0.00	0.008	1.000	50	25
− 0.07	19.85	0.53	0.015	1.000	38	19
− 0.09	26.5	0.86	0.027	1.000	35	17
− 0.103	30.6	1.00	0.035	1.000	38	19
− 0.13	39.3	1.20	0.055	0.444	—	—
(− 0.15)	(45.7)	(1.31)	(0.074)	(0.353)	—	—
(− 0.2)	(61.5)	(1.47)	(0.125)	(0.269)	—	—

...of a single layer of atoms. Curve 8 illustrates the situation, but it must be borne in mind that the intensity calculated from the third approximation is only *conditionally correct*.

The *reflecting power* of a single layer moves, in the region of selective reflection, between 0.003 and 0.03. Equivalent to this is the observation that the electrons

Footnotes: 1) F. Zwicky, Proc. Nat. Acad. Sci. 13, pp. 518—525. 1927. 2) M. Born, Ztschr. f. Phys. 38. p. 803. 1926.

— p. 127 —

...penetrate into the crystal up to the 17th to 70th atomic layer. As the limit of penetration the place was assumed at which the electron current has sunk to 10 percent of its original value. This limitation of the penetration depth of the electrons is due solely to the exponential intensity decrease in the crystal on account of the selective reflection. How far, on the other hand, the electrons are able to penetrate into the crystal in the normal case depends on their absorption

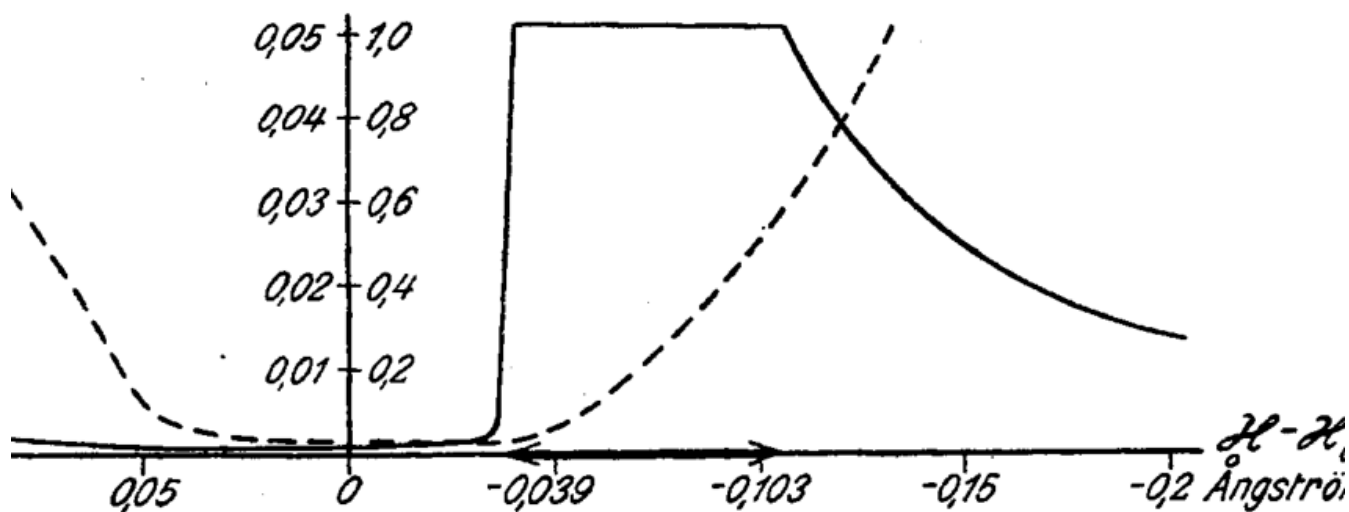
Intensity variation of the (622) reflection ——— Reflected intensity from the entire crystal S (right scale) - - - - - the same from a single layer (s) (left scale) Bold-drawn part of the $\varkappa$ -axis: region of selective reflection Fig. 8 [The figure plots the reflected intensity against $\varkappa - \varkappa_0$ in Ångström⁻¹ (axis marks 0.1, 0.05, 0, -0.039, -0.103, -0.15, -0.2): the solid curve (right scale S , 0.2–1.0) shows a flat total-reflection plateau of intensity 1.0 between $\varkappa - \varkappa_0 = -0.039$ and -0.103 and falls off on both sides, while the dashed curve (left scale s , 0.01–0.05) shows the much smaller single-layer reflecting power peaking near the edges of the selective-reflection region.]

by means of *inelastic collisions*, which we have neglected. From the success of Rupp's experiments one may conclude that this absorption is of a somewhat *smaller* order of magnitude than the intensity decrease caused by selective reflection. If, on the other hand, one evaluates the experimental intensity curve found by Davisson and Germer for the (331) reflection (at 54 volts), one finds, on the assumption that the electrons are simply absorbed in proportion to the path traversed in the crystal, that 54-volt electrons penetrate into the crystal approximately up to the 6th layer. But here absorption through inelastic collisions, intensity decrease through selective reflection, and disturbance through inhomogeneous direction of incidence are intertwined, and moreover the penetration depth certainly decreases very strongly with electron energy.

— p. 128 —

In our theory of electron diffraction *two* important factors are still *lacking*, which would have to be taken into account in exact *intensity calculations*: the influence of the *thermal motion* of the atoms, which, judging by the evidence of experiment and by the successes of Houston's theory of conductivity¹⁾, makes itself felt in a similar way as the Debye factor in X-ray interferences; and the *inelastic collisions*, through which the electron drops out of the number of those moving with the original velocity, and the intensity of the reflected beam is diminished. The study of these two phenomena would be required in particular also for a theory of the *continuous background* in the experiments of Davisson and Germer.

Apart from this, the agreement of theory with experiment, insofar as the effects are observable at all, seems quite good: the *refractive index* has the theoretically required variation with wavelength, and the *mean potential* to be obtained from the refractive index is in good agreement with the theoretically calculated one. The *resolving power* also comes out of the observed order of magnitude. The *reflecting power* does have a plausible value, but is not yet exactly comparable with experiment. The *shift* of the selectively reflectable wavelengths due to the participation of the potential variation in the determination of their magnitude is — apart from experimental errors — responsible for the *fluctuations* of the value of V_0 calculated from the experiments about its mean value. For an exact numerical comparison, an improved calculation of the atomic fields must be undertaken. Nothing conclusive can be said from the experiments about the higher Fourier coefficients of the potential, whereas one of the *most important results* of the theory is the *determination of the work function* W_a of the Richardson effect.



Intensitätsverlauf der (622)-Reflexion

- Reflektierte Intensität vom ganzen Kristall S (rechte Skala)
 - - - - - desgleichen von einer Schicht (s) (linke Skala)

Stark ausgezogener Teil der κ -Achse: Gebiet der Selektivreflexion

Fig. 8

vermittels unelastischer Stöße ab, die wir vernachlässigt haben

Fig. 8. Intensity curve of the (622) reflection.

— Reflected intensity from the whole crystal S (right scale);

- - - - the same from a single layer s (left scale).

Heavily drawn part of the κ -axis: region of selective reflection.

I would like to express my deepest thanks to Prof. Sommerfeld for suggesting this work and for his continued interest and encouragement during its progress.

Summary

According to wave mechanics, the diffraction of

Footnotes: 1) W. Houston, Ztschr. f. Phys. 48. p. 449. 1928.

— p. 129 —

electrons by crystals observed by Davisson and Germer, Rupp, Thomson, etc., is treated. In the first approximation, the observed discrepancy between X-ray and electron diffraction can be explained by a refractive index. From the observed refractive indices one may infer, for nickel, an accelerating potential $V_0 = 14.8$ volts which the crystal exerts on electrons (§ 1). The dynamics of the field of the de Broglie waves in the crystal (§ 2) and the conditions at its surface (§ 3) are developed. With the simplifying assumption that, besides the primary beam, only *one* strong diffracted beam occurs (§ 4), the dispersion equation can be set up. From it, in far-reaching analogy to Ewald's dynamical theory of X-ray reflections, the more precise position of the reflection maxima and the "resolving power" of the crystal (§ 5) as well as the reflected intensity (§ 6) can be determined. The approximation is then carried one step further (§ 7).

On the basis of the developed dynamical theory of electron diffraction, the characteristic deviations from X-ray reflection are discussed (§ 8). In connection with the very small penetrating power compared with X-rays of the same wavelength, electron diffraction deviates from X-ray diffraction in a characteristic way. The extinction laws of X-ray reflections, too, can be violated under certain circumstances. § 9 treats the case, realized by Davisson and Germer, of reflection in several directions symmetric with respect to the incident beam. The mean crystal potential acting on the electrons, responsible for the refractive index, is found on purely theoretical grounds, on the basis of wave-mechanical assumptions about the structure of the crystal atoms, to be nearly of the same magnitude as follows from the experiments (§ 10). The relation to the work function in the Richardson effect is explained. Then the Fourier coefficients of the potential variation (§ 11) are calculated, and finally with their help, in an example, the numerical values of the reflecting and resolving power, etc., are computed (§ 12). The experimental results are thereby in satisfactory agreement.

(Received July 9, 1928)

Printing Errors in the 1928 Original

Rechecked against the clearer 75-page *Annalen der Physik* facsimile – 17 September 2026

Certification scope. The mathematical layer of this English edition has now been checked page by page against the clearer supplied facsimile (original journal pp. 55–129). The audit covers all main equation numbers (1)–(94), all lettered and repeated numbered occurrences, displayed unnumbered formulas, inline mathematical notation, and the numerical/math tables. The rendered source page image, not the PDF text layer or OCR, is the authority for glyph decisions.

The companion audit ledger contains 158 numbered-equation occurrences and 2,661 indexed math-line/inline-math occurrences. A source-printed anomaly is therefore no longer an “unresolved” item: it receives a PASS when the English edition faithfully reproduces what Bethe actually printed, with an editorial note explaining the apparent source error.

Seven source-print anomalies rechecked at high resolution

1. **Original p. 69, Eq. (16) – summation index. PASS – source print confirmed.** The source clearly prints a summation subscript g while the vacuum-wave amplitudes and propagation vectors in the summand carry the index m . On the next page the boundary-condition sums are over m . The printed g is therefore very likely a typesetting error for m . The English body deliberately reproduces the printed g and flags it rather than silently emending Bethe.

2. **Original pp. 80–81, Eq. (41) – missing square. PASS – source print confirmed.** The source clearly prints

$$|\gamma_h| < v_0.$$

However Eq. (41b) contains γ_h^2 , and with $v_0 = 8\pi^2 meV_0/h^2$ the derivation makes the dimensionally consistent condition $\gamma_h^2 < v_0$. Thus Eq. (41) almost certainly has a source typesetting error. The English equation remains faithful to the print; the audit note records the likely intended form.

3. **Original p. 94, Fig. 7 – curve-label subscripts. PASS – source print confirmed.** The prose immediately above Fig. 7, consistent with Eq. (72), uses

$$(\sigma_{11}\tau_{22} - \sigma_{12}\tau_{12})^2,$$

whereas the label written on the solid curve in Fig. 7 is visibly

$$(\sigma_{11}\tau_{22} - \sigma_{22}\tau_{11})^2.$$

This is a real mismatch in the 1928 printing. The previous checklist described it imprecisely as a difference in “reflection indices”; that wording is corrected here. The English edition reproduces each source location as printed.

4. **Original p. 102, Eq. (37b) – two internal formula anomalies. PASS – source print confirmed.** The third line visibly has the denominator $2\kappa^2 - k_g^2$, while the final V_{22}^{ii} numerator is visibly an unsquared v_{g-hi} . Both differ from the pattern generated by Eq. (35a): the weak-beam denominator there is $\kappa^2 - k_g^2$, and the $i = j$ specialization of the preceding product would give a square in the numerator. Both items are therefore very likely source typesetting errors. They are retained exactly as printed in the English transcription and explicitly documented.
5. **Original pp. 109, 117–118, Eq. (92) – Laguerre subscript. PASS – source print confirmed.** Original p. 109 and Eq. (91a) on p. 117 use L_{n+l}^{2l+1} , and the Waller integral quoted on p. 118 again uses L_{n+l}^{2l+1} . Eq. (92) alone visibly prints L_{n-l}^{2l+1} . The isolated $n - l$ is therefore almost certainly a source typo. The English body reproduces Eq. (92) as printed and includes a translator’s note.

6. Original p. 70 · equation number misprint. PASS · source print confirmed. The text refers to ·(19)· where the context and numbering sequence clearly indicate Eq. (18). The English body reproduces ·(19)· as printed and includes a translator’s note.

7. Original p. 126 · table number inconsistency. PASS · source print confirmed. The text refers to ·Tab. 5· where the actual table is numbered ·Tabelle 4·. The English body reproduces ·Table 5· as printed and includes a translator’s note.

Corrections confirmed in the current body

The following previously identified transcription errors have been corrected and were rechecked against the clearer scan:

- Fraktur propagation vector $\mathfrak{f}$ restored where earlier copies had $\mathfrak{k}$; Eqs. (18)–(19) are confirmed as $\mathfrak{f}$ notation.
- Original p. 92 before Eq. (68): $2\pi g_n$ confirmed.
- Original p. 92, Eq. (70): the first τ_{22} denominator is $-4\pi(\mathfrak{g}, \mathfrak{C}_0 - \pi\mathfrak{g})$.
- Original p. 103 after Eq. (79): the German is *anzunehmen*; “assumed” is correct. The earlier audit change to “excluded” was erroneous and has been reversed.
- Original p. 104, Eq. (80): π restored in $\mathfrak{C}_0 + \pi\mathfrak{g}$; Eq. (81)’s final subscript is clearly g_3 ; Eq. (82) uses Bethe’s ϑ glyph.
- Original pp. 109–117: the atomic/effective nuclear-charge symbol is italic z ; the affected x/μ -like misreadings have been corrected.
- Original p. 121: the unnumbered asymptotic Fourier coefficient has denominator $(1 + \alpha^2)^{2n}$.

Status of this edition

The mathematical notation and the complete prose translation have been audited against the supplied German facsimile: every numbered formula and every indexed mathematical occurrence has been checked. Transcription and translation errors identified in the audit have been corrected. Verified source-print anomalies are preserved as printed and documented, not silently modernized.

Für die beobachtete Diskrepanz gibt es nun von vornherein zwei Erklärungsmöglichkeiten: Eine Veränderung der Kristalldimensionen oder aber der Wellenlänge der einfallenden Elektronen. Davisson und Germer zogen zunächst rein phänomenologisch die erste Erklärung heran¹⁾ und berechneten aus ihren Versuchen, daß die Abstände der zur Oberfläche parallelen Netzebenen in der Nähe der Kristalloberfläche auf 70–95 Proz. ihres normalen Wertes im Kristallinneren kontrahiert sein müßten, um der Abweichung in den selektiv reflektierbaren Wellenlängen Rechnung zu tragen. Natürlich kann einer solchen, mit der Elektronengeschwindigkeit außerordentlich

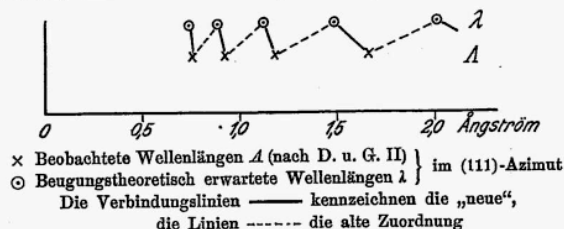


Fig. 1

stark (und noch dazu ganz unregelmäßig) variierenden Kontraktion des Kristalls eine Realität nicht zukommen, und der Kontraktionsfaktor war wohl auch von Davisson und Germer von vornherein nur als phänomenologische Beschreibung ihrer Versuche gedacht.

Die andere mögliche Annahme besteht darin, daß die Wellenlänge der de Brogliewelle, die außerhalb des Kristalls λ beträgt, innerhalb desselben einen anderen Wert, λ' , besitzt, d. h. daß dem Kristall ein Brechungsindex

$$(2) \quad \mu = \frac{\lambda}{\lambda'}$$

für de Brogliewellen zugeschrieben werden muß. Ein solcher Brechungsindex ist zunächst rein phänomenologisch von Eckart²⁾ eingeführt worden. Verfasser³⁾ hat unabhängig davon

1) Eine Arbeit von Patterson (Nature 120, S. 46. 1927) beschäftigt sich theoretisch mit der Möglichkeit eines Kontraktionsfaktors.

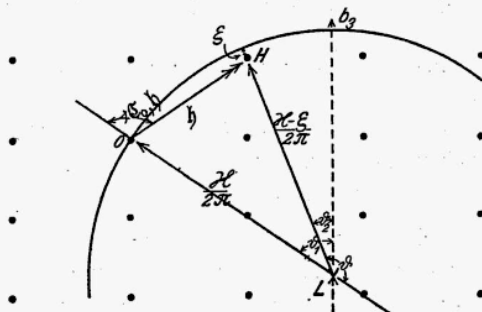
2) C. Eckart, Proc. Nat. Acad. Washington 12, S. 460. 1927.

3) H. Bethe, Naturwissenschaften 15, S. 787. 1927 (zitiert als „Bethe I“).

bezüglich der beiden Komponenten parallel zur Oberfläche übereinstimmt:

$$A \mathfrak{L} = \alpha t$$

Der Ausbreitungspunkt ist also bereits auf eine ganz bestimmte, durch den Punkt $-\frac{\alpha\tau}{2\pi}$ gelegte *Normale* zur Kristalloberfläche (am reziproken Gitter) festgelegt. Auf dieser Parallele existiert *ein Punkt L*, den wir mit Ewald als *Lauepunkt* bezeichnen, für welchen der *Ursprung* des reziproken Gitters *genau* auf die



L Lauepunkt; O Nullpunkt des reziproken Gitters; H „Reflexionspunkt“ (h_1, h_2, h_3). b_3 Richtung der Kristallnormale (geom. Ort für Laue- und Ausbreitungspunkt). $\frac{x}{2\pi}$, $\frac{x-\zeta}{2\pi}$: vom Lauepunkt aus gezogene Ausbreitungsektoren

Fig. 2

Ausbreitungskugel zu liegen kommt. In diesem Fall wird der Gitterpunkt $\mathfrak{h}$ einen ganz bestimmten Abstand von der Ausbreitungskugel besitzen, den wir als *Anregungsfehler* bezeichnen. Bezeichnen wir den Fahrstrahl vom Lauepunkt zum Ursprung des reziproken Gitters mit $\frac{\mathfrak{C}}{2\pi}$, so gilt also:

$$(32) \quad \mathfrak{C} = A \mathfrak{L} + \gamma_0 n; \quad \gamma_0 = \sqrt{\kappa^2 - A^2}, \quad |\mathfrak{C}| = \kappa$$

$$(32a) \quad |\zeta + 2\pi \mathfrak{h}| = \kappa - \zeta.$$

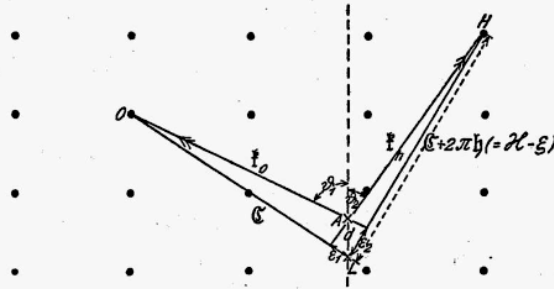
ζ stellt den Resonanzfehler ε_2 dar, für den Fall, daß ε_1 Null ist. ζ steht in einer einfachen Beziehung zu κ , wenn wir in Über-

Nunmehr gehen wir an die Behandlung des Gleichungssystems (18). Wir nehmen dazu, wie schon in § 2 S. 63 angedeutet, die Amplituden der beiden Wellen mit den Ausbreitungsvektoren

$$(34a) \quad \begin{cases} \mathfrak{k}_0 = \alpha t + (\gamma_0 - d) n, & k_0 = \kappa - \varepsilon_1 \\ \mathfrak{k}_h = \mathfrak{k}_0 + 2\pi \mathfrak{h}, & k_h = \kappa - \varepsilon_2 \end{cases}$$

als groß gegenüber allen anderen an und nennen sie der Hervorhebung halber c_1 und c_2 . Dann berechnen wir alle übrigen Amplituden in der Weise aus den Gleichungen

$$(18) \quad \psi_g (\kappa^2 - k_g^2) = - \sum v_g \psi_{g-s},$$



Anpassung d und Resonanzfehler ε_1 und ε_2

Fig. 3

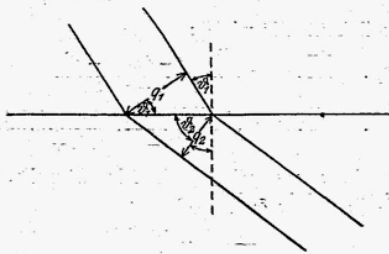
daß wir in den rechts stehenden Summen nur die Glieder berücksichtigen, welche die beiden großen Amplituden c_1 und c_2 der beiden Hauptwellen enthalten. Damit erhalten wir für die „kleinen“ Amplituden die Werte

$$(35) \quad \psi_g = - \frac{c_1 v_g + c_2 v_{g-h}}{\kappa^2 - k_g^2}$$

die wir nunmehr in diejenigen beiden Gleichungen des Systems (18) einsetzen können, welche sich auf die beiden Hauptwellen beziehen. Es sind dies die Gleichungen:

$$\begin{aligned} \psi_0 (\kappa^2 - k_0^2) &= 2c_1 \kappa \varepsilon_1 = - \sum \psi_s v_{-s} \\ \psi_h (\kappa^2 - k_h^2) &= 2c_2 \kappa \varepsilon_2 = - \sum \psi_s v_{h-s} \end{aligned}$$

Nun interessiert uns aber weniger der mathematische Ausdruck für die verschiedenen Elektronenwellen, sondern vor allem die Zahl der in der Zeiteinheit durch den Querschnitt jedes Strahls transportierten Elektronen, die „Stromstärke“ S . Diese berechnet sich als Produkt der Elektronengeschwindigkeit mit dem Querschnitt des Elektronenstrahls und der Zahl der pro Volumeinheit vorhandenen Elektronen. Die Elektronengeschwindigkeit ist der Wellenzahl direkt (der Wellenlänge umgekehrt) proportional, der Querschnitt des Elektronenstrahls ist direkt proportional dem Cosinus des Winkels zwischen ihm und der Normalen auf der Kristalloberfläche (vgl. Fig. 4), das



Querschnitt der Strahlenbündel und Richtungscosinus: $\frac{q_1}{q_2} = \frac{\cos \vartheta_1}{\cos \vartheta_2}$

Fig. 4

Produkt beider Größen ist also der Normalkomponente des Ausbreitungsvektors proportional. Die Zahl der Elektronen pro Volumeinheit wird bekanntlich durch das Quadrat des absoluten Amplitudenbetrages der de Brogliewelle gegeben. Wir bezeichnen die Stromstärke des Primärstrahls im Vakuum

$$(56) \quad C_0^2 T_1 = S_0$$

und haben nun zwei Fälle zu unterscheiden:

1. *Außerhalb* des Gebietes der *Selektivreflexion*¹⁾ ist $\bar{W} > 1$, also η und ω reell. Die Intensitäten des *Primärstrahls*

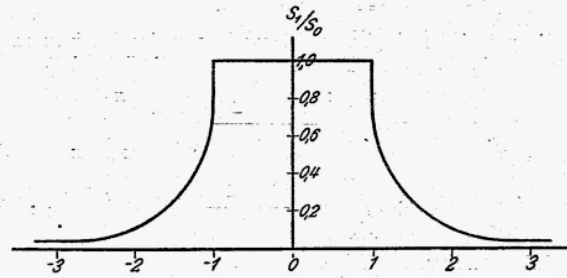
$$(56a) \quad s_1 = \frac{W^2 - \cos^2 \omega (H - \kappa)}{W^2 - \cos^2 \omega H} \cdot S_0$$

1) Die gleichen Überlegungen gelten *stets* für den Lauefall (vgl. S. 79).

$$(64) \quad \frac{i}{\Delta l} = \frac{\kappa_0}{2 \Delta \kappa} \sqrt{\frac{8}{4}} = \frac{\kappa_0^2}{2 V_{12}} \sqrt{\frac{8}{4}} \cdot \frac{1 + \cos \vartheta}{\sqrt{\cos \vartheta}}$$

und hat nach § 11 einen numerischen Wert von etwa 100 für Wellenlängen $\lambda =$ etwa 1 Å. Im übrigen entspricht der Intensitätsverlauf des reflektierten Strahls der von Ewald¹⁾ berechneten Kurve (vgl. Kurve 4).

Die hier aufgestellten Intensitätsformeln dürfen natürlich in *keiner Weise genau* genommen werden. Einmal sind die



Reflektierte Intensität in Abhängigkeit vom Selektionsfehler W

Fig. 5

Größen V_{11} , V_{12} , V_{22} durchaus nicht, wie hier angenommen, von der Wellenzahl κ unabhängig. Daneben geht aber sicher ein großer Teil der Intensität durch die hier vernachlässigten „schwachen Strahlen“ des dynamischen Wellensystems verloren, ein noch erheblicherer scheidet durch unelastische Stöße aus der Zahl der überhaupt beobachteten Elektronen aus. Dazu kommt, daß der einfallende Elektronenstrahl durchaus nicht exakt aus Elektronen einer bestimmten Richtung besteht. Dieser Umstand *verbreitert* natürlich das Wellenlängengebiet, auf das der Kristall mit einer bestimmten Netzebene anspricht, erheblich, und vermindert gleichzeitig die Intensität des reflektierten Strahls im Verhältnis zum einfallenden. Außerdem ist noch der Einfluß der Temperatur (Debye-faktor) unberücksichtigt geblieben, der gleichfalls den reflektierten Strahl schwächt.

1) Ewald III, a. a. O., S. 593, Fig. 19 (Braggfall).

In der „zweiten Näherung“ waren die Werte dieser Größen

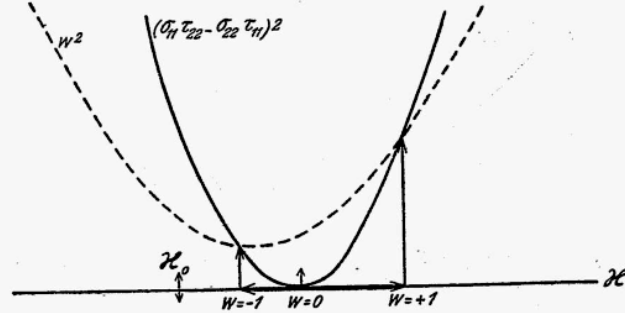
$$\begin{aligned} \sigma_{11} &= \kappa \cos \vartheta_1 & \sigma_{12} &= 0 & \sigma_{22} &= \kappa \cos \vartheta_2 \\ \tau_{11} &= \frac{1}{2} V_{11} & \tau_{12} &= \frac{1}{2} V_{12} & \tau_{22} &= \frac{1}{2} V_{22} - \kappa \zeta \end{aligned}$$

Bezeichnen wir noch in Analogie zur zweiten Näherung

$$(71) \quad w^2 = -(\sigma_{11} \tau_{12} - \sigma_{12} \tau_{11})(\sigma_{22} \tau_{12} - \sigma_{12} \tau_{22})$$

und

$$(72) \quad W = \frac{\sigma_{11} \tau_{22} - \sigma_{12} \tau_{12}}{w},$$



Das Gebiet der Selektivreflexion in der dritten Näherung:

$$\left. \begin{aligned} (\sigma_{11} \tau_{22} - \sigma_{12} \tau_{12})^2 \\ w^2 = -(\sigma_{11} \tau_{12} - \sigma_{12} \tau_{11})(\sigma_{22} \tau_{12} - \sigma_{12} \tau_{22}) \end{aligned} \right\} \text{ als Funktion der Wellenzahl } \kappa.$$

Zwischen den beiden Schnittpunkten der Kurven ($W^2 = 1$) liegt (asymmetrisch zu $W = 0$) das Gebiet der Selektivreflexion (stark ausgezogene κ -Achse)

Fig. 6

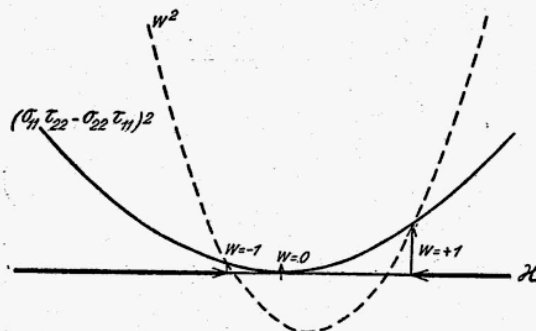
so erhalten wir die Lösung der Dispersionsgleichung in der zu (46) analogen Form

$$(73) \quad d = \frac{\sigma_{12} \tau_{12} - \sigma_{22} \tau_{11}}{\sigma_{11} \sigma_{22} - \sigma_{12}^2} + \frac{w}{\sigma_{11} \sigma_{22} - \sigma_{12}^2} (W \pm \sqrt{W^2 - 1}).$$

Es läßt sich zeigen, daß w im allgemeinen für den Braggfall reell ist, und etwa einen Verlauf hat, wie ihn Fig. 6 zeigt. Dann gibt es auch in der dritten Näherung ein Gebiet der „Selektivreflexion“ ($W < 1$). Begrenzt wird es von zwei Wellenzahlen κ_1 und κ_2 , die sich als Wurzeln der in κ quadratischen Gleichung $W^2 = 1$ ergeben. Die Intensitätsformeln des vorigen Para-

graphen behalten ihre Gültigkeit, nur ändern w und W ihre Bedeutung.

Bisweilen ist aber auch w imaginär. Dann könnte es unter Umständen vorkommen, daß w^2 und $(\sigma_{11}\tau_{22} - \sigma_{12}\tau_{12})^2$ den in Fig. 7 dargestellten Verlauf haben. Dann ist das Gebiet der Selektivreflexion ($W < 1$) in zwei voneinander getrennte Ge-



Anomaler Verlauf der in Fig. 6 dargestellten Kurven (Bezeichnungen siehe dort). Aufspaltung des Gebiets der Selektivreflexion (stark ausgezogene χ -Achse)

Fig. 7

biete geteilt. Es wäre denkbar, daß die doppelt beobachtete Reflexion (642) im {110}-Azimut bei 170 und 188 Volt auf eine derartige Aufspaltung zurückzuführen wäre.

§ 8. Beziehung zu den Kontinuums-theorien der Röntgenreflexion. Auslöschungsgesetze

Unsere hier entwickelte Theorie der Elektronenbeugung an Kristallen steht in Beziehung zu den sogenannten „Kontinuums-theorien“ der Röntgenreflexionen, welche die Fourierdarstellung der Elektronendichte (Zahl der Elektronen pro Volumeinheit) im Kristall:

$$(74) \quad \rho^E(\mathbf{r}) = \sum_{\mathbf{g}} \rho_{\mathbf{g}}^E e^{2\pi i \mathbf{g} \cdot \mathbf{r}}$$

zur Berechnung der für die Intensität der gebeugten Röntgenstrahlen wichtigen *Atomformfaktoren* benutzen. W. H. Bragg¹⁾

1) W. H. Bragg, Phil. Trans. Roy. Soc. A. 215. S. 253. 1925.

Theorie der Beugung von Elektronen an Kristallen 127

bis zur 17. bis 70. Atomschicht in den Kristall eindringen. Als Grenze des Eindringens wurde dabei die Stelle angenommen, an der der Elektronenstrom auf 10 Proz. seines ursprünglichen Wertes abgesunken ist. Diese Begrenzung der Eindringungstiefe der Elektronen rührt allein von dem exponentiellen Intensitätsabfall im Kristall wegen der selektiven Reflexion her. Wie weit dagegen im Normalfall die Elektronen in den Kristall einzudringen vermögen, hängt von ihrer Absorption

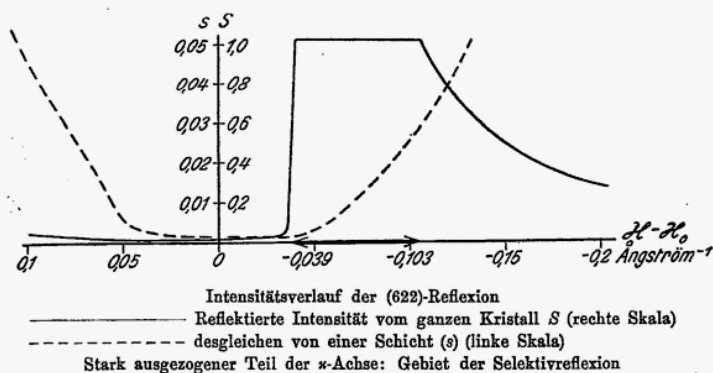


Fig. 8

vermittels unelastischer Stöße ab, die wir vernachlässigt haben. Aus dem Gelingen der Versuche von Rupp ist zu schließen, daß diese Absorption eher von etwas *geringerer* Größenordnung ist als der durch Selektivreflexion bedingte Intensitätsabfall. Wertet man andererseits die experimentelle Kurve der Intensität aus, die Davisson und Germer für die (331)-Reflexion (bei 54 Volt) gefunden haben, so findet man unter der Annahme, daß die Elektronen einfach proportional dem im Kristall durchlaufenen Weg absorbiert werden, daß 54-Volt-Elektronen etwa bis zur 6. Schicht in den Kristall eindringen. Doch sind hier Absorption durch unelastische Stöße, Intensitätsabfall durch Selektivreflexion und Störung durch inhomogene Einfallrichtung verquiekt und außerdem nimmt die Eindringungstiefe sicher sehr stark mit der Elektronenenergie ab.

Tabelle 1

Beobachtete Elektronenreflexionen (nach Davisson und Germer)
zugehörige Brechungsindizes μ und Kristallpotentiale V_0

Beobachtungsergebnisse					Theoretische Folgerungen				
Energie d. Elektr. im Vakuum E (in Volt)	Azimut	Ordnung	Bewegungswinkel Θ	Wellenlänge im Vakuum λ (in Å)	Millersche Indizes d. Reflexion ($h_1 h_2 h_3$)	Wellenlänge im Kristall λ (in Å)	Brechungsindex $\mu = \frac{\lambda}{\lambda'}$	Mittleres elektrostatisches Potential V_0 (in Volt) „Alte“ „Neue“ Zuordnung	
54	[111]	1	50°	1,670	(331)	1,494	1,118	- 18	+ 13
65	[100]	1	44°	1,520	(422)	1,353	1,123	- 21	+ 16,5
106	[111]	1	28°	1,190	(442)	1,128	1,048	- 39	+ 11,5
126	[100]	1	28°	1,093	(533)	1,040	1,047	- 44,5	+ 11
143	[110]	1	56°	1,024	(531)	1,044	(0,979)	- 43	- 7
160	[100]	2	60°	0,968	(622)	0,923	1,045	- 26	+ 14
[170]	[110]	[1]	[46°]	[0,940]	[642]	[0,871]	[1,075]	[- 34]	[+ 26]*
174	[111]	1	22°	0,928	(553)	0,893	1,038	- 56,5	+ 13
(179)	(110)	(1)	(44,5°)	(0,915)	(642)	(0,871)	(1,050)	(- 43)	(+ 17)*
181	[111]	2	55°	0,910	(551)	0,877	1,036	- 37	+ 13
[188]	[110]	[1]	[43°]	[0,894]	[642]	[0,871]	[1,024]	[- 52]	[+ 8]*
190	[100]	1	20°	0,889	(644)	0,836	1,060	- 53	+ 23
230	[100]	2	46°	0,807	(733)	0,788	1,023	- 56	+ 10
248	[111]	2	44°	0,778	(662)	0,748	1,036	- 54	+ 19
248	[110]	1	34°	0,778	(753)	0,734	1,057	- 52	+ 28
258	[111]	1	< 20°	0,763	(664)	0,739	1,032	- 71	+ 15
292	[100]	1	< 20°	0,717	(755)	0,696	1,028	- 79	+ 15
310	[100]	3	70°	0,696	(822)	0,676	1,028	- 21	+ 16
312	[100]	2	37°	0,693	(844)	0,676	1,023	- 72	+ 14
343	[111]	2	34°	0,662	(773)	0,645	1,022	- 76	+ 15
					(660)	0,676	0,971	- 58	[- 21]
347	[111]	3	62°	0,658	(771)	0,615	1,068	- 21	[+ 48]
					Mittel	0,646	1,019	- 40	+ 14
370	[100]	3	57°	0,637	(933)	0,615	1,035	- 44	+ 25
								Mittel +14,8	

*) Beobachtet sind zwei Strahlen bei 170 und 188 Volt. Zur Berechnung des Mittels von V_0 wurde 179 Volt zugrunde gelegt.

Tab. 1 enthält¹⁾ die von Davisson und Germer²⁾ beobachteten selektiv reflektierbaren Wellenlängen sowie die Richtung³⁾, in die sie reflektiert werden, sodann die Millerschen Indizes h_1, h_2, h_3 der reflektierenden Netzebene (nach der

1) Die Tabelle ist auszugsweise bereits in Bethe II enthalten.

2) Vgl. Davisson und Germer I und II, a. a. O.

3) Vgl. S. 55, Fußnote 4.

Tabelle 2

Nicht beobachtete Elektronenreflexionen

Ordnung n	Azimut	Reflexions- indizes ($h_1 h_2 h_3$)	Wellen- länge λ im Kristall	$E + V_0$ Elektronen- energie im Kristall	$\sin \theta$ (Beugungs- winkel im Kristall)	μ berechnet mit $V_0 = 14,8$ Volt)	$\sin \Theta$	$\Theta^*)$
1	[111]	220	2,03	36	0,945	1,31	1,24	—
1	[100]	311	1,85	44	0,86	1,23	1,04	ca. 90°
1	[110]	420	1,22	100	0,98 ₅	1,08	1,06	—
1	[331]	620	0,81	226	0,99 ₅	1,03 ₅	1,03	—
1	[210]	640	0,78	244	0,96	1,03	0,99	ca. 80°
2	[111]	440	1,02	144	0,95	1,05 ₅	1,00	ca. 90°
2	[100]	511	1,06	134	0,98 ₅	1,06	1,05	—
3	[111]	551	0,72	289	1,00	1,02 ₅	1,02 ₅	—
3	[100]	711	0,72	289	1,00	1,02 ₅	1,02 ₅	—

*) Ein Strich bedeutet, daß der Strahl wegen Totalreflexion nicht beobachtbar ist. Andernfalls ist der zu erwartende Ausfallswinkel Θ angegeben.

jedes Azimut die längsten nach der Gittertheorie zu erwartenden selektiv reflektierbaren Wellen: Im (111)- und (110)-Azimut müssen die längsten Wellen erster Ordnung Totalreflexion erleiden, können also gar nicht beobachtet werden, während der entsprechende Elektronenstrahl im (100)-Azimut streifend ins Vakuum austreten muß. Gerade hier haben aber Davisson und Germer einen besonders starken „grazing beam“ beobachtet.¹⁾ Nicht ganz so gut ist die Übereinstimmung mit der Theorie in den höheren Ordnungen. So sollte der an der Ebene

1) „Grazing beams“ (streifend austretende Elektronenstrahlen) sind von Davisson und Germer in jedem der drei Hauptazimute beobachtet. Ihr Auftreten wird nach Davisson und Germer im allgemeinen dadurch erklärt, daß der Weg der Elektronen von einer Atomschicht zur nächsten bei streifendem Strahlengang sehr groß ist, und daß daher ein großer Teil der von den tieferen Schichten des Kristalls gestreuten Elektronen absorbiert wird, ehe er an die Oberfläche gelangt. Den Hauptteil der Intensität des beobachteten Strahls liefern infolgedessen die an der Oberflächenschicht gebeugten Elektronen, und es ist also für das Zustandekommen eines streifend austretenden Strahls *nicht* unbedingt erforderlich, daß die von sämtlichen Schichten des Kristalls gestreuten Wellen miteinander in Phase, mit a. W., daß die Laueschen Gleichungen der Raumgitterbeugung erfüllt sind. Doch wird in diesem Falle die *Intensität* des beobachteten, streifend ausfallenden Strahls besonders groß sein, und dies ist im (100)-Azimut gerade der Fall.

Tabelle 3

Fourierkoeffizienten des Potentials und der Ladungsdichte für Nickel

G^2	$g_1 g_2 g_3$	Kern	Beiträge zu ϱ_g					ϱ_g in Elementar- ladung, pro Atomvolum	V_g in Volt	v_g in Angström ⁻²
			K	L	M^I	M^{II}	M^{III}			
3	111	28	-2	-7,8	-1,45	-4,15	-4,85	7,75	13,4	3,57
4	200	28	-2	-7,7	-1,3	-3,45	-3,85	9,7	12,6	3,31
8	220	28	-2	-7,4	-0,8	-2,15	-1,75	13,9	9,05	2,38
11	311	28	-2	-7,3	-0,55	-1,35	-0,9	15,9	7,5	1,98
12	222	28	-2	-7,2	-0,45	-1,15	-0,7	16,5	7,15	1,88
16	400	28	-2	-7,0	-0,25	-0,6	-0,15	18,0	5,85	1,54
19	331	28	-2	-6,9	-0,15	-0,3	+0,1	18,75	5,15	1,35
20	420	28	-2	-6,8	-0,1	-0,25	+0,15	19,0	4,95	1,30
24	422	28	-1,95	-6,5	-0,05	-0,1	+0,25	19,65	4,25	1,12
27	511 333	28	-1,95	-6,4	0,0	0,0	+0,3	19,95	3,85	1,02

G^2	$g_1 g_2 g_3$	ϱ_g in Elementar- ladung, pro Atomvolum	V_g in Volt	v_g in Angström	G^2	$g_1 g_2 g_3$	ϱ_g in Elementar- ladung, pro Atomvolum	V_g in Volt	v_g in Angström
					80	840	21,55	1,40	0,37
32	440	20,2	3,29	0,87	83	753 911	21,65	1,36	0,36
35	531	20,4	3,04	0,80	84	842	21,65	1,34	0,35
36	600 442	20,4	2,96	0,78	88	664	21,75	1,29	0,34
40	620	20,6	2,68	0,71	91	931	21,95	1,26	0,33
43	533	20,65	2,50	0,66	96	844	22,05	1,19	0,31
44	622	20,65	2,45	0,64	99	933 771 755	22,10	1,16	0,30
48	444	20,8	2,26	0,59	100	10, 0, 0 860	22,20	1,16	0,30
51	551 711	20,9	2,14	0,56	120	10, 4, 2	22,8	0,99	0,26
52	640	20,9	2,09	0,55	150	—	23,5	0,82	0,21
56	642	21,0	1,95	0,51	200	10,10,0 14,2,0 10, 8, 6	24,45	0,76	0,17
59	553 731	21,1	1,86	0,49	300	—	25,55	0,44	0,12
64	800	21,1	1,72	0,45	500	—	26,45	0,27	0,072
67	733	21,25	1,65	0,43	1000	—	26,85	0,14	0,037
68	820 644	21,35	1,63	0,43	5000	—	27,6	0,03	0,007
72	822 660	21,4	1,55	0,41	30000	—	28,0	0,005	0,001
75	751 555	21,4	1,49	0,39					
76	662	21,4	1,47	0,39					

§ 12. Numerische Berechnungen der Größen der Elektronenbeugungstheorie

Wir wollen die im vorigen Paragraphen gewonnenen Fourierkoeffizienten jetzt dazu benutzen, um für einen der von Davisson und Germer beobachteten „Lauepunkte“ die Be-

von 300 bis 50 Volt. Die Halbwertsbreite dürfte etwa die Hälfte davon betragen, d. h. man wird für 160 Volt mit etwa 5 Proz. Halbwertsbreite rechnen können ($\frac{\lambda}{\Delta\lambda} = 20$).

Zwicky¹⁾ hatte aus der ersten Näherung nach der Bornschen Methode²⁾ für das Auflösungsvermögen den offensichtlich viel zu hohen Wert 10^8 gefolgert. Aus unserer zweiten Näherung würde $\frac{\lambda}{\Delta\lambda} = 60$ folgen.

3. Die *Intensität* des reflektierten Strahles ist in der Tab. 5 in Abhängigkeit von der Wellenzahl κ gegeben, und zwar sowohl für einen „dicken“ Kristall wie für einen solchen, der nur

Tabelle 4
Intensitätsverlauf der (622)-Reflexion
(Dritte Näherung)

$\kappa - \kappa_0$ im $\text{\AA}^{-1}$	w in $\text{\AA}^{-2}$	W	$\kappa_0 = 6,80 \text{\AA}^{-1}$		Eindringungs- tiefe	
			Reflektierte 1 Atom- schicht	Intensität ganzer Kristall	$\text{\AA}$	Atom- schichten
(+ 0,1)	(31,7)	(2,96)	(0,042)	(0,059)	—	—
0,05	15,1	4,20	0,008	0,029	—	—
0	7,09	4,60	0,002	0,024	—	—
— 0,03	4,55	3,08	0,001	0,044	—	—
— 0,035	7,07	1,54	0,002	0,240	—	—
— 0,039	8,80	1,00	0,003	1,000	133	65
— 0,045	11,22	0,43	0,005	1,000	66	32
— 0,053	14,00	0,00	0,008	1,000	50	25
— 0,07	19,85	0,53	0,015	1,000	38	19
— 0,09	26,5	0,86	0,027	1,000	35	17
— 0,103	30,6	1,00	0,035	1,000	38	19
— 0,13	39,3	1,20	0,055	0,444	—	—
(— 0,15)	(45,7)	(1,31)	(0,074)	(0,353)	—	—
(— 0,2)	(61,5)	(1,47)	(0,125)	(0,269)	—	—

aus einer einzigen Schicht von Atomen besteht. Die Kurve 8 veranschaulicht die Verhältnisse, doch ist dabei zu berücksichtigen, daß der aus der dritten Näherung berechneten Intensität nur eine *bedingte Richtigkeit* zukommt.

Das *Reflexionsvermögen* einer einzigen Schicht bewegt sich im Gebiet der Selektivreflexion zwischen 0,003 und 0,03. Damit gleichbedeutend ist die Feststellung, daß die Elektronen

1) F. Zwicky, Proc. Nat. Acad. Sci. **13**, S. 518—525. 1927.

2) M. Born, Ztschr. f. Phys. **38**, S. 803. 1926.

Appendix: Historical Terminology and Modern Equivalents

Bethe's 1928 paper uses the terminology of the Ewald-Laue era of dynamical diffraction. This translation deliberately preserves that historical vocabulary rather than modernizing it. The table below gives, for the reader's convenience, the approximate modern (TEM / dynamical-theory) counterparts. These are aids to reading, not synonyms to be read back into Bethe's text.

German (Bethe 1928)	English (this translation)	Modern equivalent / note
Ausbreitungsvektor	propagation vector	wave vector $\mathbf{k}$
Wellenzahl $k_0 = 2\pi/\lambda_0$	wave number	magnitude of wave vector $ \mathbf{k} $
Primärstrahl / Primärwelle	primary beam / primary wave	incident beam / incident wave
reflektierter Strahl	reflected beam	diffracted beam
Resonanzfehler (ϵ)	resonance error	excitation error s_1 (approx.)
Anregungsfehler	excitation error	excitation error s_1
Reflexionsvermögen	reflecting power	(integrated) reflectivity
Selektivreflexion	selective reflection	Bragg diffraction
Elementarzelle	elementary cell / unit cell	unit cell
reziprokes Gitter	reciprocal lattice	reciprocal lattice
Laue-Gleichungen	Laue equations	Laue equations
Dispersionsgleichung	dispersion equation	dispersion relation / surface
Brechungsindex	refractive index	refractive index
Fourierkoeffizient	Fourier coefficient	cf. structure factor
des Potentials	(of the potential)	
dynamische Theorie	dynamical theory	dynamical diffraction theory
Grenzbedingungen	boundary conditions	boundary conditions
Beugung (am Kristall)	diffraction (by the crystal)	diffraction
Auflösungsvermögen	resolving power	resolving power
Oberflächengitter	surface grating	(historical usage)

Note: Bethe distinguishes Resonanzfehler from Anregungsfehler; modern electron diffraction generally uses a single excitation error s_1 . The distinction is preserved in the translation.